

GE Healthcare

LightSpeed VCT Discovery CT750 HD Pre-Installation Manual



OPERATING DOCUMENTATION



5220253-1EN
Revision 4

The information in this manual applies to the following GE Healthcare CT Scanners:

- LightSpeed VCT (with GOC6 console)
- Discovery CT750 HD

The information in this manual does NOT apply to non-fixed (mobile) installations.

IMPORTANT PRECAUTIONS

LANGUAGE

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners, oder des Patienten durch Stromschläge, oder mechanische und sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다.
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRDINJUMS

(LV)

- Šī apkālpes rokasgrāmata ir pieejama tikai anglu valodā.
- Ja klienta apkālpes sniedzējam nepieciešama informācija citā valodā, nevis anglu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkālpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkālpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT-BR)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros.

ATENÇÃO

(PT-PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE

(SR)

- Ovo servisno uputstvo je dostupno samo na engleskom jeziku.
- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge.
- Ne pokušavajte da opravite uređaj ako niste procitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

You should closely examine all packages at time of delivery. If you notice any damage, have the notation "Damage in Shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by any General Electric representative or hospital receiving agent. Whether noted or concealed, you MUST report damage to the carrier immediately upon discovery and within 14 days after receipt, and you must hold the contents and containers for inspection by the carrier. A transportation company will not pay a claim for damage if you do not request an inspection within this 14-day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

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CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment, if not properly used, may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, GE Healthcare, will gladly assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, GE Healthcare, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers: please contact the GE Healthcare Sales or Service representatives. GE personnel: please use the GE Healthcare iTrak/PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
1.0	03/17/08	Initial Release, for NPI.
2.0	8/20/08	Updatated cover art, name, tables, Interconnect drawings, cooling requirements, and service clearance requirements. Added new languages in Precautions, Completed updates to make this document compatible for LightSpeed VCT w/GOC6 and Discovery CT750 HD. Chapter 5, Section 1.0: Updated Measuring Service Clearances Chapter 14, Section 3.2 Updated GE Healthcare-supplied Cables (long).
3.0	10/08/08	Updated per FCTge40232, and per FCTge41654.
4.0	11/19/08	FCTge43607 - <i>Floor Vibration Specifications</i>

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Table of Contents

Chapter 1

Introduction 19

Section 1.0: What is Pre-Installation?	19
Section 2.0: What is Pre-Installation Work?	19
Section 3.0: Pre-Installation Tools	19
3.1 Customer Installation Check List	20
3.1.1 Pre-Installation Manual Guide	20
Section 4.0: Pre-Installation Checklist	21

Chapter 2

Installation Types 25

Section 1.0: Determining Your Installation Type	25
1.1 How to Determine the Best Installation Type for Your Site	25
1.2 Typical Installations	25
1.3 Construction Site Installations	25
1.3.1 Full Construction Site with Completed Radiology Area	26
1.3.2 Full Construction Site with Limited Delivery Access	26
1.4 Relocatable Building Installations	26
1.5 Upgrade Installations	27
1.6 Quick Installations	27
1.6.1 Requirements	27
1.6.2 Restrictions	27
1.7 Two-Step (Temporary) Installations	28

Chapter 3

System Siting Requirements 29

Section 1.0: System Siting Requirements	29
Section 2.0: Customer System Siting Requirements	30
2.1 Regulatory Requirements	30
2.2 Electrical	30
2.3 Structural	31
2.3.1 Dust and Air Quality	31
2.3.2 Environmental Influences Considered	31
2.3.3 Finished Floor Requirements	31
2.3.4 Finished Walls Requirements	32
2.4 Radiation Protection	32
2.5 Environmental	32
2.6 Options	32
2.7 Clearances	32
2.8 Network	32
2.9 Chemical Contamination	32

2.10	Delivery	33
Section 3.0: Site Readiness - GE's Responsibilities		34
3.1	Project Manager of Installation Tasks	34
3.1.1	Pre-Installation Delivery Tasks	34
3.1.2	Site Review with Customer	34

Chapter 4

Regulatory Requirements 35

Section 1.0: Regulatory Clearances		35
1.1	Regulations	35
1.2	Clearance Requirements	35
1.2.1	Minimum Regulatory Workspace Clearances by Major Subsystem	37
1.3	How to Measure	39
1.4	NEC Conduit and Duct Fill Rate	40
Section 2.0: Terms and Definitions		41

Chapter 5

Service Clearance Requirements 43

Section 1.0: Measuring Service Clearances		43
Section 2.0: Special Service Clearances		44
2.1	Cover Removal	44
2.2	Power Distribution Unit (NGPDU)	44
2.3	Console	44
2.4	Storage Cabinet	45

Chapter 6

Room Sizes 47

Section 1.0: Room Dimensions		47
1.1	Suggested Room Size	47
1.2	Typical Room Size	47
1.2.1	Minimum Room Size	47
Section 2.0: Suggested and Typical Room Layouts		49
Section 3.0: Minimum Room Layouts		51
Section 4.0: Control Room Considerations		53
4.1	Typical Control Room Layout	53

Chapter 7

System Component Dimensions 55

Section 1.0 Minimum Operating Clearances		55
1.1	Ceiling Pedestal Mount Installation	55
1.2	Injector Control Installation	55
1.3	System Operational Clearances	56
Section 2.0: Component Dimensions		57

2.1	System Dimensions	57
2.2	Table and Gantry Dimensions	59
2.3	Power Distribution Unit Dimensions (NGPDU-4).	61
2.4	Power Distribution Unit Dimensions (NGPDU-6).	62
2.5	Operator Console Dimensions and Ergonomic Specification	63

Chapter 8 Structural and Mounting Requirements 65

Section 1.0: Overview	65
1.1 Importance of Meeting Structural Requirements	65
1.1.1 Levelness, Vibration, and Floor Loading	65
1.1.2 Seismic Loading	65
1.1.3 Anchoring.	65
Section 2.0: Ceiling Requirements	65
2.1 Mounting Plate.	66
2.2 Pedestal Length Requirement	67
2.3 Electrical Box Requirement	68
Section 3.0: Minimum Floor Requirements	68
3.1 Floor Levelness Specifications	68
3.1.1 Critical Specifications.	68
3.1.2 Floor Levelness Guidelines	68
3.1.3 Measuring Floor Levelness	68
3.2 Floor Vibration Specifications	70
3.2.1 Requirements	70
3.2.2 Sources of Floor Vibration	71
Section 4.0: Floor Loading and Component Weights	72
4.1 Floor Loading and Anchoring Guidelines.	73
Section 5.0: GE-Supplied Anchoring	74
5.1 Specifications of GE-supplied Anchors	75
5.2 Requirements for Using GE-supplied Anchors	75
Section 6.0: Seismic Mounting	77
6.1 Overview	77
6.2 Center-of-Gravity Information	77

Chapter 9 Environmental Requirements 83

Section 1.0: Temperature and Humidity Specifications	83
1.1 Temperature (Scan and Control Rooms).	83
1.2 Humidity (Scan Room & Control Room)	83
1.3 Other Guidelines	83
Section 2.0: Cooling Requirements	84
Section 3.0: Altitude.	85
Section 4.0: Electro-Magnetic Interference (EMI)	86
4.1 Gantry	86

4.2	Console / Computer Equipment	86
4.3	PDU	86
4.4	EMI Reduction	86
4.5	Equipment EMI “Envelopes”	87
Section 5.0: Electro-Magnetic Compatibility (EMC)		88
5.1	General Scope	88
5.2	Electromagnetic Emission	88
5.3	Electromagnetic Immunity	89
5.4	Installation Requirements and Environment Control	91
5.4.1	Cable Shielding and Grounding	91

Chapter 10

Radiation Protection Requirements 93

Section 1.0: Shielding Requirements	93
---	----

Chapter 11

Network Requirements 97

Section 1.0: Network Connections	97
1.1 Network Type	97
1.2 Network Speed	97
1.3 Network Cable Routing	97
Section 2.0: Customer Broadband Responsibilities	97
2.1 Contact GE to Find Zone Broadband Specialist	97
2.2 Provide GE with IT Contact Information for the Site	98

Chapter 12

Power Requirements 99

Section 1.0: Introduction	99
Section 2.0: System Input Power	99
2.1 Power Source Configuration	99
2.2 Rating	99
2.3 Regulation	100
2.4 Phase Imbalance	100
2.5 Sags, Surges and Transients	100
2.6 Grounding	100
Section 3.0: Recommended Power Distribution System	101
3.1 Using a Dedicated Distribution Transformer (Recommended)	101
3.2 Using an Existing Distribution Transformer	101
3.3 System Power Requirements	101
Section 4.0: Ground System	103

Chapter 13

Interconnection Data 105

Section 1.0: Introduction	105
Section 2.0: Component Designators	105
Section 3.0: Interconnect Runs, Wiring, and Cables	106
3.1 GE Healthcare Supplied (HD Standard Length 2281840-10)	106
3.2 GE Healthcare-Supplied (HD Optional, Long Run Length 2281840-9)	108
3.3 GE Healthcare-Supplied (VCT 64 Standard Length 2281840-7)	109
3.4 GE Healthcare-Supplied (VCT Optional, Long Run 2281840-6)	110
3.5 UPS Wiring Cables	111
3.6 A1 UPS	111
3.7 Contractor/Customer-Supplied	112
Section 4.0: Contractor Supplied Components	114
Section 5.0 Primary Power Disconnect	115
Section 6.0	
Scan Room Warning Light and Door Interlock	116

Chapter 14

Delivery and Storage Requirements 117

Section 1.0: Delivery to the Facility	117
1.1 Loading Dock Deliveries	117
1.2 Ground (Non-Loading Dock) Deliveries	117
1.2.1 Liftgate Truck	117
1.2.2 Tiltbed Truck	117
1.2.3 Forklift Truck	117
Section 2.0: Delivery to the Scan Suite	118
2.1 Lifting	118
2.1.1 Stairway Deliveries	118
2.2 Floor Protection	118
2.3 Un-loading and un-packing the System	118
Section 3.0: Dollies	119
3.1 Installations within the United States	119
3.2 Zero Clearance Dollies	119
3.3 Tilting Table Dollies	119
3.4 Installations Outside of the United States	119
Section 4.0: Gantry Delivery Considerations	120
4.1 Gantry Shipping State	120
4.2 Door Openings	120
4.3 Elevator Requirements	120
Section 5.0 Table Delivery Considerations	121
Section 6.0 Console Delivery Considerations	122
Section 7.0: Storage Requirements	123
7.1 Short-term Storage (Less than Six Months)	123
7.2 Construction-Site Storage	123

Section 8.0 Extreme Temperature Delivery and Storage.	123
---	-----

Chapter 15

Handling Requirements 125

Section 1.0: Transportation	125
Section 2.0: Handling Requirements	125
2.1 Avoid Dropping.	125
2.2 Avoid Shocks and Vibrations	125
2.3 Avoid Tipping	125
2.4 Inclines and Flat-bed Truck Removal.	126

Chapter 1

Introduction

This document contains the physical and electrical data necessary for planning and preparing a site for scanner installation. The responsibility of arranging and paying for this work rests solely with the purchaser.

Section 1.0: What is Pre-Installation?

Pre-installation work comprises any site preparation that requires completion prior to the installation of the LightSpeed VCT/Discovery CT750 HD scanner. This manual guides you through the pre-installation siting and regulatory requirements. Pay special attention to these chapters. However, bear in mind that neither this document, nor the video (DVD) included with the Pre-Installation Kit, answers all of your questions. Contact your GE Healthcare Project Manager of Installation (PMI) for answers to any additional questions.

Likewise, prior to any construction or approval, General Electric Headquarters Architectural Planning must review all CT site plans, preliminary concepts, and final working drawings. Contact your GE Project Manager of Installation (PMI) for complete information regarding your site-specific room layout.

Section 2.0: What is Pre-Installation Work?

Pre-Installation work includes:

- Site renovation.
- Alterations or modifications to products not specifically included in the sales contract.
- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of AWG stranded copper interconnection wiring, conforming to the following requirements:
 - * The electrical contractor shall ring out and tag all wires at both ends.
 - * Wires shall be continuous and without splices.
 - * Ground wires shall conform to product requirements.
 - * Color-coded wires shall be used whenever possible, to enable easier identification.
- All work shall conform to IBC (International Building Code) and local building and safety codes.

Note: GE Healthcare neither provides nor installs the wires, conduits, junction boxes, or ducts illustrated in this publication, unless specifically mentioned.

Section 3.0: Pre-Installation Tools

A list of primary customer tools for successfully completing the pre-installation process for a LightSpeed VCT/Discovery CT750 HD system appears below.

3.1 Customer Installation Check List

- **LightSpeed VCT/Discovery CT750 HD Pre-Installation Manual**
Provided in CD-ROM format.
- **Pre-Installation Check List**
Located on [page 21](#) of the Pre-Installation Manual, and on the Service CD.
- **Regulatory and Service Clearance Information**
Included in the Pre-Installation Manual.
- **LightSpeed VCT/Discovery CT750 HD Floor Template (P/N 5111499)**
Included with system, and also available from your PMI. Use this to determine equipment layout and anchoring locations.
- **Site Print**
Supplied by your PMI or sales rep. Must show actual room size, location of all equipment in the finished room, all service and operating clearances, and meet all regulatory requirements.

3.1.1 Pre-Installation Manual Guide

[Table 1-1](#) shows the location in this Pre-Installation Manual of the information necessary for fulfilling each the corresponding pre-installation requirements.

Installation Site Requirement Information	
<i>: Determining Your Installation Type, on page 25</i>	<i>: System Siting Requirements, on page 29</i>
<i>: Room Dimensions, on page 47</i>	<i>Structural and Mounting Requirements, on page 65</i>
<i>Regulatory Requirements, on page 35</i>	<i>Service Clearance Requirements, on page 43</i>
<i>Radiation Protection Requirements, on page 93</i>	<i>Network Requirements, on page 97</i>
<i>Environmental Requirements, on page 83</i>	<i>Power Requirements, on page 99</i>
<i>Delivery and Storage Requirements, on page 117</i>	<i>Handling Requirements, on page 125</i>
Contractors must complete ALL WORK before the scheduled delivery date.	

Table 1-1 Locations of Site Requirement Information in this manual

Section 4.0: Pre-Installation Checklist

Required Information for Site

Complete before the scheduled delivery date.

Hospital Name as it appears on the system screens:

Network ID numbers / IP addresses

PACS: _____

Camera: _____

AW: _____

Other - Specify type & ID: _____

Other - Specify type & ID: _____

Camera setup information: _____

AW Direct Connect address: _____

Do you want HIPAA enabled? No___ Yes___

Do you want automatic downloads enabled? No___ Yes___

GE	Cust	Dates
Y N	Y N	
	☐ ☐	Have the facilities department, contractor, and GE verified the project schedule?
	☐ ☐	Will the committed site-ready date be met?
☐ ☐	☐ ☐	Does the completion date for any/all construction meet or precede the delivery date?
☐ ☐	☐ ☐	Is the Power & Ground survey complete? Date: _____ Hospital contact: _____
	☐ ☐	Is the Site-Ready visit scheduled? Date: _____
	☐ ☐	Is the delivery date scheduled? Date: _____
	☐ ☐	Is the installation date scheduled? Date: _____
☐ ☐	☐ ☐	Is the installation timing determined? A: Weekdays___ B: Weekend___ C: Quick Install___ If B or C, have all sub-contractors been notified? No___ Yes___
☐ ☐	☐ ☐	Does the delivery and/or installation date require adjustment?
	☐ ☐	Is the first-use date scheduled? Date: _____
☐ ☐	☐ ☐	Applications/Training dates scheduled? On-Site Training Date: _____ Healthcare Institute Training Date: _____

Table 1-2 Schedule Date Commitments

GE	CUST	General / Site Requirements
Y N	Y N	<i>Must be completed five (5) weeks before scheduled delivery date.</i>
☐ ☐	☐ ☐	Were final drawings approved and distributed to the contractors?
☐ ☐	☐ ☐	Are final drawings “signed off” to approve equipment layout and orientation?
☐ ☐	☐ ☐	Do the actual room dimensions match those on the final drawings?
	☐ ☐	Has the radiologist health physician reviewed and approved the room layout and shielding requirements?
	☐ ☐	Have any additional requirements or questions about the installation been discussed with GE? List: _____ _____ _____ _____
	☐ ☐	Is there a person assigned to review and verify that all installation requirements are met? Name: _____
	☐ ☐	Have the specific site requirements been discussed with the contractors? Refer to the GE final drawings specifications. (See Table 1-1 .)
	☐ ☐	Has the responsibility of cabling, installing, and interfacing any accessories not on the order been discussed?
	☐ ☐	Are all third-party vendors identified, notified and scheduled? (Examples: Netcom, Medrad, etc.)
	☐ ☐	Have all regulatory requirements been met per <i>Regulatory Requirements</i> , on page 35?
	☐ ☐	Will existing network, broadband, and camera cable drops reach new locations and will they meet the requirements and function with LightSpeed VCT/Discovery CT750 HD? If not, what are the requirements? List: _____ _____ _____

Table 1-3 Site Planning

GE	Cust	Equipment
Y N	Y N	<i>Must be completed five (5) weeks before scheduled delivery date.</i>
☐ ☐	☐ ☐	Has the order been reviewed for completeness and compatibility with existing equipment? Typical equipment: Remote monitors ____ AW relocation ____ Cardiac option ____ Injectors ____
☐ ☐		Are interfaces to existing and/or new accessories ordered and planned for accordingly?
☐ ☐	☐ ☐	Have the following peripheral locations been included in the site drawings? EKG monitor____ Injector control____ Laser camera ____ UPS ____ 2 nd Monitor____
☐ ☐	☐ ☐	Will GE Healthcare provide additional services per contract negotiations?
☐ ☐		Are correct length cables on order?

Table 1-4 Equipment Compatibility

GE	Cust	Network Installation and Setup
Y N	Y N	<i>Must be completed five (5) weeks before scheduled delivery date.</i>
☐ ☐	☐ ☐	Have IP addresses and Host Names been obtained? No___ Yes ___
☐ ☐	☐ ☐	Will a network camera be used? No___ Yes ___
☐ ☐	☐ ☐	Required: Is the network installed, are network jacks installed, and is the entire network tested?
☐ ☐	☐ ☐	Required: Broadband VPN installed/setup?
☐ ☐	☐ ☐	Required: Are network software options ordered ___ HIS RIS option ___ DICOM print ___ AW ___
	☐ ☐	Optional: Has modem option ordered? ___ (Requires a site escalation)
☐ ☐	☐ ☐	Optional: Is the LightSpeed VCT/Discovery CT750 HD service telephone line identified and installed for InSite? (<i>Electrical, mechanical, etc.</i>)

Table 1-5 Network Connections

GE	Cust	Other
Y N	Y N	<i>Must be completed before the scheduled delivery date.</i>
	☐ ☐	Arrangements made in the schedule to allow adequate time for remodeling, if required (such as wall, floor, or ceiling repair work, painting, other cosmetic finishes)
	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to the installation of the new equipment?
☐ ☐	☐ ☐	Is de-installation of existing equipment required? No___ Yes ___ Removal date _____
☐ ☐	☐ ☐	Is there a trade-in of existing equipment? No ___ Yes ___ GoldSeal _____
☐ ☐	☐ ☐	Delivery route identified and verified with the proper hospital personnel? No___ Yes ___ Elevators and doors checked for size and weight constraints? No___ Yes ___
☐ ☐	☐ ☐	Have appropriate arrangements been made with traffic for delivery? No___ Yes ___
☐ ☐	☐ ☐	Will acceptance/performance testing or bio-medical testing be required? No___ Yes ___ Date: _____
☐ ☐	☐ ☐	Are trash and/or recycling bins available for the removal of papers, boxes, etc. during the installation? No___ Yes ___

Table 1-6 Miscellaneous Tasks

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Chapter 2

Installation Types

Section 1.0: Determining Your Installation Type

1.1 How to Determine the Best Installation Type for Your Site

Discuss installation options with your PMI to determine which of the installation types listed below best fits your site and schedule.

1.2 Typical Installations

Typical installations occur at established sites with finished, dust-free, occupancy-ready scan suites. The rooms range from suggested to minimum room sizes, and have NO ongoing construction on-site. A typical installation allows customers flexibility for room upgrades and site improvements. Upgrades and improvements may require additional planning prior to system delivery, especially when involving:

- Seismic approval
- Floor structural improvements
- HVAC improvements
- Electrical Improvements
- Review of scan room shielding requirements by a qualified radiological health physicist.

As with any installation, the final site design for a typical installation must meet all service and regulatory requirements detailed in this manual.

1.3 Construction Site Installations

A *construction installation* describes installations at sites without an occupancy permit, often with ongoing construction. In general, construction sites fail to meet the recommended specifications for delivery of the system. GE Healthcare does not recommend construction installations, as they can result in delays, increased costs, and possible damage to the system. When construction-site delivery proves unavoidable, the installation falls into one of two categories:

- Full construction site with completed radiology area
- Full construction site with limited delivery access

Review the following categories to determine which most closely matches the condition of the planned installation site.

1.3.1 Full Construction Site with Completed Radiology Area

This type of site consists of a finished, dust-free, occupancy-ready radiology suite. While there is no remaining construction in or around the scan suite area, there may be ongoing construction in other areas. At the time of delivery such sites feature:

- Dust control measures deployed in the radiology suite area.
- Scan suite access limited to a single entrance (see [Figure 2-1](#)).
- Radiology suite sealed off from the remaining construction area.
- Operational HVAC, with a positive air pressure within the radiology suite.

In addition, the radiology suite at such a site REMAINS in a dust-free, occupancy-ready state after delivery and throughout the remaining construction phase.

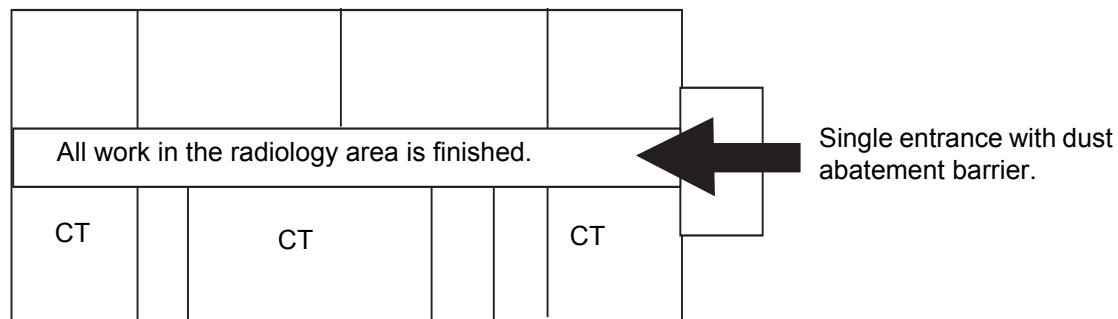


Figure 2-1 Full construction site with completed radiology area

1.3.2 Full Construction Site with Limited Delivery Access

This type of site allows delivery during ongoing construction of the radiology suite area. At such sites, delivery occurs prior to site completion, but the product remains stored until a finished, dust-free, occupancy-ready radiology suite area is ready. This type of site requires the LightSpeed VCT/Discovery CT750 HD to be delivered in a sealed package with dollies. Delivery to the storage area may require a lift truck or riggers. Installation work begins only when the site reaches the completed, dust-free, occupancy-ready radiology suite requirement.

Note: If delivery requires vertical or horizontal lifting, the PMI adds the necessary identifier to the order.

1.4 Relocatable Building Installations

A relocatable building is made in a factory and delivered to the site of its permanent location. Relocateable buildings qualify as fixed sites and must satisfy all of the requirements of a fixed site. The CT scanner and table must be mounted on a solid concrete floor. Any other floor type installations must be designed by the customer’s structural engineer and meet all GE Healthcare’s specifications listed in this manual.

Refer to [Structural and Mounting Requirements on page 65](#) of this manual for further information.

1.5 Upgrade Installations

Upgrade installations occur after the installation of another system. A change in the customer's needs requires the installation of additional equipment at the same site. For example, adding a PET system to an existing CT system.

To proceed with an upgrade installation, the customer's room size must be large enough to accommodate the new product without violating the regulatory and service requirements of the new product. When planning for an upgrade installation, siting requirements of the new equipment may exceed those of your existing system. Requirements needing additional consideration include:

- Floor thickness
- Room shielding
- Additional electrical capacity
- Increased cooling capacity
- Scan room shielding requirements

The final site design must include a room layout showing the equipment room with the recommended room size dimensions. All upgrade installations must meet all service and regulatory requirements detailed in this manual.

1.6 Quick Installations

Quick Installations involve sites requiring minimum room improvements. These installations typically consist of a weekend de-installation and room prep completion, with a next-business-day delivery and installation.

1.6.1 Requirements

A site must meet a number of requirements to qualify for a Quick Installation, including:

- Existing electrical disconnect device, wire size, and grounds must meet all requirements referenced in [Chapter 3, Section 2.2 on Page 30](#).
- Existing structural specifications met, including floor thickness, and all requirements referenced in [Chapter 3, Section 2.3 on Page 31](#).
- Existing HVAC capacity and regulation must meet all requirements referenced in [Chapter 3, Section 2.5 on Page 32](#).
- Existing CT suite must meet all regulatory and minimum size requirements referenced in [Chapter 3, Section 2.7 on Page 32](#).
- Existing facility must accommodate delivery and meet all delivery requirements referenced in [Chapter 3, Section 2.10 on Page 33](#).
- Existing facility must meet all scan room shielding requirements referenced in [Chapter 11](#).

Consult your Project Manager of Installation (PMI) for information about any additional requirements.

1.6.2 Restrictions

The following restrictions govern Quick Installations:

- Quick Installs require a new room print that accurately reflects the rooms targeted for upgrade.
- You CANNOT re-use existing floor anchors from a non-VCT system.
- New floor anchors must be a minimum of 102 mm (4 in.) from any existing floor penetrations.
- Rooms not meeting the minimum requirements for the final product must undergo an upgrade/enlargement prior to installation.

1.7 Two-Step (Temporary) Installations

Two-Step installations are the temporary installation of one system in a site, with the intention of upgrading the site to another system at a later date. The following restrictions apply to two-step installations:

- Must comply with ALL siting requirements necessary for the upgraded or final system. This includes the recommended room size and all electrical, structural, and HVAC requirements.
- All requirements referenced in [Chapter 3, Section 1.0](#) and [Chapter 3, Section 2.0](#) apply to these installations.
- The customer is responsible for verifying compliance with all requirements.
- Rooms not meeting minimum requirements for the final product must undergo sufficient upgrading/enlargement.

Note: Temporary installations include all systems installed at a site for a period ranging from two weeks to six months.

Chapter 3

System Siting Requirements

Section 1.0: System Siting Requirements

The requirements listed in this manual apply to all fixed-site customer installations, including installations within relocatable buildings. The following requirements represent the MINIMUM that a site must meet before beginning ANY new or replacement system installation. All parties should review these requirements to ensure that the site:

- Meets all service requirements.
- Meets all regulatory requirements.
- Meets all minimum structural, flooring, and vibration requirements.
- Meets minimum HVAC requirements.
- Meets minimum electrical requirements.
- Meets all network requirements.
- Meets radiation protection requirements.
- Meets all operational clearances.
- Includes all finished doors, floors, windows, ceilings, and walls, with all plumbing and cabinets already installed. ([Finished Floor Exception 1 on page 31](#) and [Finished Floor Exception 2 on page 32](#) may apply. [Finished Walls Requirements on page 32](#) may apply.)
- Does not have ANY continuing construction in the scan room OR neighboring suite areas.
- Conforms to the final GE Healthcare site print, which must be kept ON-SITE and must show all items intended for the finished room.

Note: Each site receives a LightSpeed VCT/Discovery CT750 HD Quick Start Kit from its PMI. Use the Pre-Install Checklist in the Pre-Installation Manual to confirm that the site meets all of the requirements listed above. GE Healthcare recommends completing all work to meet these requirements PRIOR to starting installation.

Section 2.0: Customer System Siting Requirements

This section provides a breakdown of customer tasks crucial for ensuring proper site preparation, regardless of whether planning for a replacement system at an existing site, or designing a new scan room for a first system.

Installation cannot proceed until verification of site-readiness occurs. A site is ready **ONLY** when it meets ALL delivery, regulatory, system, network, radiation protection, and operational requirements, as well as requirements for any options. The purchaser is responsible for completing all work necessary to install the system, and includes:

- Completion of all items in [Section 2.3, Structural](#) (recommended before installation begins).
- PMI verification that ALL items on the Pre-Installation Checklist are completed.
- Review and preparation of all site-ready items.

To ensure timely delivery and installation, GE Healthcare recommends that the customer complete all necessary work and schedule a site-ready visit prior to the delivery date.

To confirm that the site meets all requirements, you may need to employ these and other contractors:

- Structural Engineer and/or Architect
- HVAC Contractor
- Electrical Contractor
- Qualified Radiological Health Physicist
- Cleaning Services



NOTICE **SERVICE NOTICE:** An improperly prepared site—one that is in a state of construction—can result in a delayed installation date and/or damage to the system.

2.1 Regulatory Requirements

Verify that the site conforms to all of the following:

- The room meets all regulatory clearance requirements.
- The room meets all minimum size requirements.
- The site print is on-site, reflects actual room size and layout, and has received final approval.
- No grounded walls are found in regulatory clearance areas.
- The room meets all local codes.

2.2 Electrical

- Install the correct size junction boxes with covers at locations shown in the installation plan.
- Install appropriate conduits and duct work for system cables. If the suite houses additional components, determine the necessary considerations and complete the connections.
- Install a power supply of correct voltage output and adequate kVA rating.
- Install local disconnects, including proper over-current protection. This includes the A1 main disconnect with Lock-out and Tag-out (LOTO) installation.

2.3 Structural

- Install “steelwork” or other suitable support work for mounting equipment from walls or ceilings.
- Review structural requirements, including:
 - Floor vibration
 - Floor levelness
 - Floor thickness,
 - Any seismic considerations, if applicable.
- Complete all suite and room renovations and modifications prior to delivery.

2.3.1 Dust and Air Quality

Ensure that the scan suite area is free of all dust, and not subject to ANY ongoing construction, including the installation of cabinets, hanging doors, and ceiling tiles.



NOTICE

SERVICE NOTICE: Because the system's air-intake is near the bottom of the gantry (as shown in [Figure 7-3](#)) and draws in air through a filter in the gantry heater assembly, fine dust—like that created during room construction or renovation—can clog this and other filters found on the DAS, tube, and operator console. If this occurs, dust may become deposited throughout the gantry, table, operator console, and PDU electronics. Once inside the unit, removal becomes impossible, resulting in potential DAMAGE to electronic components and EARLY SYSTEM FAILURE. Consequently, the scanner is the LAST item installed in the scan suite area.

TYPES OF DUST TO AVOID

Ensure that NO construction occurs in or immediately around the scan suite area that results in:

- Concrete dust
- Drywall dust
- Ceiling tile dust
- Wood sawdust or shavings
- Dust tracked into the CT suite from adjoining rooms

Failure to take appropriate precautions to protect the system against these types of dust may result in DAMAGE to the system and early SYSTEM FAILURE.

2.3.2 Environmental Influences Considered

CT systems are designed with commercial components that are sensitive to air contaminants like sulfide, chloride and nitrates. It is the responsibility of the purchaser to ensure that the levels of these contaminants are low (Class 1). See IEC60654-4 for air quality guidelines.

2.3.3 Finished Floor Requirements

Installation requires a finished floor in the scan and control rooms. The scan room must be level by 6 mm (1/4 in) over the table and gantry area to be acceptable. You cannot use shims to level the floor. Eight or more floor covering openings that are 101.6 mm (4 in) in diameter are made to ensure the table and gantry rest on a solid surface. These floor penetrations can be sealed if required. These requirements apply to all installation types.

FINISHED FLOOR EXCEPTION 1

For sites replacing their scan room floor covering after the table and gantry are installed, the floor can be clean-finished with dust-free concrete. The finish floor in the scan room requires no dust-producing operations when applying final floor covering.

FINISHED FLOOR EXCEPTION 2

Facilities under new construction that have a finished radiology area with a single controlled-access and dust abatement barrier, can have a finished concrete floor in the scan room. The finished concrete floor in the scan room requires no dust-producing operations when applying final floor covering.

2.3.4 Finished Walls Requirements

Finished walls inside the scan and control rooms must be painted at the time of installation. This requirement applies to all installation types.

A finished walls exception is made for the following condition:

In new construction and upgraded facilities, a primer coat of paint is acceptable for equipment installation. However, the final coat of paint must be applied using a brush of some type (e.g. roller or bristle). The final coat of paint cannot be applied using a spray method.

2.4 Radiation Protection

A qualified radiological health physicist should verify that the scan room's radiation shielding provides adequate radiation protection for the planned system.

2.5 Environmental

Review HVAC requirements, including system environmental controls and patient comfort needs. Make sure the site provides an HVAC system capable of maintaining the recommended temperature and humidity specifications at the time of installation.

2.6 Options

Confirm the following:

- All customer installation options reviewed and final locations determined.
- All GE-supplied installation options reviewed and final locations determined.
- The laser camera should be on-site at the time of system installation.

2.7 Clearances

- Review operational clearances to verify whether daily use items fit (e.g. beds and carts).
- Consider clearances for emergency medical equipment.
- Ensure that all storage cabinets and sinks appear on the site print in their proper locations.
- Confirm that adequate space exists in the scan suite for delivery and installation of all replacement parts following installation of the system.

2.8 Network

Ensure that network communication is in place and active.

2.9 Chemical Contamination

Never install wet film processors in the same room as the scanner, as this may result in possible contamination of Lightspeed Series components. Chemicals utilized by such processors can contribute to increased equipment failures and downtime, and decreased reliability.

When siting this equipment, consider the effects that contact with these chemicals and the resulting fumes might have on human subjects in proximity to them. In addition, film processor equipment installation must meet all manufacturer requirements (e.g. ventilation specifications) as well as all applicable local, state, and national codes.

2.10 Delivery

- Determine room dimensions and verify that doorways adequately accommodate the system.
- Verify the existence of an accessible, dust-free, non-construction-zone route to the scan suite that accommodates delivery.
- Identify elevators, doorways, and hallways that can accommodate delivery.
- Provide floor protection, if needed.
- Request rigging, if needed.

Section 3.0: Site Readiness - GE's Responsibilities

3.1 Project Manager of Installation Tasks

The GE Healthcare Project Manager of Installation (PMI) assists the purchaser in meeting all system siting requirements.

3.1.1 Pre-Installation Delivery Tasks

The PMI also performs the following pre-installation delivery tasks:

- Determines the delivery type: ground, dock, or tilt-bed truck.
- Determines if delivery requires tilt dollies or riggers; orders dollies and lifting crates, as needed.
- Determines if the delivery requires the use of floor protection.
- Determines if ground delivery requires the use of a tilt-bed truck, and informs GE Transportation of the need for a tilt-bed truck.

3.1.2 Site Review with Customer

A site-ready visit should occur prior to the delivery date. This visit verifies that the site meets all system siting requirements and confirms that installation can proceed. During the site-ready visit, a GE representative confirms that the site meets all of the required site-ready conditions including floor levelness, and delivery route readiness. Lifting options and construction site packaging must be ordered prior to delivery and cannot be added on-site.

Chapter 4

Regulatory Requirements

Section 1.0: Regulatory Clearances

1.1 Regulations

Federal regulations and national standards that determine the minimum clearances for U.S. installations include:

- 29 CFR 1910 (OSHA)
- NFPA 70E (STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE)
- NFPA 101 (LIFE SAFETY CODE)
- AMERICANS WITH DISABILITIES ACT



NOTICE

All systems installed within the United States and United States territories, and within U.S. government facilities, regardless of country, must comply with all United States Federal and local regulations. All systems installed outside the United States must comply with either the national, state, or local regulatory clearance requirements for the country in which the installation occurs, or U.S. Federal regulations, whichever is greater.

1.2 Clearance Requirements

A map of clearance requirements necessary for U.S. *regulatory* compliance is provided in [Figure 4-1, on page 36](#).

Note: A similar map of detailed dimensional clearance measurements necessary for safe *servicing* of the system is provided in [Figure 5-1, on page 43](#).

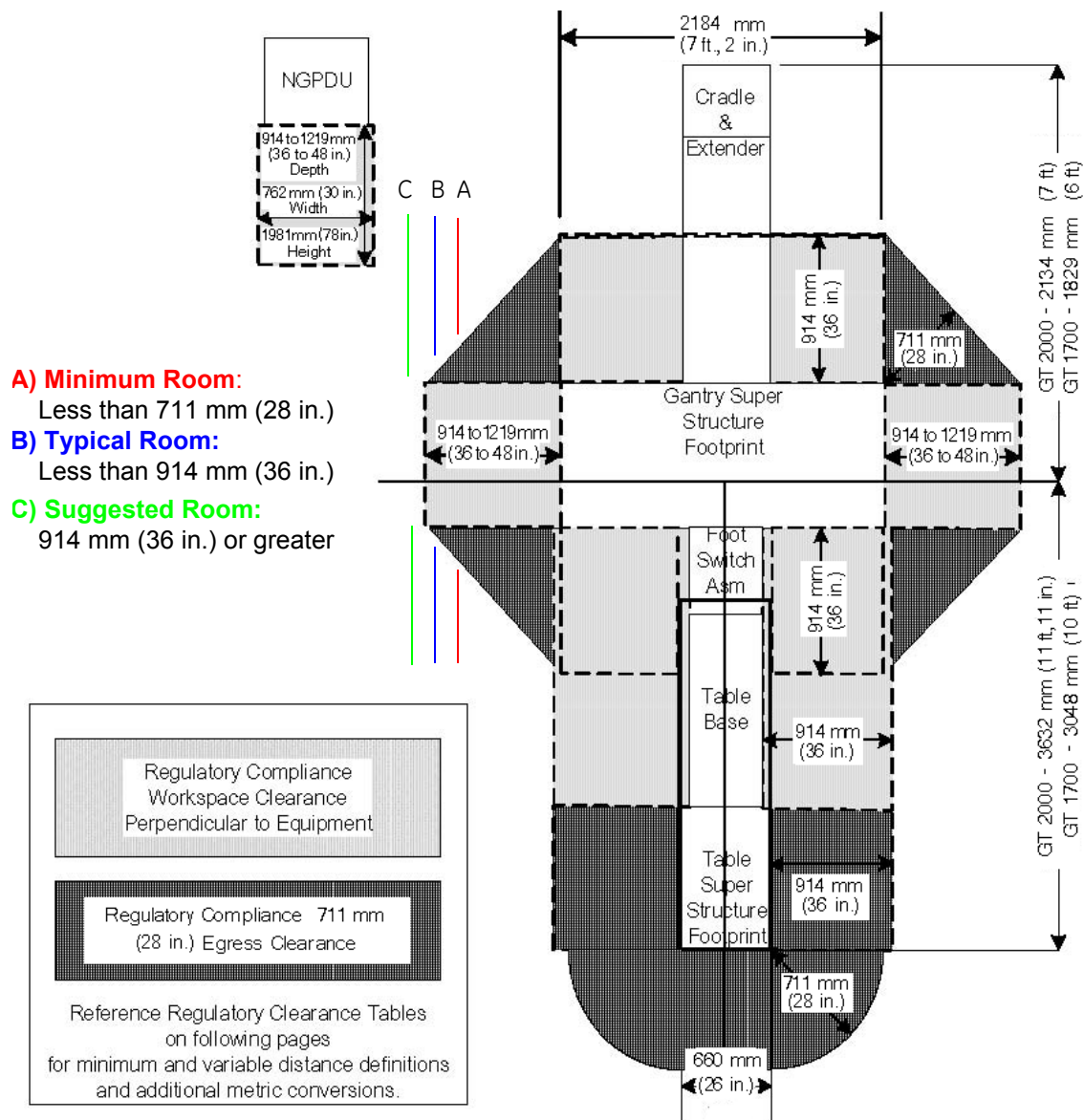


Figure 4-1 Regulatory Clearance Requirements for the LightSpeed VCT/Discovery CT750 HD System

1.2.1 Minimum Regulatory Workspace Clearances by Major Subsystem

Note the following when referring to the tables below:

- These requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed with live parts exposed.
- The customer **MUST** maintain the required regulatory clearance distances and may **NOT** use these areas for storage. This applies during normal system operation as well as during service inspection and maintenance.
- Direction of Service Access refers to a direction perpendicular to the surface of the equipment serviced.

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front and Rear of Console)	N/A (No exposed live part hazards.)	
Service Access Width (Front-Back of Workspace)		Refers to the width of the working space in front of equipment. 762 mm (30 in.) min or the equipment width, whichever is greater.
Head Clearance	1981 mm (78 in.)	Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater.

Note: Distances are measured to the finished covers.

Table 4-1 Console – Minimum workspace Clearances

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of PDU)	914 mm (36 in.)	1219 mm (48 in.) if exposed live parts of 151 - 600 volts are present on both sides of the workspace with the operator between. 1067 mm (42 in.) if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Front of Workspace)	762 mm (30 in.)	Refers to the width of the working space in front of equipment. 762 mm (30 in.) min or the equipment width, whichever is greater.
Head Clearance	1981 mm (78 in.)	Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater.

Table 4-2 NGPDU – Minimum Workspace Clearances

Note: For the Gantry and Table, distances are measured from the finished covers

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of the workspace with the operator between. 1067 mm (42 in.), if the opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of Workspace)	762 mm (30 in.)	Refers to the width of the working space in front of equipment. 762 mm (30 in.) minimum or the equipment width, whichever is greater.
Head Clearance	1981 mm (78 in.)	Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater.

Table 4-3 Gantry – Minimum Workspace Clearances

Workspace Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Table Head)	N/A	
Direction of Service Access (Table Sides)	914 mm (36 in.)	*Can reduce to 711 mm (28 in.), provided the local team obtains written and signed approval from the local AHJ (Authority Having Jurisdiction). GE must have the signed document on file.
Direction of Service Access (Table Foot)	711 mm (28 in.)	457 mm (18 in.) minimum for Front Gantry Cover removal, only if an unobstructed egress space of 711 mm (28 in.) exists around the equipment for room exit, and no trip hazards exist along the path of egress.
Service Access Width (Left-Right of Workspace)	762 mm (30 in.)	Refers to the width of working space in front of equipment. 762 mm (30 in.) minimum or the equipment width, whichever is greater.
Head Clearance	1981 mm (78 in.)	Refers to the height of the workspace measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. 1981 mm (78 in.) or the height of the equipment, whichever is greater.

Table 4-4 Table – Minimum Workspace Clearances

1.4 NEC Conduit and Duct Fill Rate

Full operation, service, and safety of the system requires the maintenance of sufficient regulatory and service clearances around equipment.

Cable length is an important consideration in room layout. The LightSpeed VCT/Discovery CT750 HD system ships with standard (short) length cables, with a set of longer cables available as an option. Refer to the electrical page of your GE site print for the specific requirements of your site.

The following rules govern cable usage for the system:

- When possible, use the rear cable cover assembly to let cables enter the gantry from the rear.
- Do not cut or otherwise shorten long cables.
- Do not store excess cable length behind the operator console, gantry, or PDU.
- Store excess cable in wall or floor ducts, if desired, provided that sufficient space exists. Refer to NEC code to determine cable fill rates for conduits and ducts.
- All installed systems shall comply with NEC 70-E Electrical Regulations governing conduit or duct fill.

Section 2.0: Terms and Definitions

CLEARANCES

Clearances are the clear space or distance between or around objects and equipment, governed by all applicable safety, service, and regulatory requirements and representing the lowest margin of freedom permissible for equipment siting.

DIMENSIONS

Dimensions are the length, width, depth, and height of equipment.

EGRESS

An egress is the single path of exit from within any room. Egress must conform to these requirements:

- OSHA requires a minimum width of 711 mm (28 in.) of continuous and unobstructed space, free of trip hazards, along the entire path of the exit.
- The path of exit may contain ONLY OSHA-approved ramps.
- All doors used to improve egress MUST lead to an exit.
- The path of exit CANNOT include the surface floor raceway in areas that would require lifting to remove the front or rear gantry covers for service.
- There must be a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table.
- Determine the amount of clearance by measuring from the table center line out to any obstruction.

Note:

Gantry front cover removal requires the use of tilting dollies that move side-to-side to reach a park position of 457 mm (18 in.) at the foot of the table. If the planned egress route is not at the back of the table, ensure that this distance exists at the back of the table to park the gantry front cover. Maintain a continuous width of 3200 mm (126 in.), equaling 1600 mm (63 in.) clearance on each side of the table to allow removal of the front cover.

If not planning to park the gantry front cover at the back of the table, be sure to maintain at least 305 mm (12 in.) of clear space along the back of the table to allow movement of the gantry front cover along the back of the table.

(PRE-INSTALLATION) ESCALATION

Pre-installation escalation is the process used to consult CT Engineering, the Design Center, or EHS to resolve pre-installation issues related to siting concerns and requirements.

GROUNDING WALL

A grounded wall is any wall with electrical conductivity to earth. Conductive materials generally found in walls include masonry, concrete, and tile. Treat as grounded additional elements commonly found in walls, including but not limited to:

- Medical gas ports and plates
- Metal doors and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 main disconnect panel
- Equipment Emergency Off panels
- Industrial equipment (such as air conditioners and vents)
- Expansion joints
- Surface raceway
- Exposed wall conduits
- Floor outlet boxes
- Floor HVAC boxes
- Floor medical gas

Common wall components NOT constituting grounded elements include:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks
- Ceiling tile grids

HEAD CLEARANCE

Head clearance represents the height dimension of the workspace, measured from the floor at the front edge of the equipment to the ceiling or any overhead obstructions. It requires a minimum of 1981 mm (78 in.), or the height of equipment, whichever is greater.

MINIMUM

Minimum indicates the lowest limit permitted by law or other authority.

SERVICE ACCESS WIDTH

Service access width refers to the width of the working space in front of the equipment, and requires a minimum of 762 mm (30 in.) or the width of the equipment, whichever is greater.

WORKSPACE

The *workspace* represents a three-dimensional box of space required for safe inspection or service of energized equipment. It consists of depth, width, and height, with the depth dimension measured perpendicular to the direction of access. U.S. regulation requires a minimum depth of 914 mm (36 in.). Additional conditions can increase the minimum requirement. For example, FCT defines *workspace* as the envelope of the component superstructure, measured for the NGPDU with the front panel removed, and measured for the gantry and table with the external covers removed.

Chapter 5

Service Clearance Requirements

Section 1.0: Measuring Service Clearances

The regulatory envelopes defined in [Figure 5-1](#) provide enough space to safely allow CT System servicing. System Servicing requires:

- Sufficient space to remove the covers from the system.
- One service engineer to be able to accomplish all service component replacement tasks without needing special tools or equipment.
- ALL room layouts to provide service space and access around the table to the gantry right side. This is needed for replacement procedures that require components that ship in large boxes, such as the tube, detector, and HV tank.

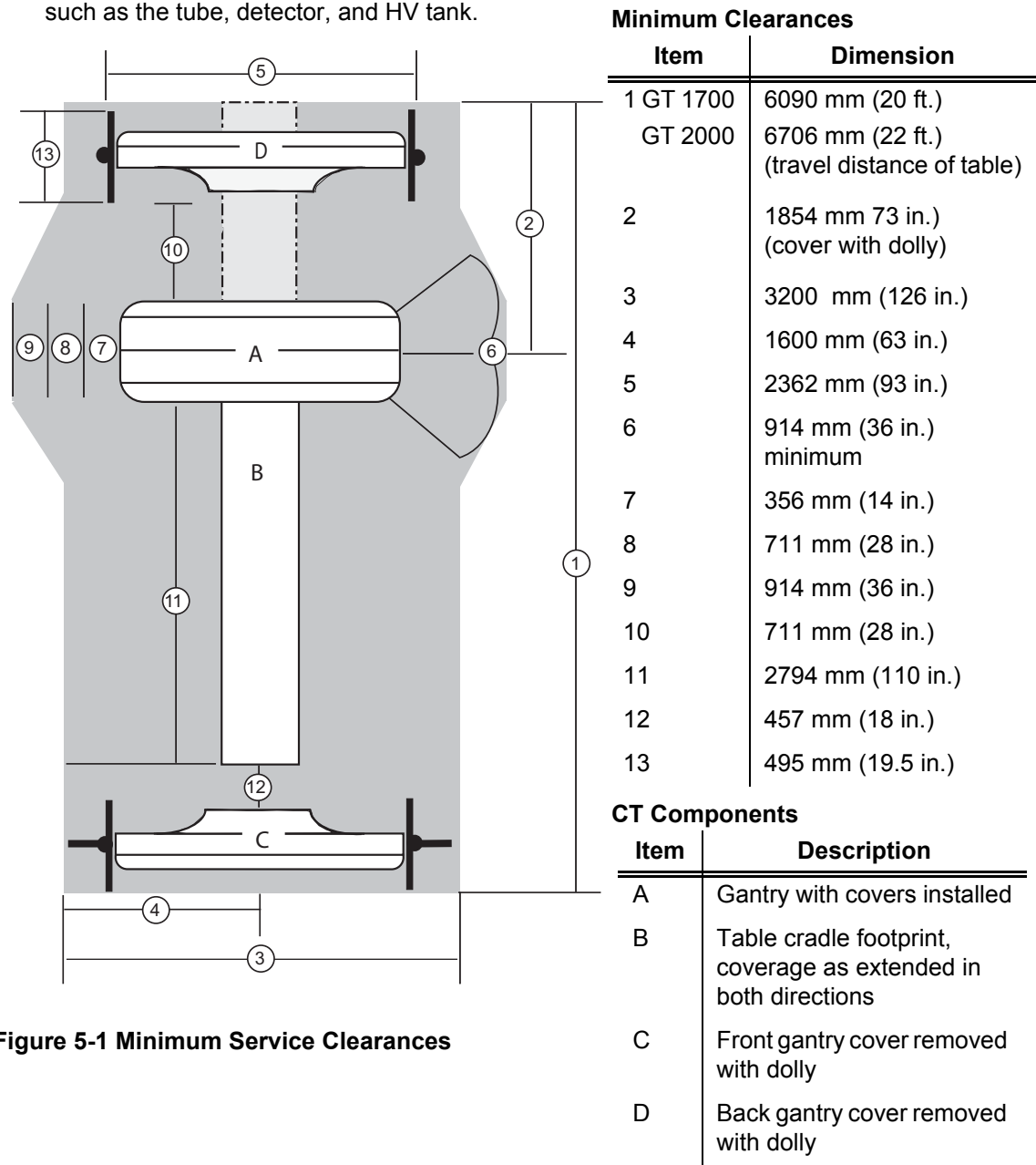


Figure 5-1 Minimum Service Clearances

Section 2.0: Special Service Clearances

Note: When calculating service clearances, refer to [Figure 5-1](#) for all service clearance needs.

2.1 Cover Removal

- Gantry front cover removal requires the use of the Tilting Cover Dollies and a minimum clearance space of 3200 mm (126 in.) to maneuver the cover, as illustrated in [Figure 5-1](#). The dollies allow the service engineer to separate the cover from the gantry, tilt it 90 degrees, roll it to the foot end of the table, and then tilt it an additional 90 degrees, so that it is upside-down relative to its normal system-mounted condition. After removal, the service engineer must then move the gantry front cover to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover, with service dollies installed, requires a clearance width of 2388 mm (94 in.) and a depth of 914 mm (36 in.) for removal, as shown in [Figure 6-2](#). Sufficient space to allow the service engineer to move the cover either straight back or to one side of the table to satisfy the minimum service clearances shown in [Figure 5-1](#) must be maintained. The rear cover with dollies cannot extend past the allowable clearance space within the room (see [Figure 5-1](#)). If the system is not sited straight (it is positioned diagonally), full service space is still required. The PMI and customer should discuss this consideration and make the necessary provisions.
- The scan room must offer sufficient space to allow adequate egress during service operations that require both front and rear cover removal. If the customer and PMI have any concern that site will not provide adequate space for egress under these conditions, they should discuss these requirements and make the necessary provisions to accommodate this event.
- A single service engineer can safely perform servicing of the table. Ensure sufficient clear space to maintain egress clearances with the table covers or cradle removed.
- A single service engineer can safely perform servicing of the system. Ensure sufficient clear space to maintain egress clearances with covers or cradle removed.
- A tube change box is 737 mm x 1524 mm (29 in. x 5 ft.), with the handles extended. The box rolls like a wheelbarrow and must have access to the right side of the gantry. It is the PMI's responsibility to demonstrate that the tube change box can be positioned in the tube change area next to the gantry and that the front and rear covers can be removed.

2.2 Power Distribution Unit (NGPDU)

When positioning the Power Distribution Unit (NGPDU), consider regulatory compliance, as defined in [Chapter 4, Section 1.0](#), Regulatory Clearances. See [Table 4-2](#) in that section.

2.3 Console

The operator console does not present an exposed live parts hazard. However, the site shall maintain a working space at all times with a minimum depth of 1219 mm (48 in.), extending the full width of the operator console for service activity.

The console is on wheels. As some service activities require access to the rear of the console, be sure to maintain sufficient space for moving the console to allow rear service access.

See [Figure 6-5](#) for a typical control room layout.

2.4 Storage Cabinet

GE Healthcare provides a storage cabinet for storing all supplied service equipment (see [Table 5-1](#)). Situate this storage cabinet within the scan room suite area to allow easy service access. The dimensions of the cabinet measure 457 mm D x 914 mm W x 1067 mm H; ~ 41 kg (18 in. D x 36 in. W x 42 in. H; ~90 lbs).

Item	Size	Weight (total)	
QA Phantom (water filled)	20 cm x 15 cm (7.9 in. x 5.9 in.)	5.5 kg	12 lb
35CM Phantom	35 cm x 7 cm (13.8 in. x 2.8 in.)	8.2 kg	18 lb
Phantom Holder	25 cm x 25 cm (9.8 in. x 9.8 in.)	3.6 kg	8 lb
FE Box (Purple)	30 cm x 38 cm x 30 cm (11.8 in. x 15 in. x 11.8 in.)	6.8 kg	15 lb
Rear Cover Dollies	158 cm x 82 cm (62.2 in. x 32.3 in.)	11.4 kg	25 lb
Front Cover Dollies	85 cm x 20 cm and 85 cm x 15 cm (33.5 in. x 7.9 in. and 33.5 in. x 5.9 in.)	15.9	35 lb
Install Support Kit (box)	30 cm x 30 cm x 38 cm (11.8 in. x 11.8 in. x 15 in.)	9.1 kg	20 lb
Three Piece Tube Hoist Assembly	77 cm x 8 cm and 38 cm x 15 cm (30.3 in. x 3.1 in. and 15 in. x 5.9 in.)	9.1 kg	20 lb
Balance Weight Kit		33 kg	73 lb

Table 5-1 Equipment Stored in Storage Cabinet

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Chapter 6

Room Sizes

Section 1.0: Room Dimensions

System Configuration	Suggested Room Size	Typical Room Size	Minimum Room Size
LightSpeed VCT or Discovery CT750 HD with GT 1700	7772 mm x 4420 mm ¹ (25 ft, 6 in. x 14 ft, 6 in)	6096 mm x 4115 mm ¹ (20 ft x 13 ft, 6 in)	6096 mm x 3556 mm ² (20 ft x 11 ft, 8 in)
LightSpeed VCT or Discovery CT750 HD with GT 2000	7772 mm x 4420 mm ¹ (25 ft, 6 in. x 14 ft, 6 in)	6706 mm x 4115 mm ¹ (22 ft x 13 ft, 6 in)	6706 mm x 3556 mm ² 22 ft x 11 ft, 8 in.

¹All regulatory requirements apply.

²All regulatory requirements apply, with the addition of no energized left-side service.

Table 6-1 Suggested Room Size Dimensions

1.1 Suggested Room Size

The suggested room configuration offers the most flexibility for future upgrades. It provides both ample workspace and space to add millwork and still meet all regulatory requirements. When local regulations require a sink in the scan room, this room size also provides sufficient space for a sink. This room size accommodates the needs of larger hospitals and medical teaching facilities, where patients may require transportation to the scan area in beds, gurneys, and larger wheelchairs and where they may require the assistance of larger medical care teams. Likewise, this room offers adequate access for crash carts and other emergency medical equipment on both sides of the table. The suggested size supports all service activities, including tube change, and accommodates all future two-step installations.

1.2 Typical Room Size

The typical room configuration represents what is most commonly found at installation sites. It offers adequate workspace and may provide adequate space for a sink, but allows only LIMITED space to add millwork and still meet all regulatory requirements. The size of this room accommodates the needs of larger clinics and medium-sized hospitals, where patients may require transportation into the scan area using gurneys and wheelchairs, and where they may require the assistance of small medical care teams. It provides access for crash carts and other emergency medical equipment on only one side of the table.

The typical room size supports all service activities, including tube change, and may offer compatibility with SOME future upgrades and two-step installations.

1.2.1 Minimum Room Size

The minimum room configuration represents the smallest functionally acceptable space for this product and represents the type of room often found at doctor's offices and smaller clinics and outpatient facilities. Due to its limited size, and to functional and regulatory requirements, this room usually provides only LIMITED workspace, and leaves to NO space to add in-room millwork and sinks and still meet the necessary regulatory and service requirements. This room can accommodate the transportation of patients into the scan area using wheelchairs, and provides access for crash carts and other emergency medical equipment on only one side of the table.

Sites considering a minimum room size may not have been designed with the structural requirements necessary to support the system and consequently may require upgrading prior to installation.

Customers considering a minimum room size should discuss their workspace requirements and future upgrade plans with their PMI, as the size and layout of these rooms often eliminates them from any future upgrade considerations and offers NO compatibility with future two-step installations.

If using the square meters (square footage) to determine regulatory compliance, please note that the front and rear cover clearances are wider than the regulatory clearance along the table length, and that the cover park position is behind the table in the home position.

Note: Sites must provide sufficient space to allow the removal of the rear cover, which is on wheels, from behind the gantry during service operations.

**CAUTION**

Operational Caution: In a minimum room layout (356 mm to 686 mm [14 in. to 27 in.]) the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, this room provides only limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall.

Section 2.0: Suggested and Typical Room Layouts

Figure 6-1 and Figure 6-2 show the recommended and typical room layouts, both with and without in-room cabinets. You need to know the locations for medical gas, surface duct work, or other items that make a grounded wall.

Note: Your room layout may meet the recommended or typical room requirements but appear different than Figure 6-2 or Figure 6-5. Your salesperson can provide a detailed room layout for your site.

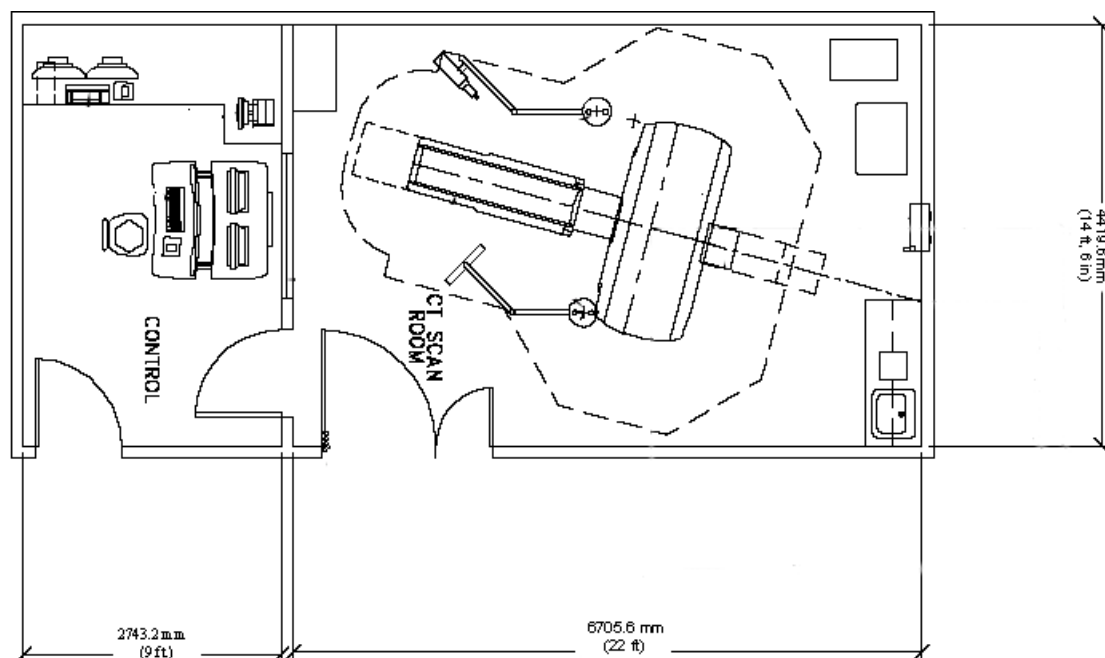


Figure 6-1 Room Layout with Cabinets

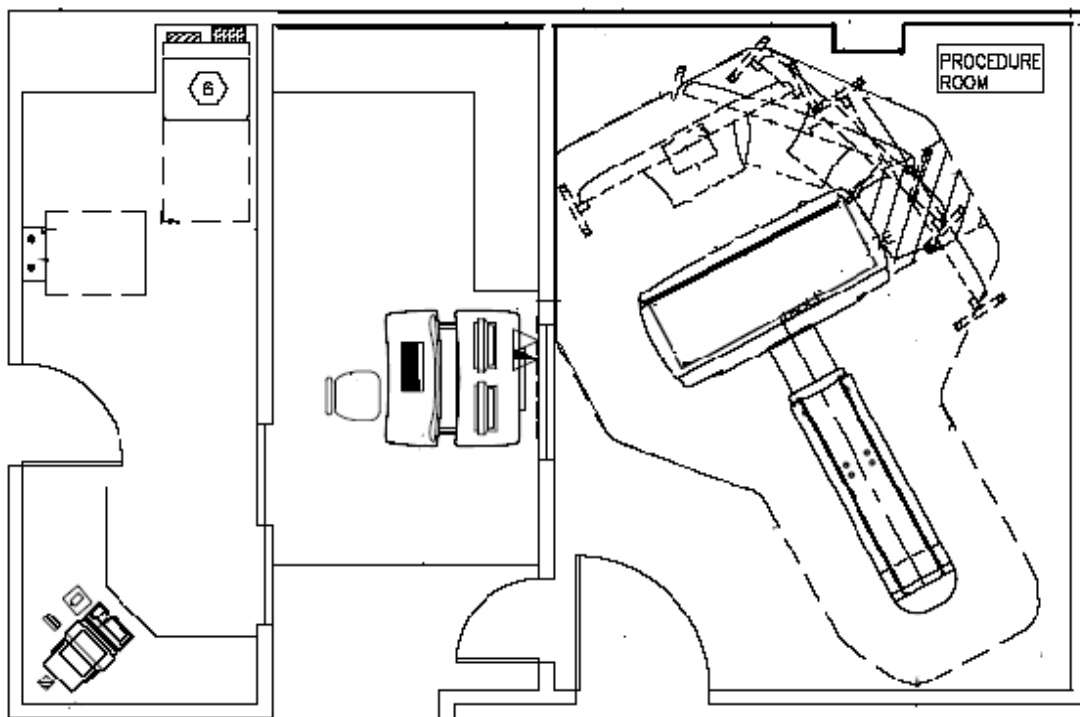


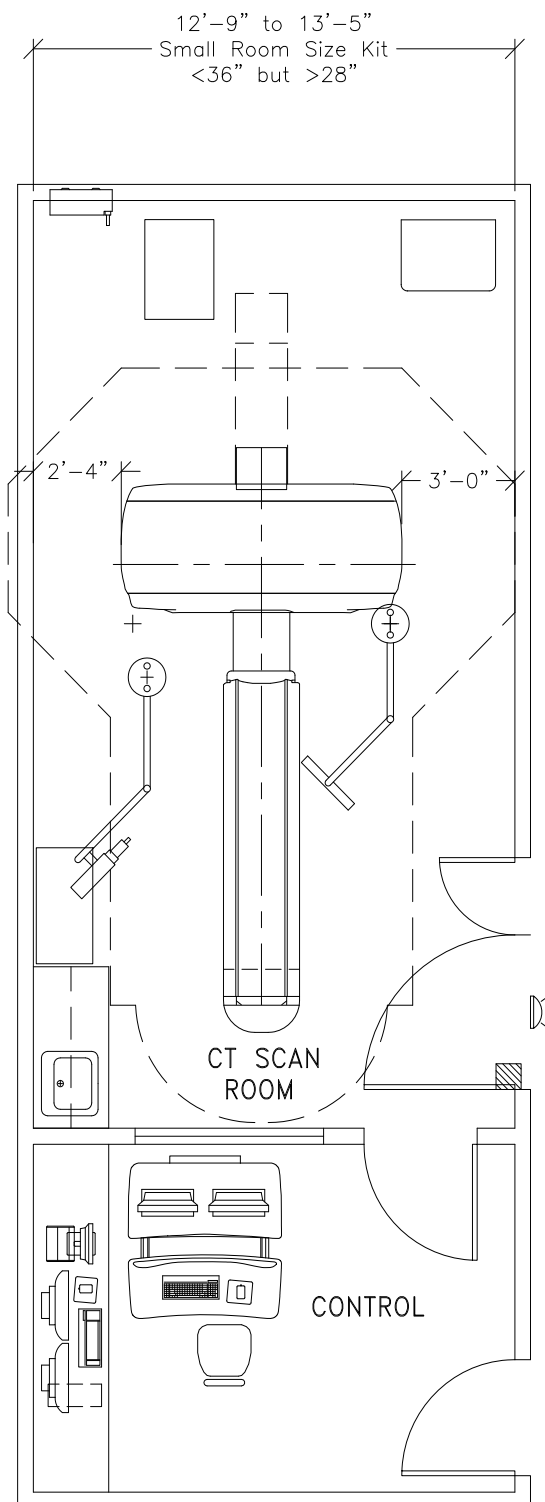
Figure 6-2 Room Layout without Cabinets

Note: The room depicted in [Figure 6-2](#) shows ample space for the removal of both front and rear covers.

Section 3.0: Minimum Room Layouts

Room A provides less than 914 mm (36 in.), but greater than 711 mm (28 in.) of space, measured from covers to left-side wall; it offers sufficient service, egress, and workspace around the gantry.

Figure 6-3 Minimum Room Size - Layout A



Room B provides less than 711 mm (28 in.), but greater than 256 mm (14 in.) of space on the gantry left side, measured from covers to left-side wall, compromising service, egress, and workspace on the gantry's left side.

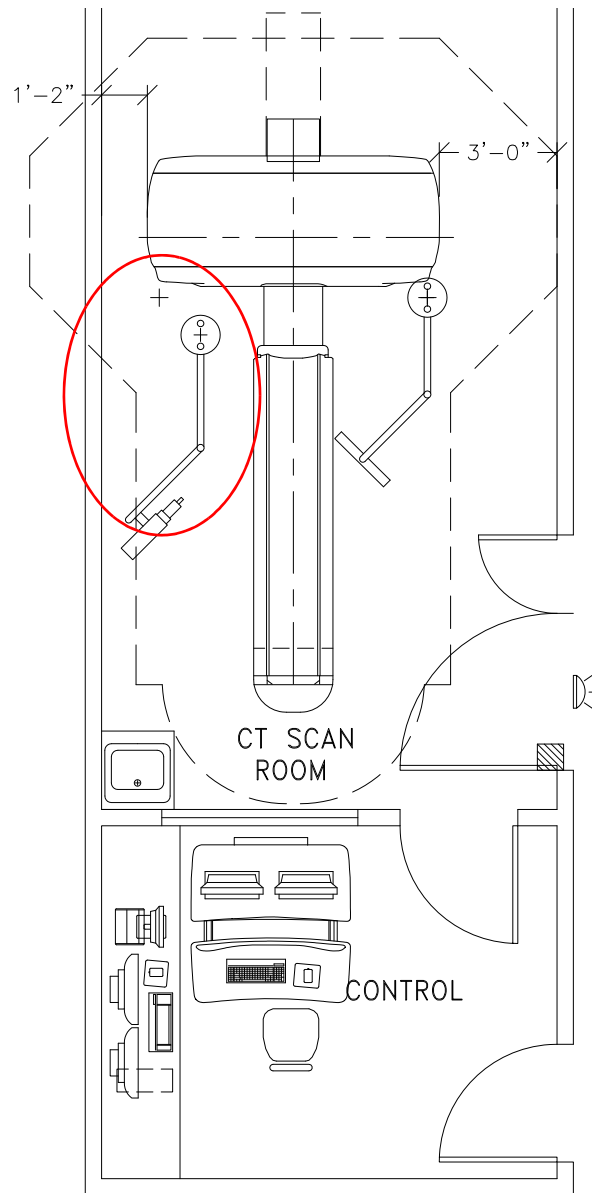


Figure 6-4 Minimum Room Size - Layout B

Section 4.0: Control Room Considerations

- The control room must provide an operating environment suitable for the operator console electronics and the operator's working comfort. See [Chapter 9, Environmental Requirements](#).
- The operator console must remain in the same configuration as shipped. Do not dismantle the console, or remove or rearrange its components.
- Do not position the monitor desktop or user desktop components on a counter-top, or mount the operator console cabinet remotely.
- Cable lengths must remain as shipped (cables cannot be cut or extended to mount the monitor on the customer's counter).
- As the console requires adequate venting, maintain 152 mm (6 in.) of clear, unobstructed space on all sides of the console to allow the four fans located on the rear of the console to exhaust air to both the left and right.
- Provide a suitable work area within reach of the operator console for the placement of the injector control. Injector controls differ in dimensions depending on the brand selected.
- A PACS, workstation, image printer, or filming device may appear in the operator console control room area. These devices or other components, though having a direct link to the operator console via network or ethernet cable, shall NOT receive power from the CT operator console. If you are using additional devices or components, consider additional room power and network connections when reviewing the operator console workspace.

4.1 Typical Control Room Layout

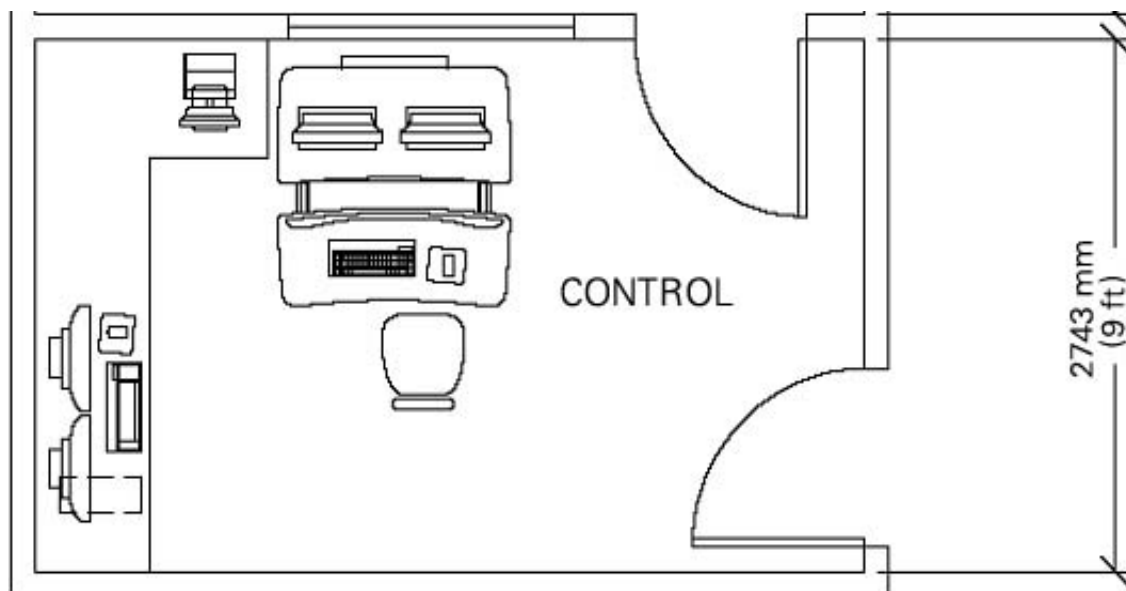


Figure 6-5 Typical Control Room Layout

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Chapter 7

System Component Dimensions

Section 1.0 Minimum Operating Clearances

The sections in this chapter provide the minimum dimension and operating clearance information for each category of components listed. Be sure that the site conforms to each of these specifications.

1.1 Ceiling Pedestal Mount Installation

The distance from the floor to the lowest point of the ceiling pedestal mount for the Injector or Monitor CANNOT measure LESS than 2134 mm (84 in.).



NOTICE Failure to maintain a distance of at least 2134 mm (84 in.) from the floor to the lowest point of the Injector or Monitor ceiling pedestal mount may pose a safety hazard.

1.2 Injector Control Installation

Minimum dimensions and clearances include the following requirements for the injector control:

- Provision of a suitable work area for placement of the injector control, within reach of the operator console.
- Wall mounted, ceiling mounted, and pedestal units require routing of cables from the gantry area to the operator console area. The supplied cable measures 15.2 m (50 ft).
- Injectors require an AC power source that is powered from the console. The IEC power cord is supplied with the injector.
- Available mounts come in several different lengths and configurations. Refer to the injector documentation for detailed installation instructions.

1.3 System Operational Clearances

The clearances listed in [Table 7-1](#) govern system operation; be sure that the site maintains each of these clearances.

System Operation	mm	inches
Ceiling Pedestal mount (optional) Lowest point to floor injector or monitor	2134 mm	84 in.
Finished ceiling to floor (suggested)	2743 mm	108 in.
Finished ceiling to floor (minimum)	2134 mm	84 in.
Table to maximum extension head end with extender from Center Line	2030 mm	80 in.
Table extension head end with extender to obstruction	152 mm	6 in.
Table in lowest position. with cradle at home position to Center Line	3209 mm	126.5 in.
Back of Console to wall	152 mm	6 in.
Back of PDU to wall	152 mm	6 in.

Table 7-1 Minimum Dimensions and Operational Clearances

Section 2.0: Component Dimensions

2.1 System Dimensions

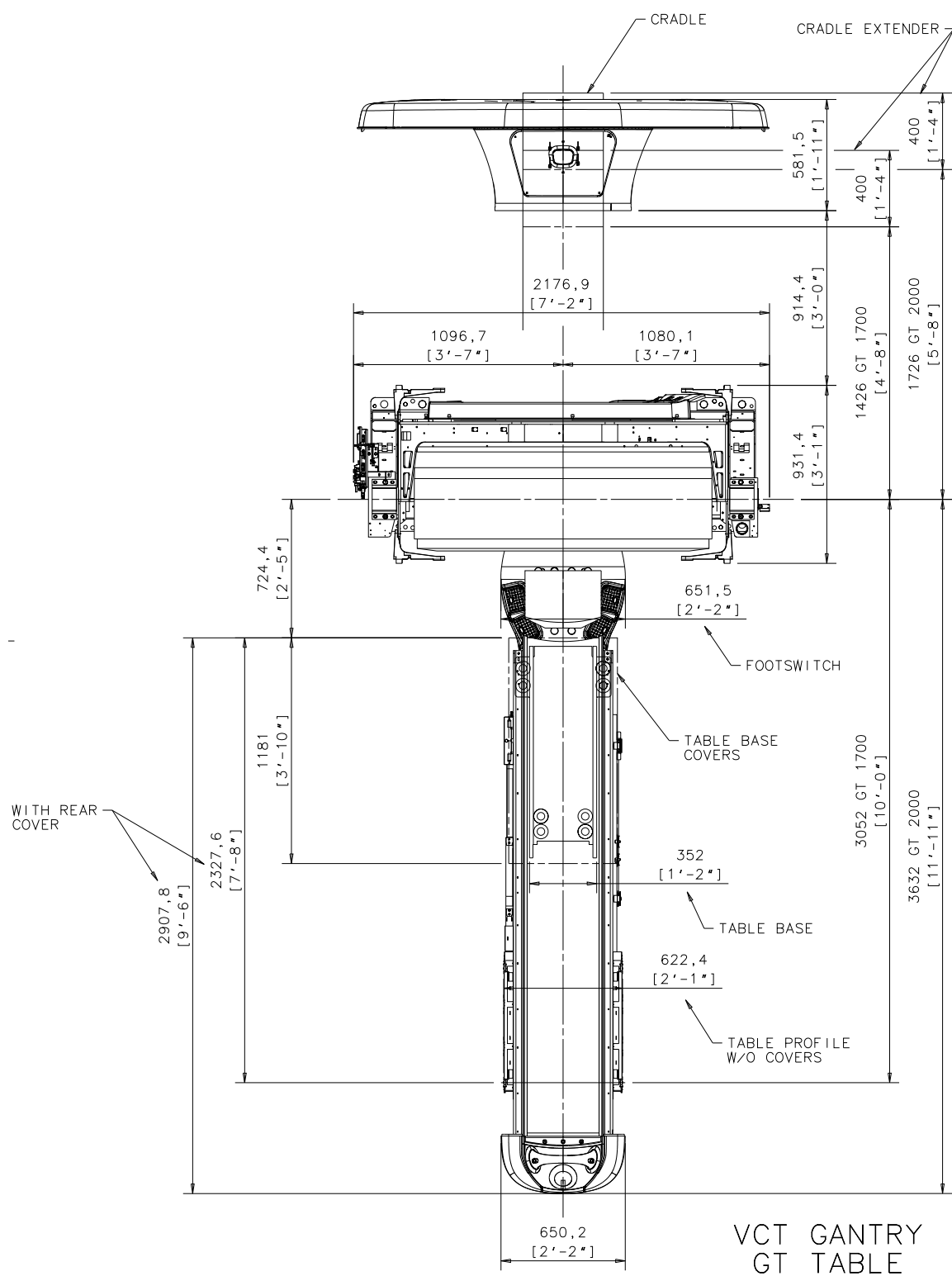


Figure 7-1 LightSpeed VCT/Discovery CT750 HD Dimensions without Cover

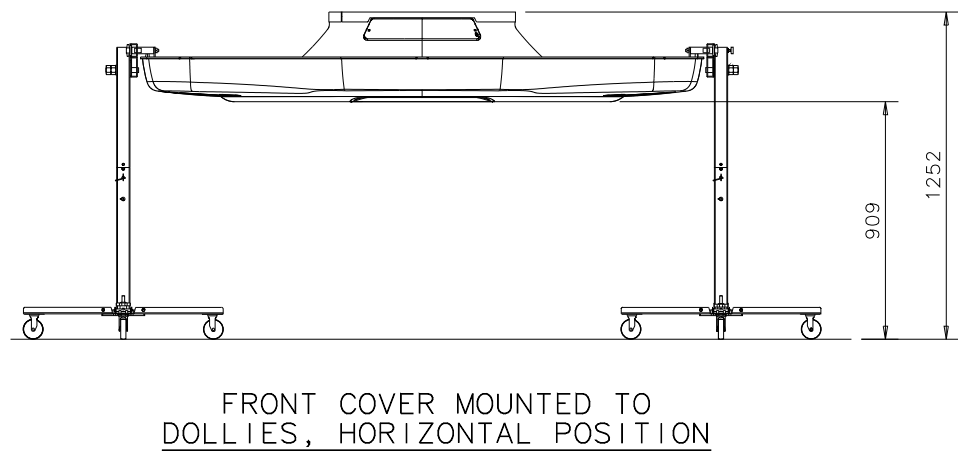
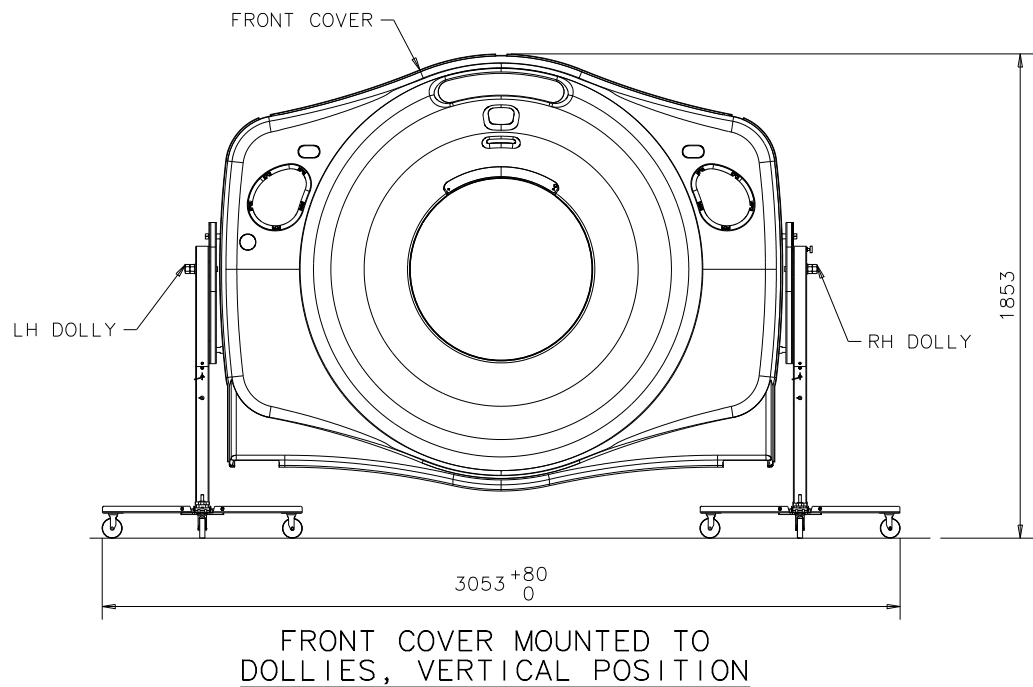


Figure 7-2 Front Cover with Service Dolly Dimensions

Description	Width		Depth		Height	
	mm	inch	mm	inch	mm	inch
Remote Color Monitor (LCD)	413	16.25	203	8	406	16

Table 7-2 Dimensions of Accessories

2.2 Table and Gantry Dimensions

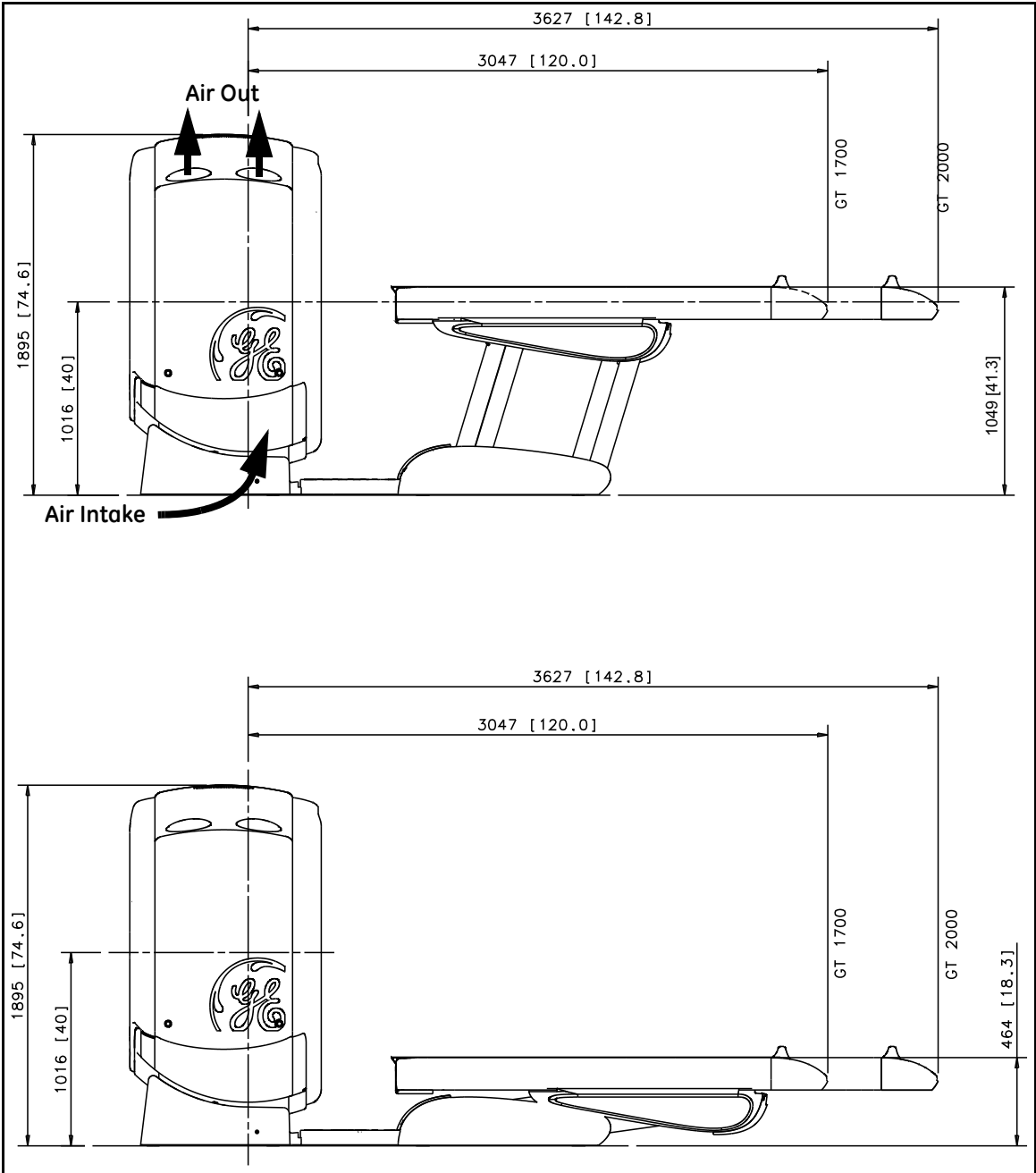


Figure 7-3 GT 1700 and GT 2000 Table and Gantry (Side View)

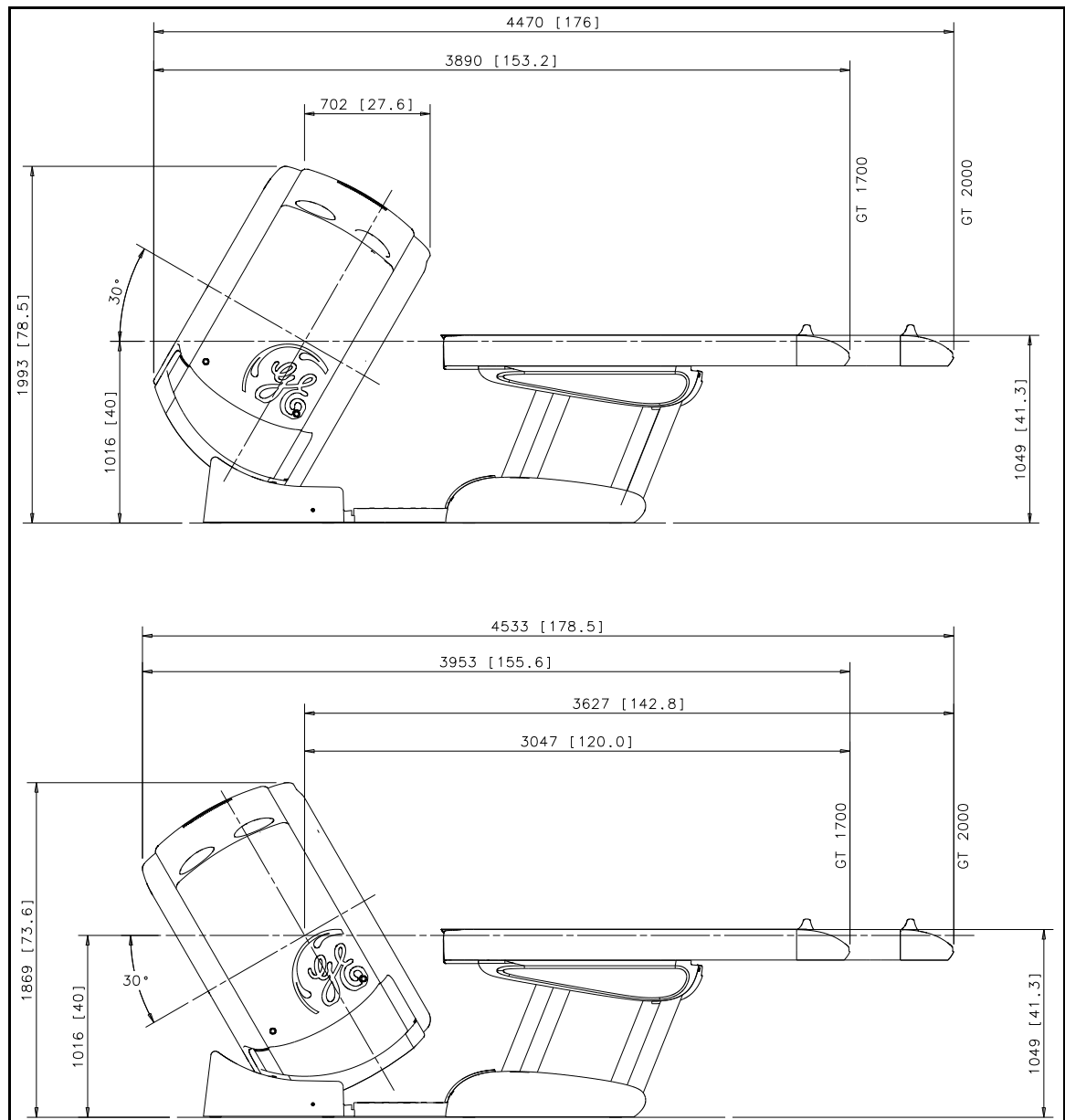


Figure 7-4 Gantry Tilted +30° (top) and -30° (bottom) - GT 1700 and GT 2000 Table Options

2.3 Power Distribution Unit Dimensions (NGPDU-4)

PDU dimensions, air intake/exhaust, seismic bracket locations, and service areas appear below.

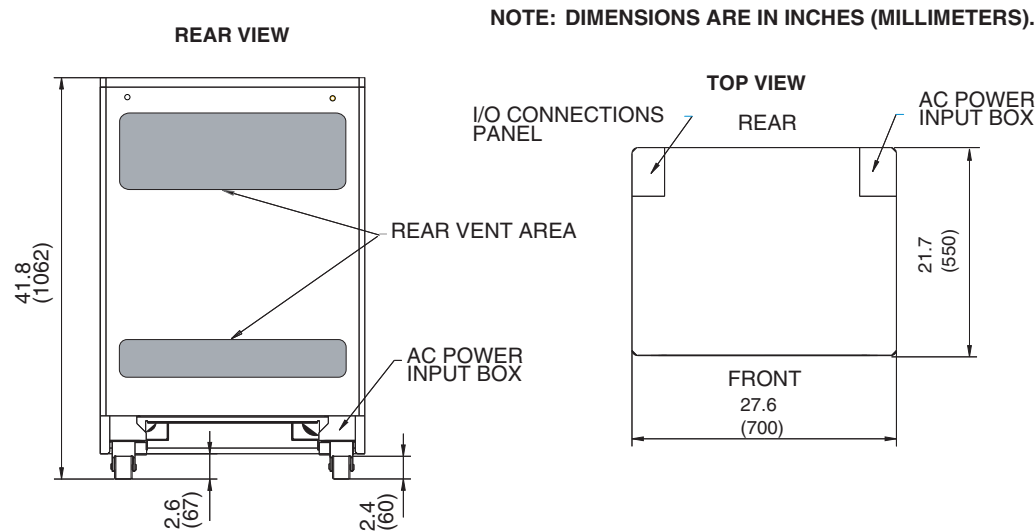


Figure 7-5 Power Distribution Unit (NGPDU-4)

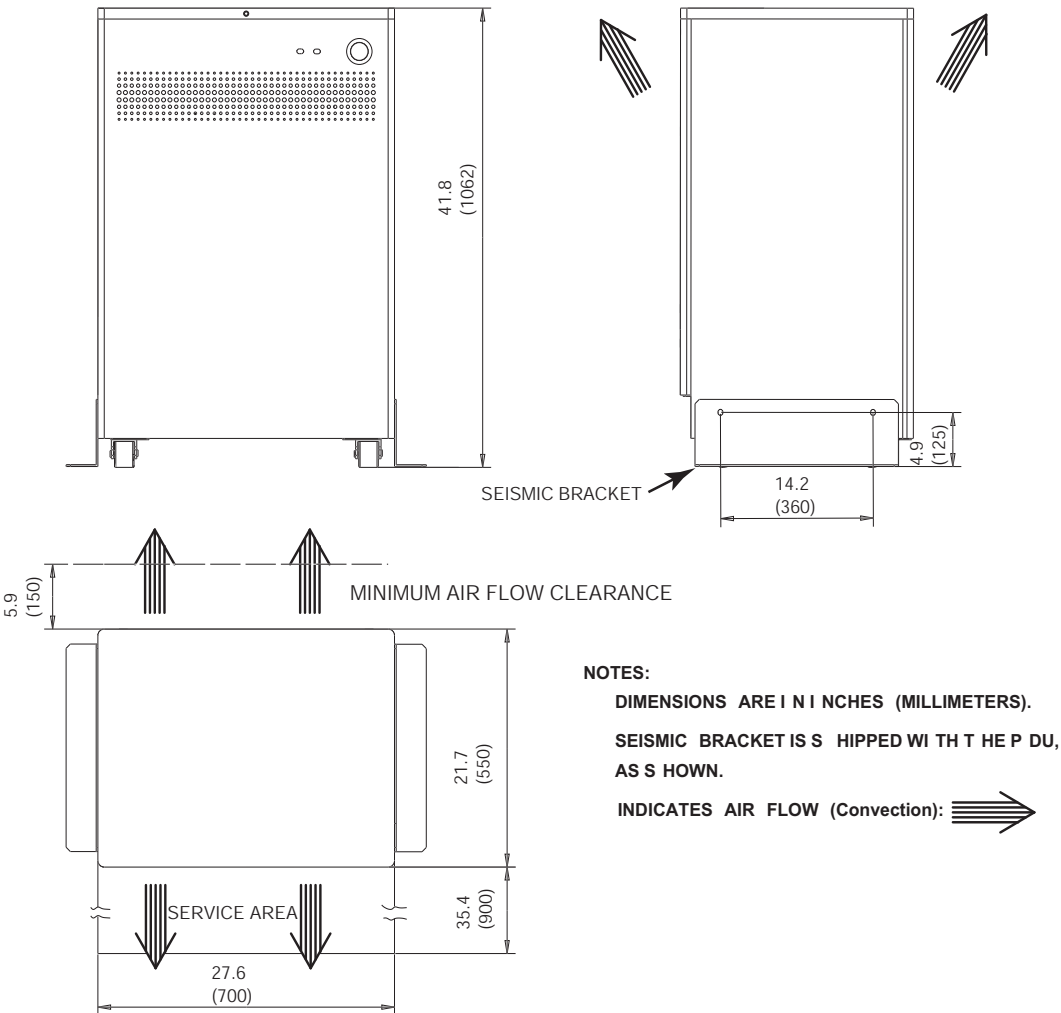


Figure 7-6 Power Distribution Unit (NGPDU-4)

2.4 Power Distribution Unit Dimensions (NGPDU-6)

PDU dimensions, air intake/exhaust, seismic bracket locations, and service areas appear below.

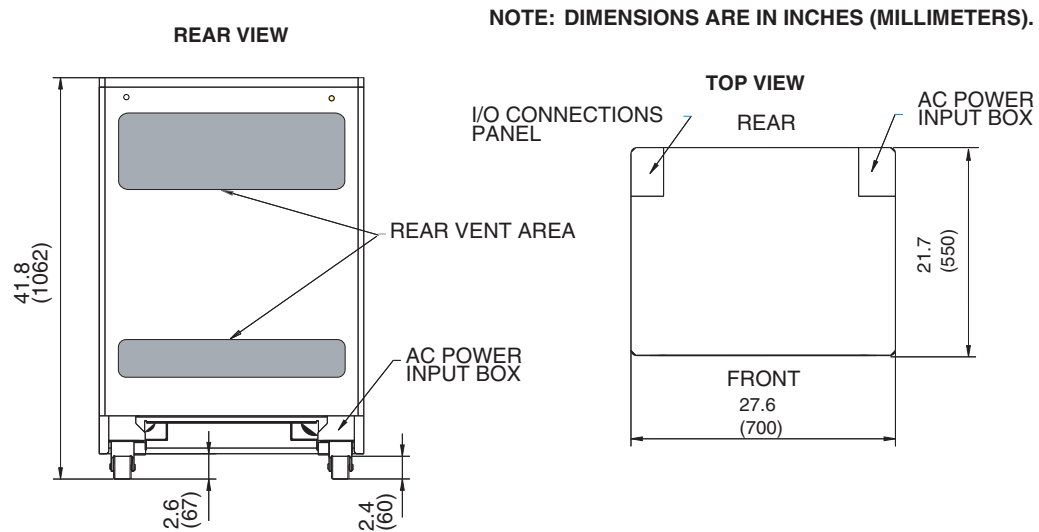
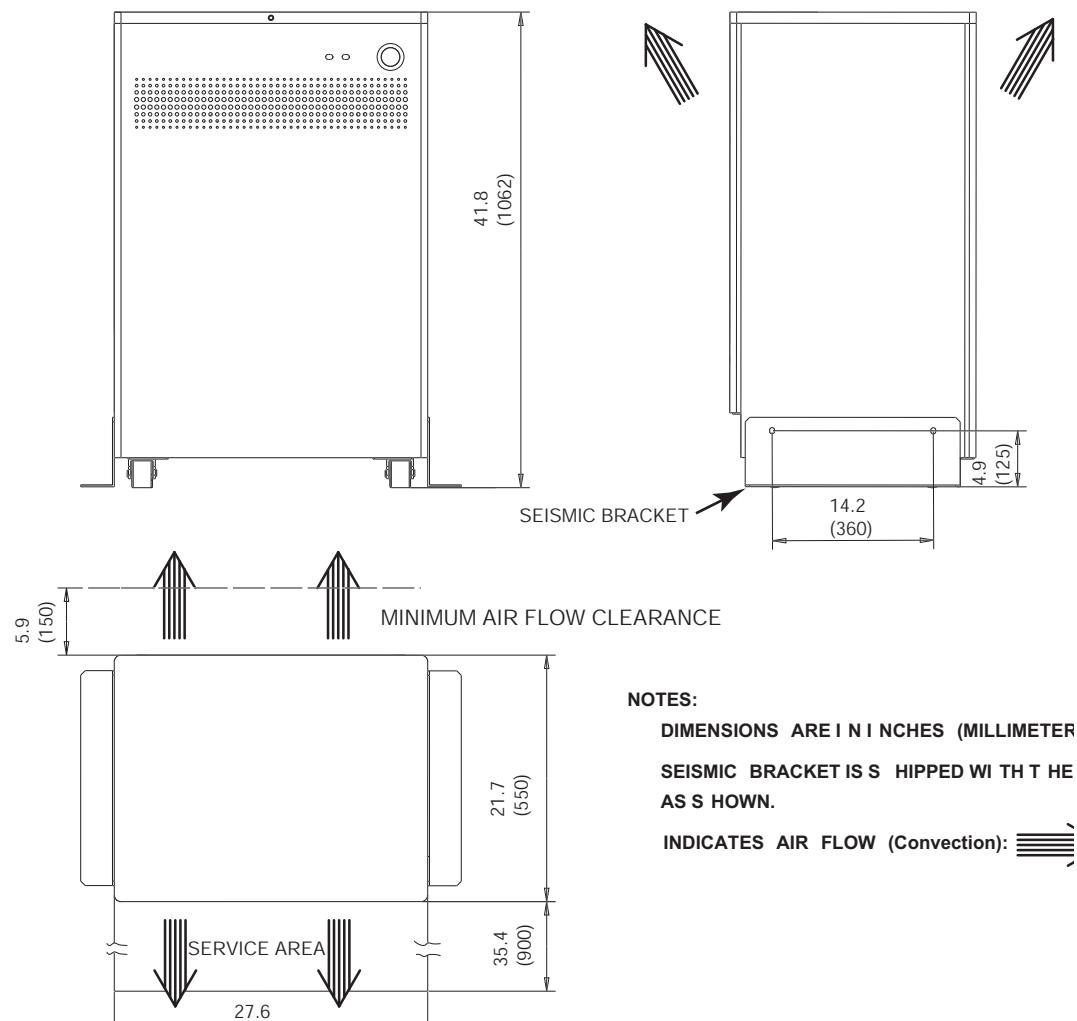


Figure 7-7 Power Distribution Unit (NGPDU-6)



NOTES:

DIMENSIONS ARE IN INCHES (MILLIMETER)

SEISMIC BRACKET IS SHIPPED WITH THE AS SHOWN.

INDICATES AIR FLOW (Convection):



Figure 7-8 Power Distribution Unit (NGPDU-6)

2.5 Operator Console Dimensions and Ergonomic Specification

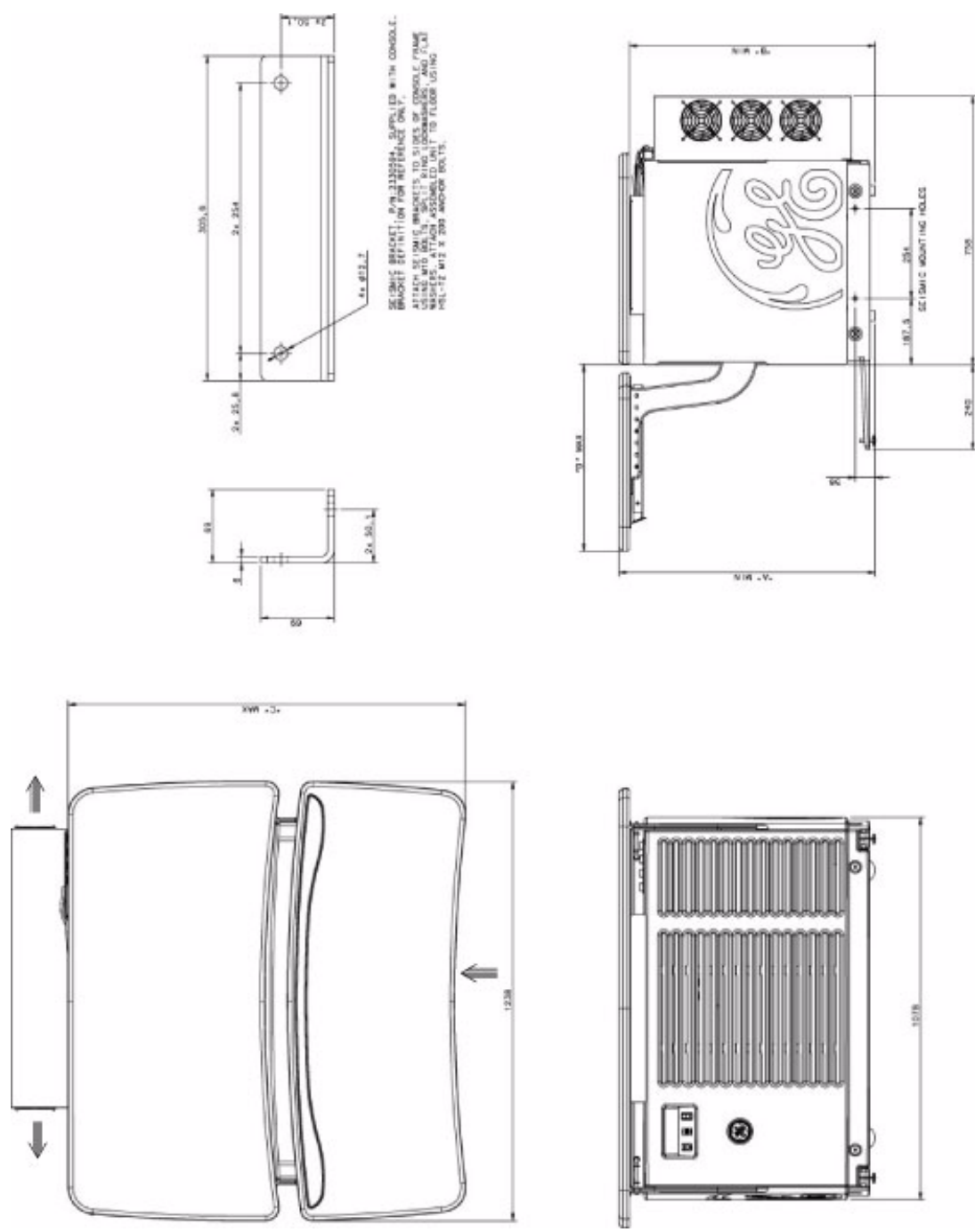


Figure 7-9 Operator's Console

Dimension	Minimum - mm (in)	Maximum - mm (in)	Clearance
A	680 (26.7)	883 (34.7)	Minimum clearance between the backside of the console and the wall is 52 mm (6 in) .
B	680 (26.7)	807 (31.7)	
C	1020 (40.1)	1228 (48.3)	
D	426 (16.7)	634 (24.9)	

Table 7-3 Console Dimensions

Chapter 8

Structural and Mounting Requirements

Section 1.0: Overview

1.1 Importance of Meeting Structural Requirements

System performance specifications require close consideration of the customer's floor properties. The information in this chapter provides critical information and guidelines that the customer or PMI should communicate to the architect, structural engineer, and contractor prior to construction or renovation. Failure to properly evaluate the customer's floor and ceiling properties may result in limited performance and possible safety hazards.

1.1.1 Levelness, Vibration, and Floor Loading

All floors, whether configured to use the recommended GE-supplied anchoring system or an equivalent anchoring method, must meet the requirements for LEVELNESS, VIBRATION, and FLOOR LOADING listed in [Section 3.0 on Page 68](#).

1.1.2 Seismic Loading

Local laws and building codes in some areas may require the customer's contractor and structural engineer to consider seismic loads. [Section 6.0 on Page 77](#) provides the information necessary for the customer's contractor and structural engineer to complete the proper seismic calculations.

1.1.3 Anchoring

[Section 5.0 on Page 74](#) lists the information necessary for the customer's contractor or structural engineer to properly implement the GE-supplied anchoring system, if appropriate for the site. Please note that local laws, building codes, seismic considerations, and building or structural limitations may require the use of anchoring methods other than the GE-supplied anchoring system. In such cases, responsibility for providing an equivalent anchoring method rests solely with the customer's contractor or structural engineer.

Consult your architect, structural engineer, contractor, or PMI to resolve any questions.



NOTICE

Responsibility for providing an approved support structure and mounting method for all floor types, other than those listed in this chapter, rests with the purchaser. General Electric accepts no responsibility for any failure of the support structure or anchoring method, including seismic mounting and anchoring. GE accepts no responsibility for methods other than those listed.

Section 2.0: Ceiling Requirements

The minimum ceiling height above the table and gantry shall measure at least 2134 mm (84 in.) or the minimum distance allowed by local laws and codes, whichever is greater, when measured from the floor to the finished ceiling or to the ceiling pedestal mounts of any ceiling-mounted components. The purchaser or their contractor shall complete the installation of all pedestals for ceiling-mounted components. The PMI will provide the necessary bolt hole information upon request.

The support structure for a ceiling-mounted option using a Mavig pedestal, requires a flush ceiling mounting plate. This flush ceiling mounting plate must be designed by a structural engineer and installed by a qualified contractor prior to the CT installation.

2.1 Mounting Plate

The pedestal-ceiling mount requires a flush ceiling mounting plate that is structurally supported to handle the weight of the load as shown in [Figure 8-1](#). You can order a Mavig-designed ceiling plate (p/n E8007PJ / E8004NG). This 356 mm x 356 mm (14 in. x 14 in.) plate has pre-drilled 1/2 in. mounting holes in each of the four (4) mounting corners.

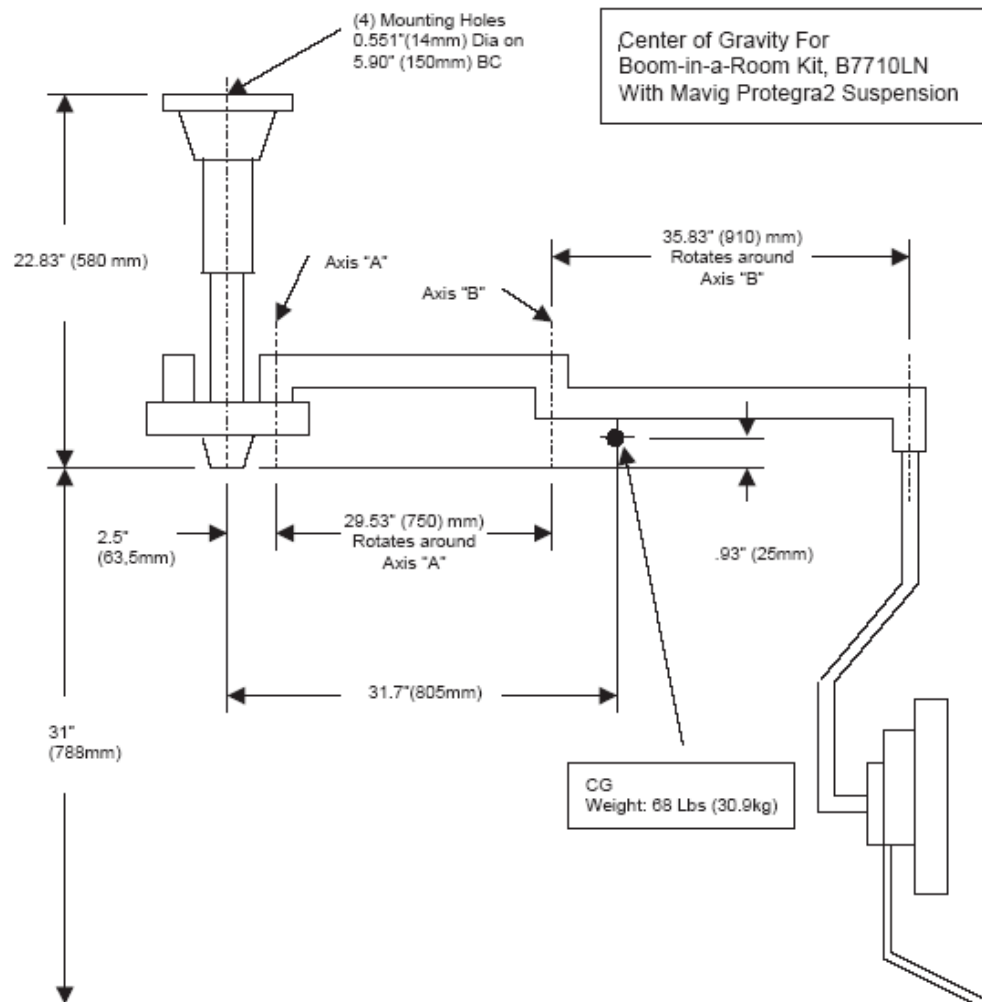


Figure 8-1 Center of Gravity for Boom-in-a-Room Kits, shown on the site print

If a structural contractor designed an equivalent plate, use the supplied template to drill the required four 0.39 in. holes in a pattern as shown in [Figure 8-2](#). An additional 0.32 in. hole is required for the pedestal-ceiling mount safety chain.

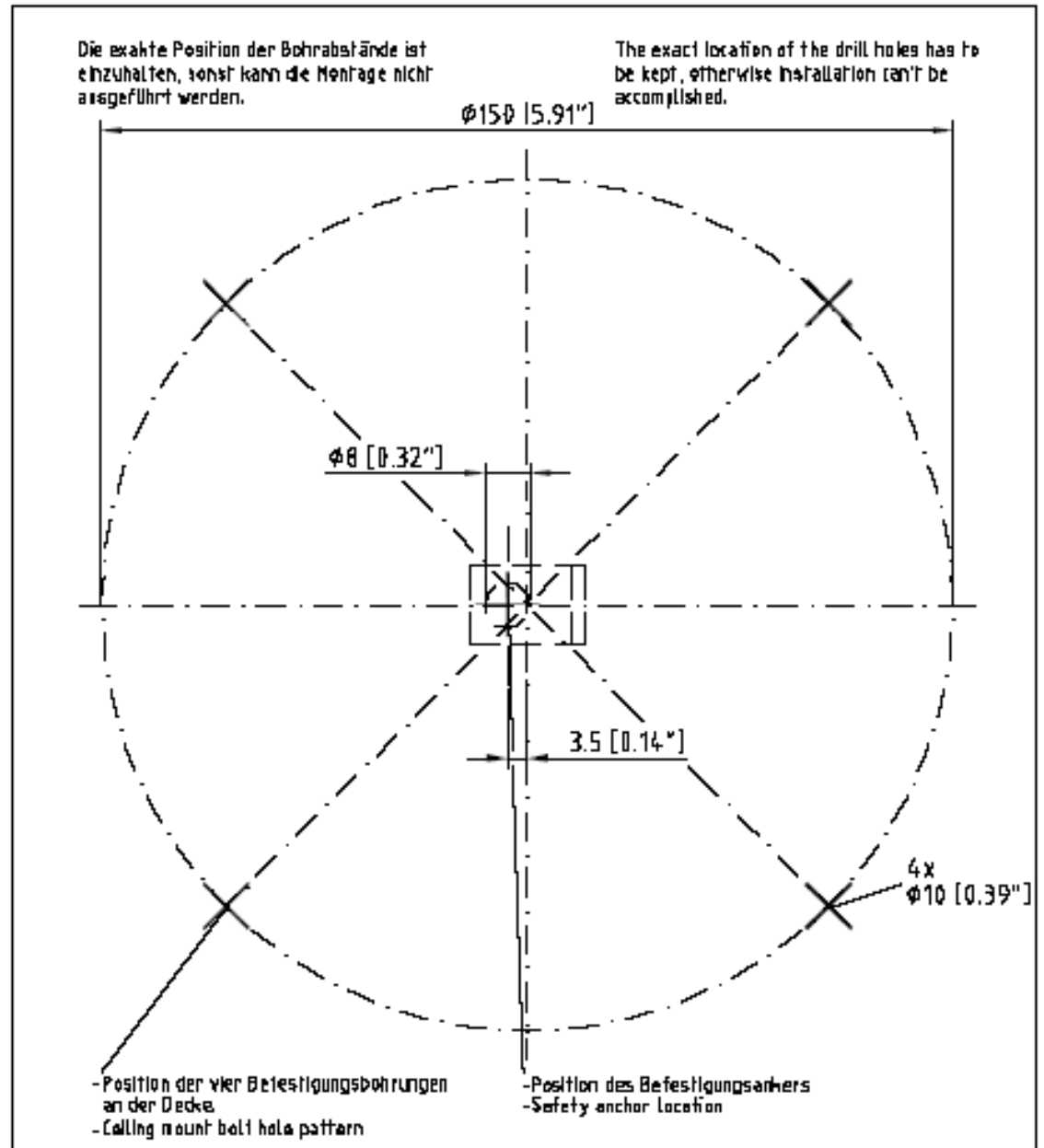


Figure 8-2 Mounting hole pattern for Mavig ceiling pedestal base

2.2 Pedestal Length Requirement

The pedestal-ceiling mount assembly must be installed at least 2134 mm (84 in.) from the lowest point to the finished floor.

2.3 Electrical Box Requirement

A 152 mm x 152 mm x 102 mm (6 in. x 6 in. x 4 in.) or equivalent ceiling box is required to be flush mounted next to the ceiling plate. There should be two (2) conduits exiting into the box and the box grounded to the mounted plate. The electrical box cover plate must be flush mounted to the finished ceiling and with provision to add a 102 mm x 102 mm (4 in. x 4 in.) centered GE-supplied electrical cover plate.

- Note:
- Additional mounting information is available on the Mavig website. Refer to the Protegra 2 Installation manual.
 - Seismic information is also available on the same website.

Section 3.0: Minimum Floor Requirements

3.1 Floor Levelness Specifications

3.1.1 Critical Specifications

Accurate patient positioning during scanning depends on proper alignment of the gantry and the table. The floor levelness specifications in [Table 8-1](#) ensure that the table and gantry height adjusters have enough range to allow proper leveling of the system.

Specification	Metric (minimum)	English (minimum)
Levelness	6 mm maximum variance over 3048 mm	1/4 in. maximum variance over 10 ft

Table 8-1 CRITICAL SPECIFICATIONS for Floor Levelness

3.1.2 Floor Levelness Guidelines

Consider the following factors when determining floor levelness:

- Factors that can disturb the levelness of a weak floor, including:
 - Moving weights such as gurneys or heavy personal equipment.
 - Changes in the system's center-of-gravity when the table moves, as the table can carry a patient load of up to 227 kg (500 lbs).
- Resilient tile, carpeting, or equivalent that may yield or compress over time. At sites with such floor coverings, be sure to cut away the tile or carpeting where the table and gantry adjusters touch the floor to expose the stable base material upon which to seat the adjusters.
- Floor shims are NOT PERMITTED.
- Refer to the steps listed in [Section 3.1.3](#) and to [Figure 8-3](#) to check whether the floor of the scan suite meets the floor levelness specifications for the system.

3.1.3 Measuring Floor Levelness

- 1.) Using the GE Floor Template (P/N 5111499) to establish the room layout and system location, locate the table and gantry anchor holes.

Note: To order a GE Floor Template (P/N 5111499), contact the GE Project Manager of Installation.

- 2.) Place the gantry template on the floor and align it according to the GE site print.
- 3.) Place the table template over the top of the gantry template, and align the scan and table centerlines.
- 4.) Secure the templates to the floor.
- 5.) Use a laser to check the levelness of the floor across the entire area covered by the template, as shown in [Figure 8-3](#).

Note: If the floor is not level, your system cannot be properly aligned.

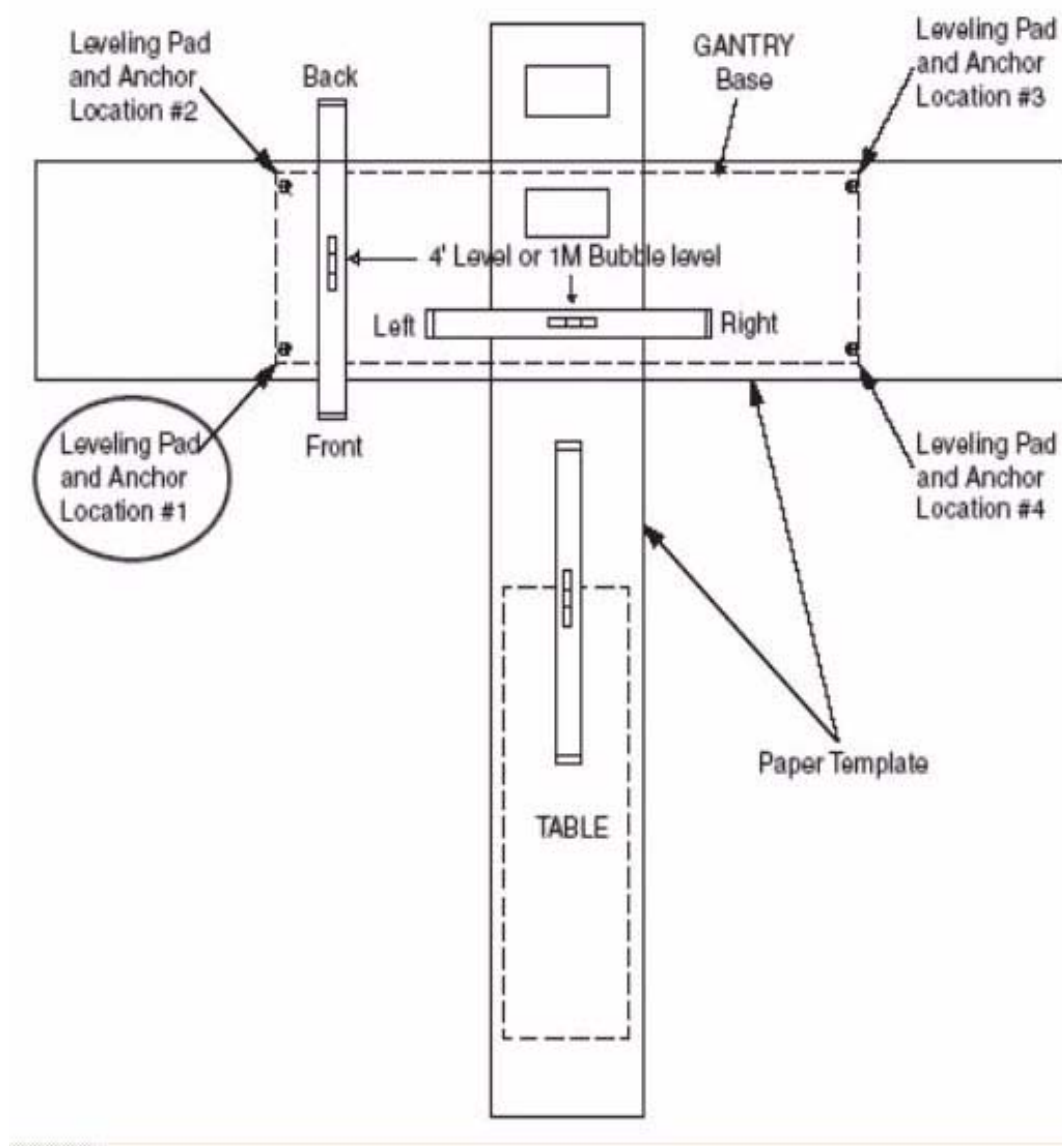


Figure 8-3 Determining Floor Levelness

3.2 Floor Vibration Specifications

3.2.1 Requirements

CT systems are sensitive to vibration and may display limited performance if exceeding the vibration limits listed below. The band of frequencies in which systems exhibit the most sensitivity appears at or near the resonant frequencies of the gantry and the patient table, the latter of which varies depending on patient mass and location. These frequencies fall within the following ranges:

- Patient Table: 2 – 10 Hz
- Gantry: 8 – 14 Hz

Floor vibration from any intermittent or continuous source, such as walking, running, exercising, mechanical equipment, and traffic, must not exceed the levels shown in [Figure 8-4](#) or [Figure 8-5](#), as represented by the solid line labeled *CT Scanner/Table*. These figures compare this limit to the limits of what the AISC (American Institute of Steel Construction) and the ISO (International Organization for Standardization) call Class A (VC-A) and Class B (VC-B).

Note: In [Figure 8-4](#) and [Figure 8-5](#) the symbol μ represents 10^{-6} .

The preferred format for measuring vibration is velocity versus frequency, as shown in [Figure 8-4](#). However, should it prove necessary to measure acceleration and there is no means to convert the measured data to velocity, then use the equivalent acceleration limit shown in [Figure 8-5](#), derived from the velocity spectrum.

Frequency [Hz]	Velocity [$\mu\text{m/s}$, rms]
4	100
10	40
12.5	40
16	50
80	50

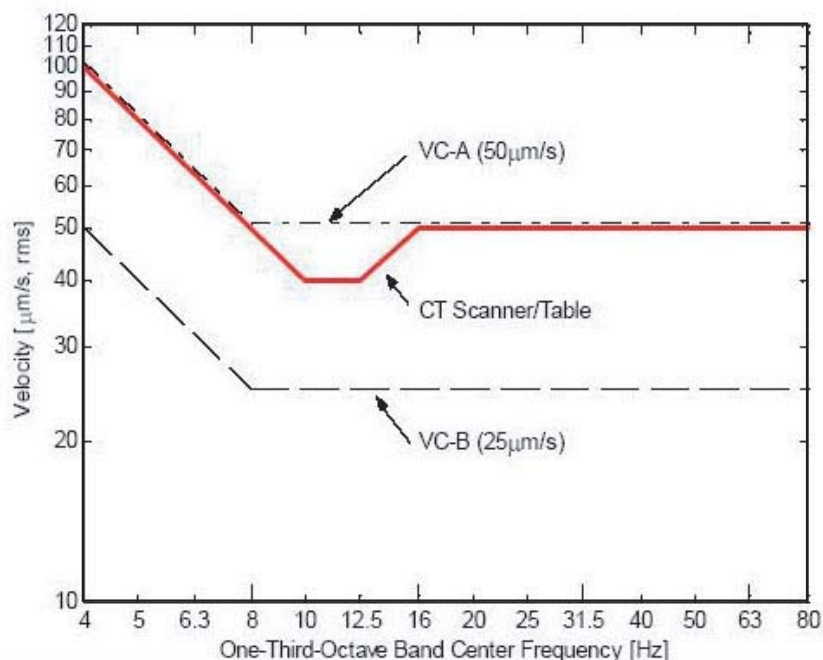


Figure 8-4 Allowable floor vibration in velocity units compared to ISO class A & B limits

Frequency [Hz]	Acceleration [mm/s ² , rms]
4	2.5
10	2.5
12.5	3.1
16	5
80	25

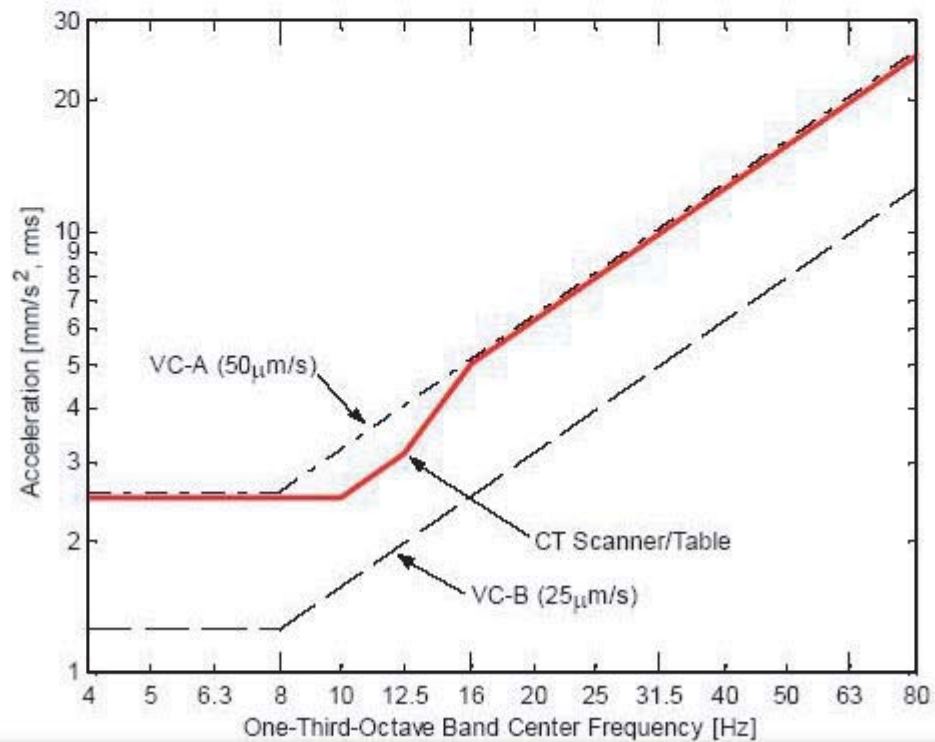


Figure 8-5 Allowable floor vibration in acceleration units compared to ISO class A & B limits

3.2.2 Sources of Floor Vibration

Consider that vibrations strong enough to affect the floor may emanate from the following sources in and around the scanning facility, requiring possible isolation of the floor or structure from them:

- Hospital power plants housing pumps, motors, air handling equipment, or air conditioning units
- Nearby rooms with exercise equipment or where running, jumping, or exercising occurs
- Hallway foot traffic
- Elevators
- Parking lots
- Roadways
- Subways
- Trains
- Heliports

Section 4.0: Floor Loading and Component Weights

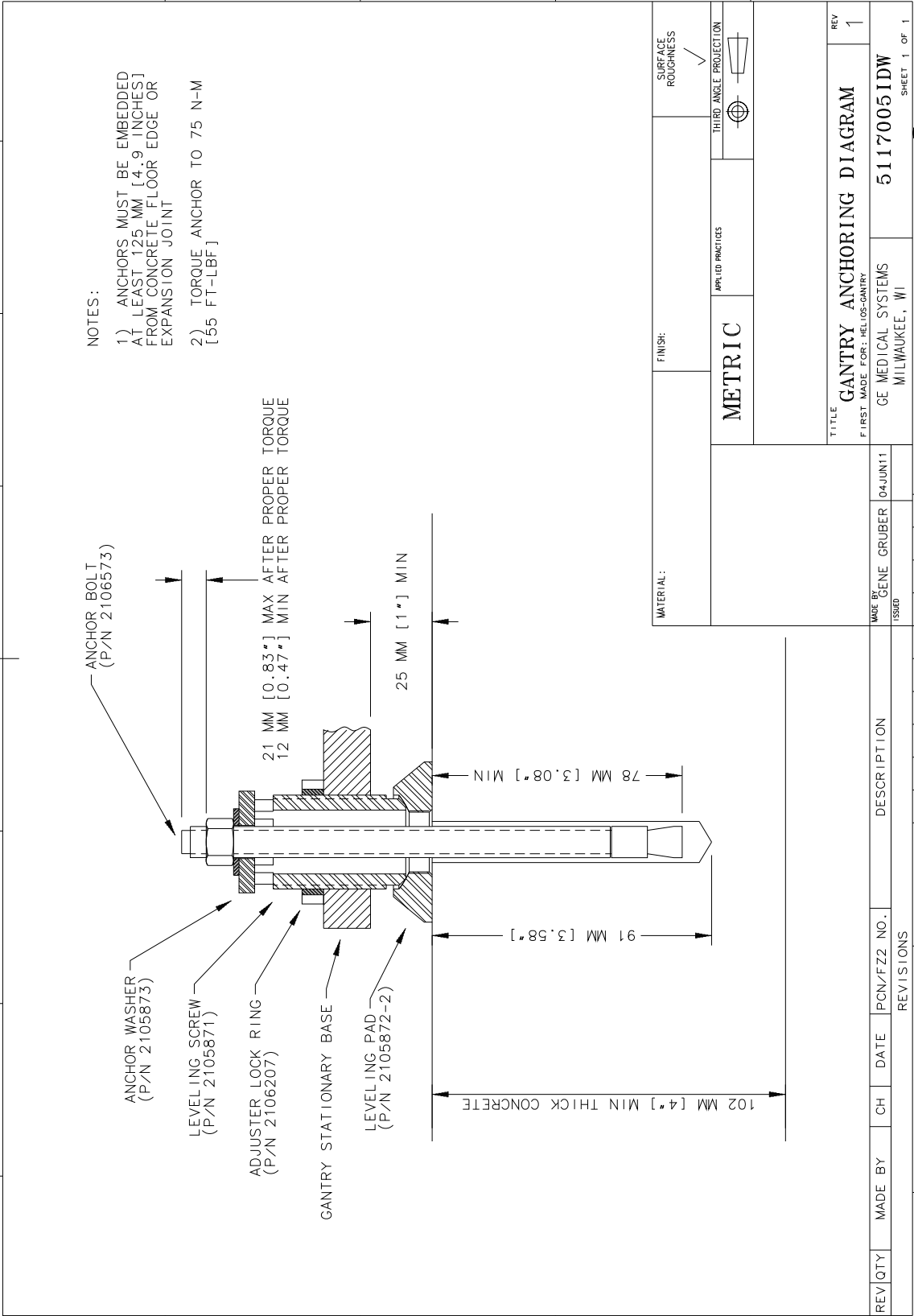
The customer's contractor and structural engineer should use the information in [Table 8-2](#) to help determine if the floor structure in the scan suite possesses sufficient strength to support the weight of the system.

System Component	NET Weight kg (lb)	Overall Width x Depth mm (in.)	Max Uplift Load N (lb)	Max Compressive load N (lb)	Supports mm (in.)
Gantry <i>with covers</i>	~1850 (~4050)	2267 X 1007 (89.25 X 39.65)	0	4895 (1100)	Four round 64 mm (2.5 inch) pads in rectangular pattern.
Dollies (each)	114 (250)		0	4895 (1100)	
Gantry Top Cover (each)	11 (25)				
Gantry Side Cover (each)	11 (25)				
Gantry Front Cover	43 (95)				
Gantry Rear Cover	45 (100)				
GT 1700 Table <i>without patient</i>	400 (880)	650 x 2436 (25.6 x 95.9)	2690 (605)	5350 (1200)	Four round 64 mm (2.5 in.) pads
GT 1700 Table <i>with 227 kg (500 lb) patient</i>	625 (1375)	650 x 2436 (25.6 x 95.9)	2690 (605)	5350 (1200) ¹	Four round 64 mm (2.5 in.) pads
GT 2000 <i>without patient</i>	568 (1250)	650 x 2910 (25.6 x 114.5)	2630 (591)	5210 (-1170)	Four round 64 mm (2.5 in.) pads
GT 2000 <i>with 227kg (500 lb) patient</i>	795 (1750)	650 x 2910 (25.6 x 114.5)	2630 (591)	5210 (1170) ¹	Four round 64 mm (2.5 in.) pads
Footswitch Assembly (GT)	15 (33)				
Power Distribution Unit NGPDU-4 (VCT)	~365 (~800)	711 x 559 (28 x 22)	0	1070 (240)	Four casters
Power Distribution Unit NGPDU-6 (HD)	~365 (~800)	711 x 559 (28 x 22)	0	1070 (240)	Four casters
Console -GOC6 with VCT	195(430)	1238 x 1372 (49 x 54)	0	820 (184)	Four casters
Console -GOC6 with HD	238(524)	1238 x 1372 (49 x 54)	0	820 (184)	Four casters
LCD Monitor (each)	10 (22)				
UPS Powerware 9355	281 (620)	330 x 1219 (13 x 48)	0	2157 (485)	Four casters

¹**Note:** Loads provided for table support with patient in worst-case-scenario positioning.

Table 8-2 LightSpeed VCT/Discovery CT750 HD Component Weight and Floor Loading Data

4.1 Floor Loading and Anchoring Guidelines



Follow the floor loading and anchoring guidelines below when preparing a site for system installation:

- The table and gantry require secure anchoring to the scan room floor. The power distribution unit and the console sit on the floor with casters; anchoring of these components to the floor is optional, unless required because of seismic considerations.
- The total floor load of a LightSpeed VCT or Discovery CT750 HD system with a GT2000 table and no UPS weighs approximately 3229 kg (7104 lbs).
- When carrying the heaviest possible patient, the table-gantry-footswitch assembly represents a concentrated load within the scan room that can weigh up to 2660 kg (5852 lbs).
- Anchors mount through the table and gantry supports. Use the floor template or its dimensions to locate the table and gantry support positions within the scan room, making sure that any anchors that pass through the supports clear all structural beams and interferences in the floor.
- If a loading analysis determines that the gantry and table position should change relative to their position on the GE site print, be sure to take into account the clearance requirements in [Chapter 4, Regulatory Requirements](#) and [Chapter 5, Service Clearance Requirements](#) when determining an appropriate location for the scanner.
- Hospitals and scanning facilities throughout the world may utilize a variety of floor types, and the disposition of different floor types may necessitate additional planning to adequately accommodate the scanner:
 - Wood floors often require substantial reinforcement. GE does not recommend using wood floors.
 - Temperature variation in blacktop or marble floors may allow anchor movement and pullout. GE does not recommend using these floors.
 - GE recommends using concrete floors with a minimum thickness of 102 mm (4 in.) when using GE-supplied anchoring or any other equivalent anchoring method.



NOTICE

Responsibility for providing an approved support structure and mounting method for all floor types other than the GE-recommended floor rests with the purchaser. General Electric accepts no responsibility for any failure of the support structure or anchoring method, including those used for seismic mounting. GE accepts no responsibility for methods other than those listed.

Section 5.0: GE-Supplied Anchoring

GE supplies anchors for mounting the table and gantry. The console and power distribution unit do not require anchoring to the floor. It is the responsibility of the customer to have a structural engineer and trained contractor use either the GE-supplied anchoring method or to provide an equivalent anchoring method to mount the table and gantry to the floor.

Consult your architect, structural engineer, contractor, or PMI to resolve any questions.



WARNING

POTENTIAL FOR PATIENT INJURY!

AN IMPROPERLY SECURED TABLE MAY TIP, DISLODGING THE PATIENT. PATIENT SAFETY DURING SYSTEM OPERATION REQUIRES PROPER ANCHORING OF SYSTEM COMPONENTS.

5.1 Specifications of GE-supplied Anchors

Table 8-3 lists the specifications of GE-supplied anchors for the scanner. For a detailed view, including dimensions and additional specifications, see Figure 8-6.

GE-Supplied Anchors	
Part Number	2106573
Description	Hilti Kwik Bolt II
Diameter	12.7 mm (0.5 in.)
Length	203 mm (8 in.)

Table 8-3 GE-Supplied Anchor Specifications

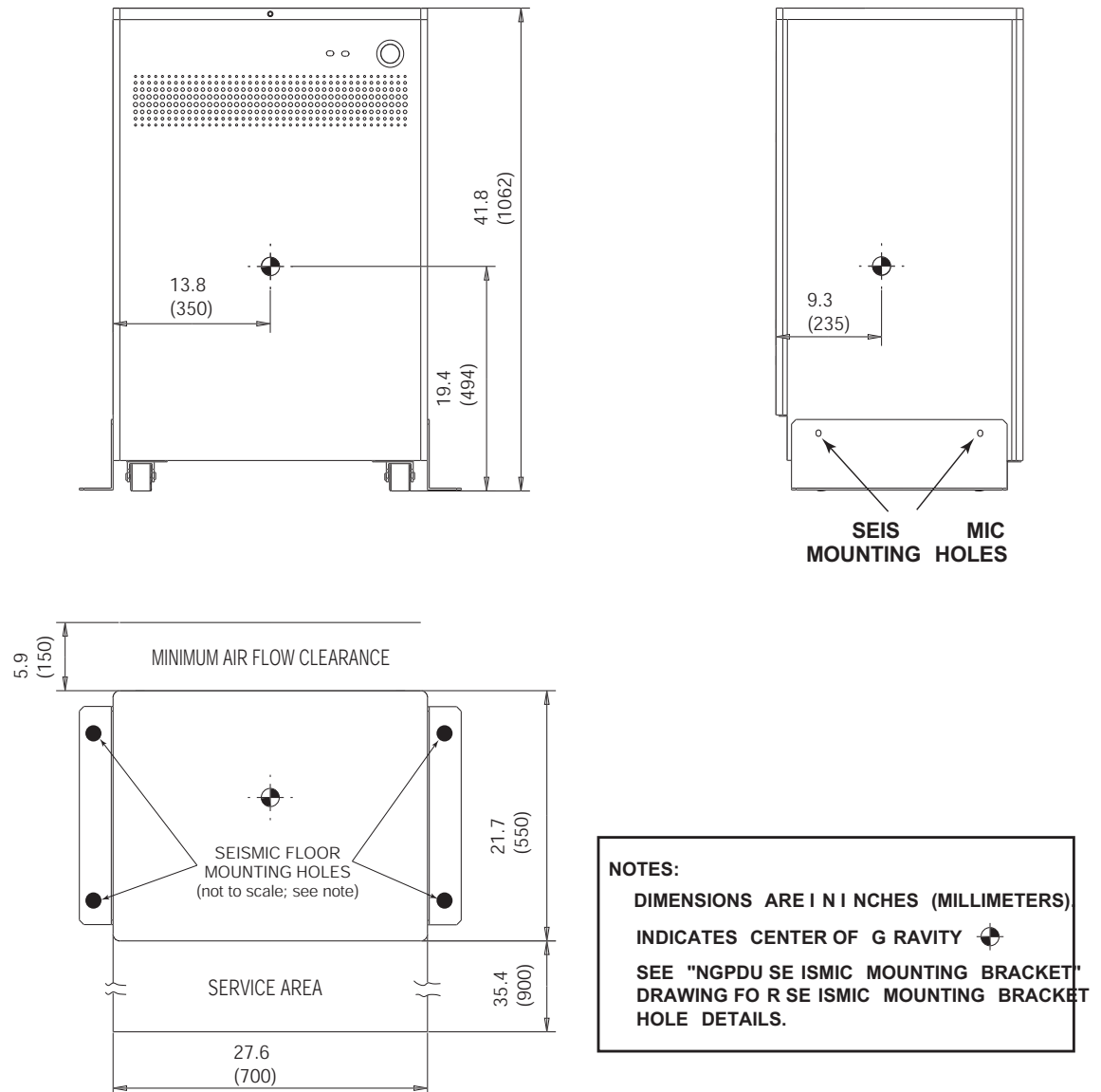
5.2 Requirements for Using GE-supplied Anchors

- Use of GE-supplied anchors (P/N 2106573) shall adhere to the following requirements:
- Use the GE-supplied anchors ONLY when mounting components on concrete floors.
 - Adhere to all anchoring requirements listed in Table 8-4.
 - Any anchors showing more than 21 mm (~0.9 in.) of thread above the torqued nut requires the installation of a second anchor in the closest adjacent mounting location. The second anchor shall meet the same requirements in Table 8-4.
 - Non-seismic installations must use a minimum of four (4) anchors to mount the gantry and four (4) anchors to mount the table.
 - Fully engage the Adjuster Lock Rings (P/N 2106207) with at least one full thread showing below the notched portion on the Adjuster Screw.

Note: The table does not have the Adjuster Lock Rings shown in Figure 8-6.

Mounting Requirements	Table	Gantry
Minimum Floor Thickness	102 mm (4 in.)	102 mm (4 in.)
Recommended Drilling Depth	95 mm (3.75 in.)	95 mm (3.75 in.)
Average Anchor Embedment	89 mm (3.5 in.)	89 mm (3.5 in.)
Minimum Anchor Embedment	76 mm (3 in.)	76 mm (3 in.)
Available Alternate Anchor Locations	Yes	Yes
Shipped Anchor Size	203 mm (8 in.)	203 mm (8 in.)
Alternate Anchoring Methods	Yes (see notes, below)	Yes (see notes, below)
Floor Levelness	6 mm (.25 in.) maximum variance over 3048 mm (10 ft)	6 mm (.25 in.) maximum variance over 3048 mm (10ft)

Table 8-4 Table and Gantry Anchoring Requirements

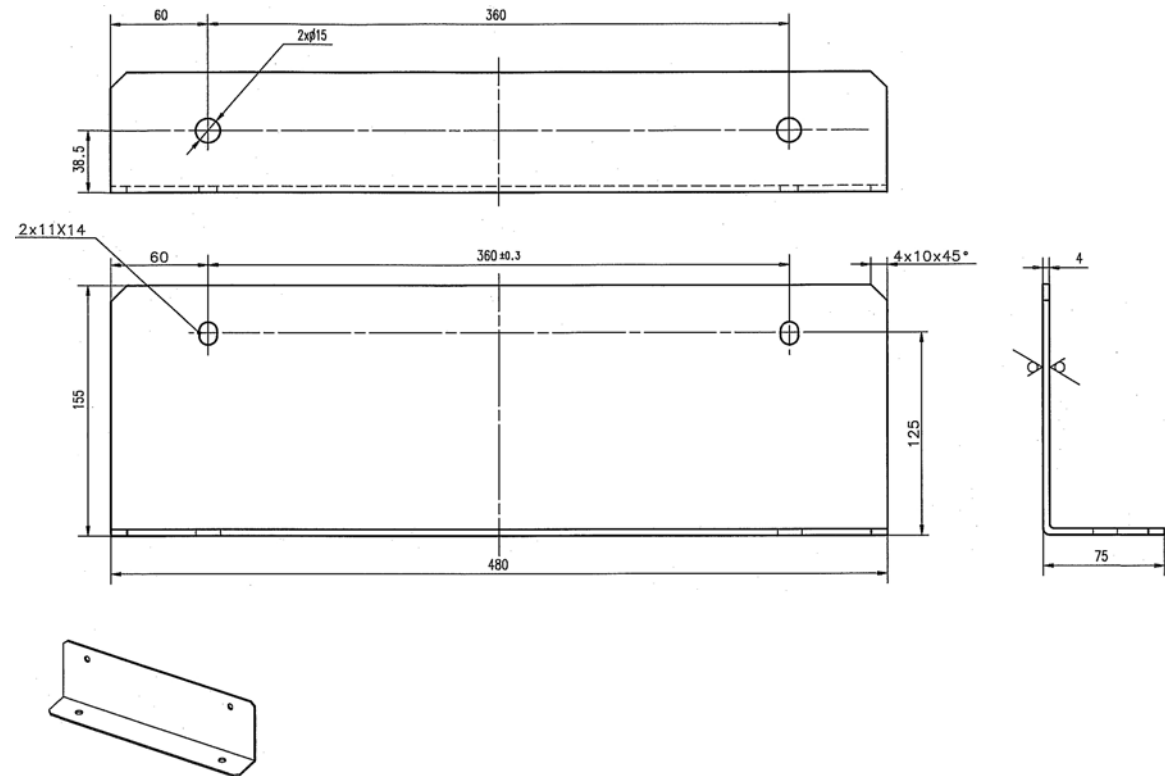


Section 6.0: Seismic Mounting

6.1 Overview

Refer to the guidelines in this section when mounting the system in seismic zones:

- Responsibility for proper seismic mounting rests with the customer. Refer to all applicable laws and codes for your locality.
- GE provides an optional Seismic Kit (R4390JC), which includes brackets to allow anchoring of the PDU (power distribution unit) and console to the floor.
- GE-supplied anchors may not meet local seismic laws and codes. Use them only if a qualified structural engineer approves them for use in local seismic applications.
- The customer's contractor often supplies a state-certified print or equivalent, showing seismic installation instructions.
- Consider seismic requirements for ceiling-mounted fixtures and refer to the appropriate installation instructions for ceiling-mounted fixtures. .



6.2 Center-of-Gravity Information

The information in the following figures provides the customer's contractor and/or structural engineer with center-of gravity information to assist in seismic calculations for the system:

- Gantry: [Figure 8-7](#)
- Table: [Figure 8-8](#)
- Power Distribution Unit: [Figure 8-9](#) (NGPDU Seismic Mounting Bracket)
- Operator's Console: [Figure 8-10](#)

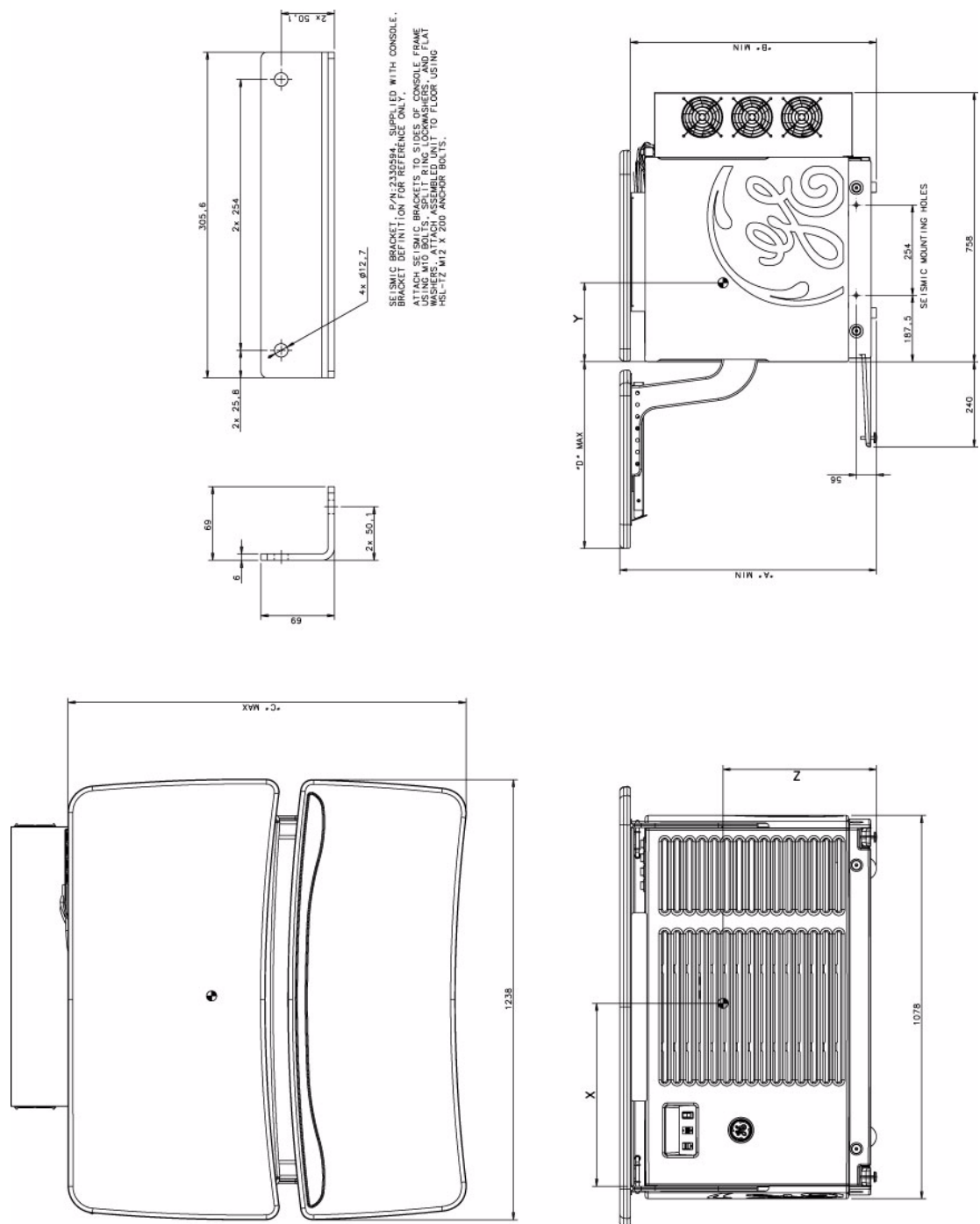


Figure 8-6 GOC6 Center-of-Gravity

Physical ID	Weight - Kg (lb)	X - mm (in)	Y - mm (in)	Z - mm (in)
5212920-100 (HD)	236 (520)	551 (21.6)	222 (8.7)	432 (17)
5212920-300 (VCT)	195 (430)	557 (21.9)	214 (8.4)	397 (15.6)

Table 8-5 Center-of-Gravity Location

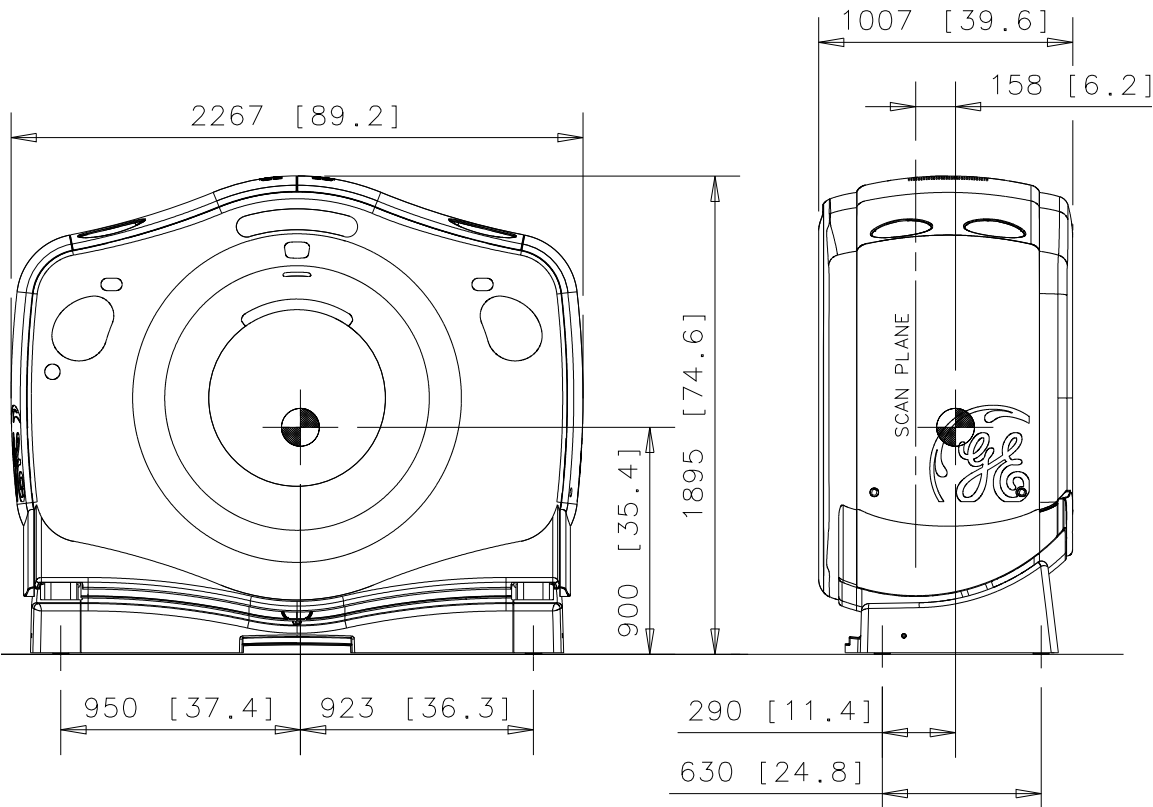


Figure 8-7 Gantry Center-of-Gravity

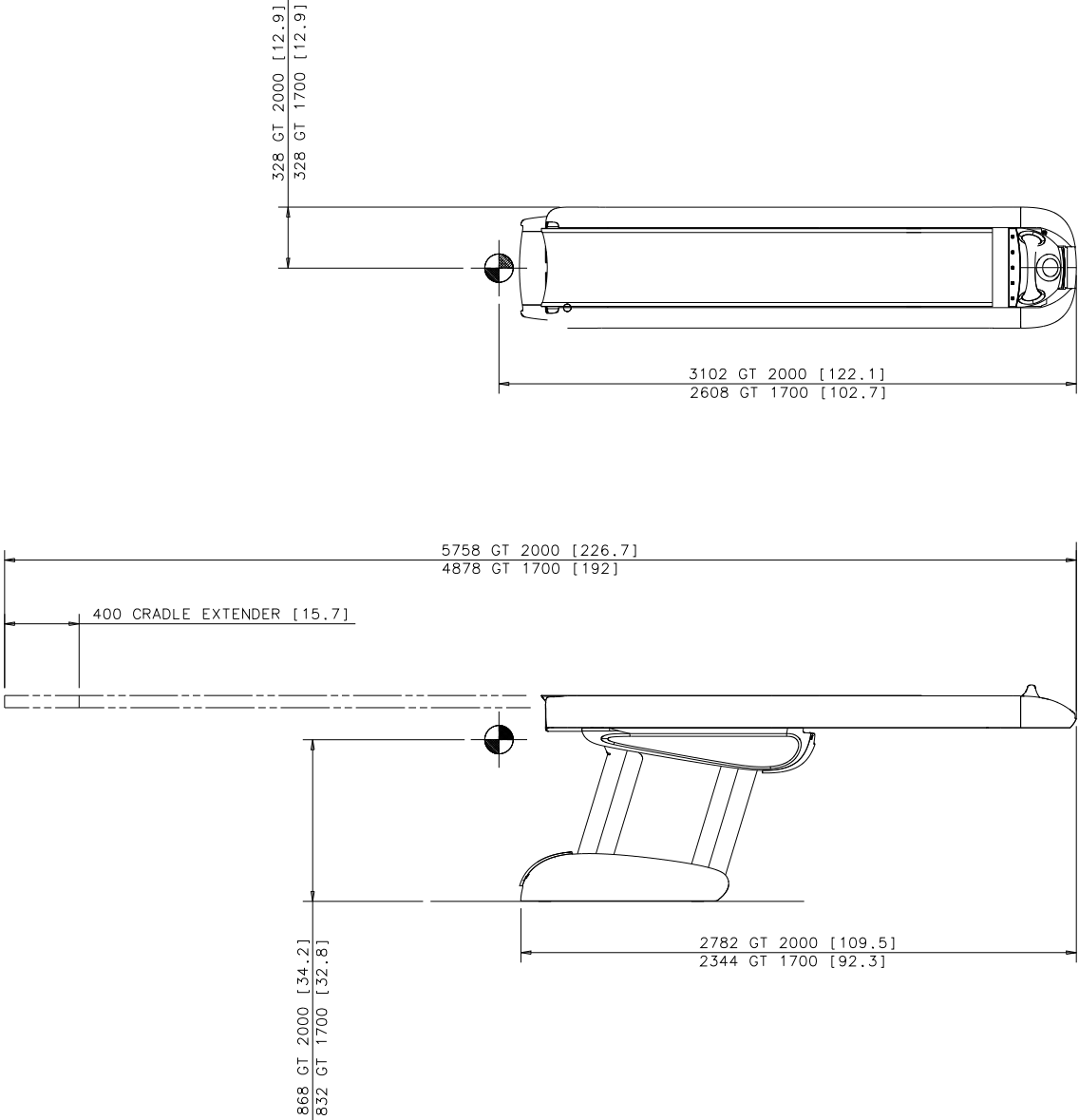


Figure 8-8 Table Center-of-Gravity

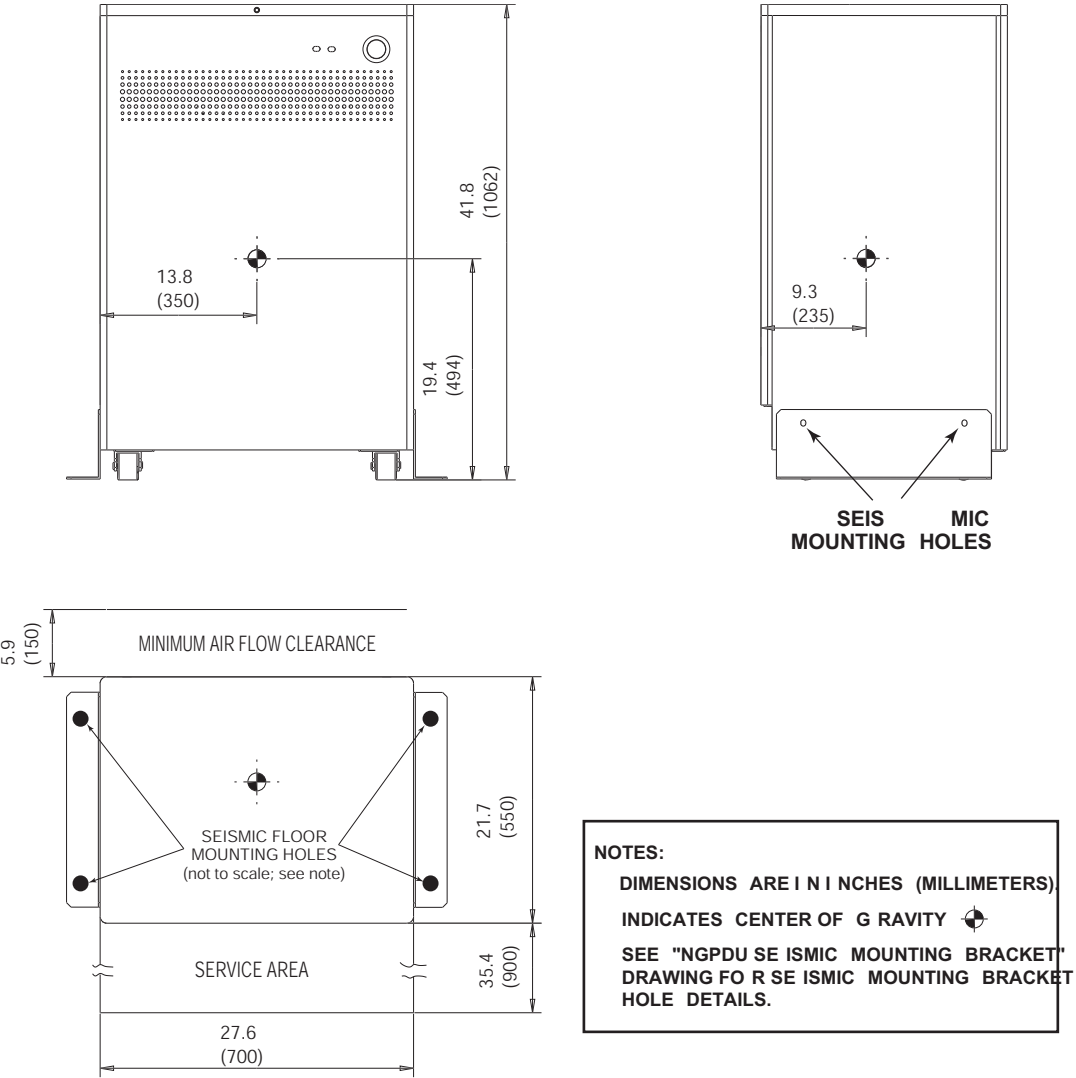


Figure 8-9 PDU Center-of-Gravity NGPDU-4 / NGPDU-6

Chapter 9

Environmental Requirements

Ensure the operational readiness and proper system calibration of HVAC prior to installation. Maintain the environmental conditions listed below at ALL times, including over nights, weekends, and holidays. Shut down the CT system if air conditioning is not working. When shutting down the system for major repair, you may also shut down the air conditioning.

Section 1.0: Temperature and Humidity Specifications

Environmental specifications apply to the table, gantry, power distribution unit, and operator console.



NOTICE

Exceeding environmental specifications may adversely affect system operation and image quality.

1.1 Temperature (Scan and Control Rooms)

Maximum allowable ambient room temperature:	26° C (79° F)
Recommended ambient room temperature:	22° C (72° F)
Minimum allowable ambient room temperature:	18° C (64° F)

Table 9-1 System Temperature Limits

Note: Be certain to account for ANY cooling equipment cycle control range, ensuring that the maximum and minimum ambient room temperatures do not exceed those shown in [Table 9-1](#) during room thermal cycling. For example, if the HVAC is capable of ± 2° C control, then the limits would be 20° C - 24° C to maintain absolute limits.

1.2 Humidity (Scan Room & Control Room)

Maximum allowable non-condensing relative humidity:	60%
Minimum allowable non-condensing relative humidity:	30%

Table 9-2 System Humidity Limits

1.3 Other Guidelines

- Accurate determination of hospital room environmental conditions may require the temporary installation of a temperature and humidity recorder near the location designated for system installation. Record temperature and humidity readings before and after installation to verify the site's true environmental conditions.
- Consider heating, ventilating, air conditioning (HVAC) needs, and redundancy (back-up). An air conditioner with two compressor units rather than one, may prevent system downtime. A redundant (back-up) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 2.0: Cooling Requirements

Use [Table 9-3](#) to assist in cooling requirements planning. Gantry operation requires over half of the cooling utilized by your scanner. Contact an HVAC specialist to determine optimal placement of the thermostat and all HVAC vents, bearing in mind that:

- Gantry air INTAKE occurs across the BOTTOM of the gantry.
- Gantry air EXHAUST occurs across the TOP of the gantry.

System Component	Max BTU/HR	Max Watt
1. Gantry maximum (See Note 1)	43,200	12,600
2. Table GT (VT)	1030	300
3. Power Distribution Unit (-6 / -4)	3400	1000
Scan Room Subtotal (see Notes 1 and 2)	47,100	13,800
4. Operator Console/Computer	6570	7845
VCT/ GOC6 and HD/GOC6 includes the following items: VDig w/ 8 drives, LCD Monitor (x2), Media Tower and Injector		
Control Room Subtotal (w/ 3 IGs)	8,825	2,592
Recommended System Total (See Note 2)	54,840	16,000
Options		
Remote Color Monitor (LCD)	425	125
UPS	2900	850
ROOM TOTAL (See Note 2)		

NOTE 1: Maximum heat output reached at tube change.

NOTE 2: Heat output does not include heat from room lighting, personnel, or non-CT equipment.

Table 9-3 System Heat Output

Refer to [Figure 7-3](#), [Figure 7-6](#), and [Figure 7-9](#) for component air flow requirements.

Figure 9-1 and Figure 9-2 show the recommended placements of the thermostat and HVAC vents (intake and output) for the scan and control rooms.

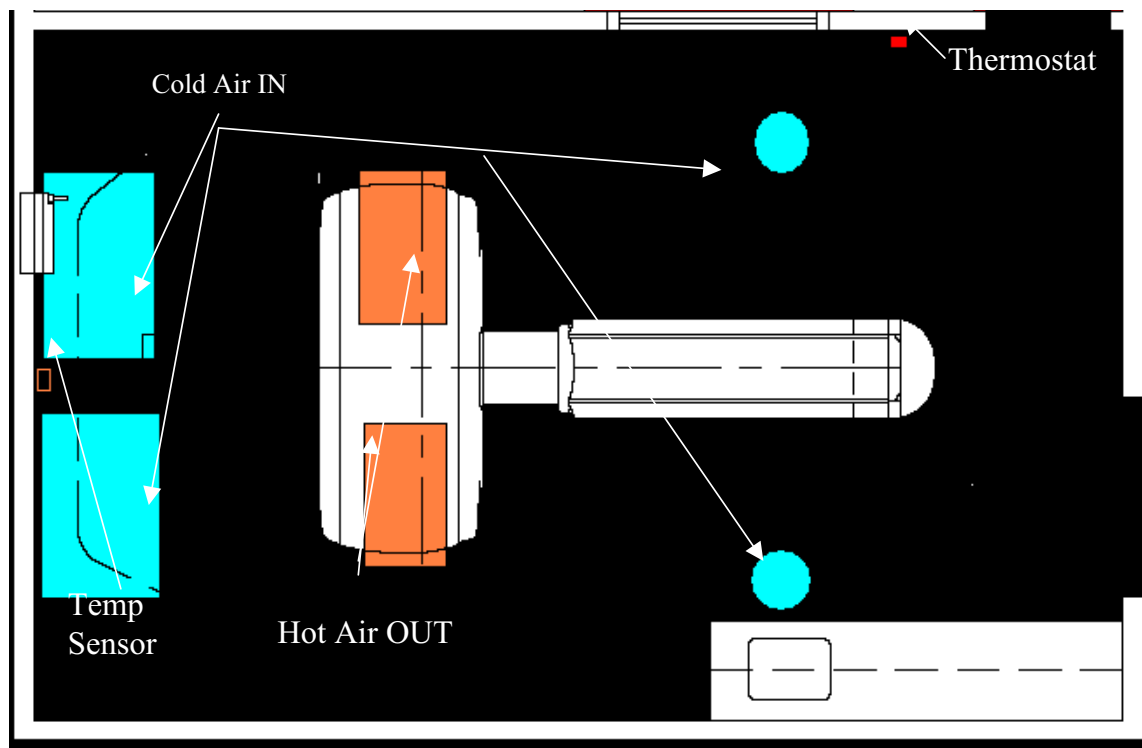


Figure 9-1 HVAC Air Vent Placement in Scan Room

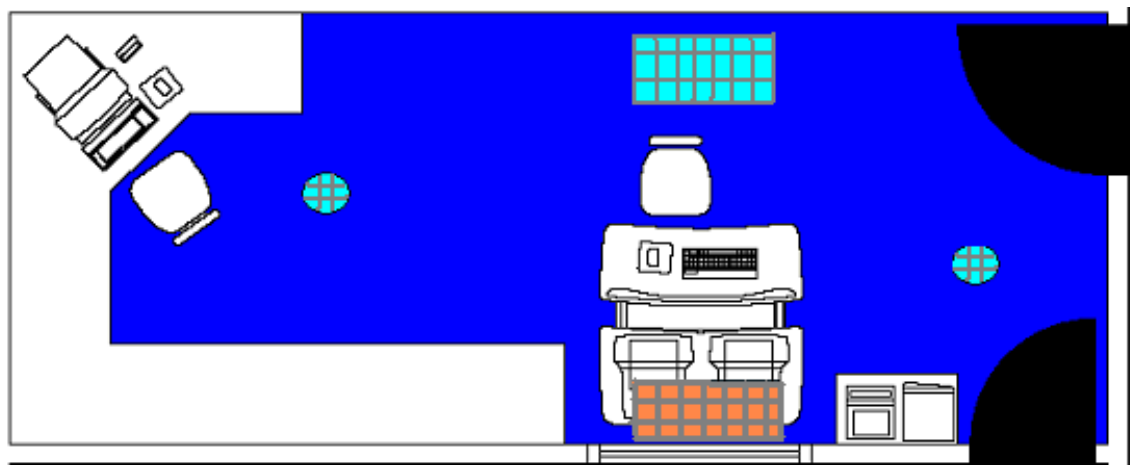


Figure 9-2 HVAC Air Vent Placement in Control Room

Section 3.0: Altitude

The system shall meet all functional and performance specifications when placed in a room that is at an elevation of -150 m to 2400 m (-492 ft to 7,875 ft) above sea level.

Section 4.0: Electro-Magnetic Interference (EMI)

4.1 Gantry

Locate the gantry in ambient static magnetic fields of less than 10^{-4} tesla (500 milligauss) to guarantee the specified imaging performance. Ambient AC magnetic fields must measure below 10^{-6} tesla (10 milligauss) peak.

4.2 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity (see [Figure 9-3](#)).

4.3 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do NOT place sensitive electronics (e.g. the operator console or computer equipment) within 1 m (3.28 ft) of the Power Distribution Unit in any direction, including above or below it (see [Figure 9-3](#)).

Note: The classification of sensitive electronics does NOT include the UPS.

4.4 EMI Reduction

If you know of or suspect the presence of fields of excessive EMI, consult GE Healthcare Sales & Service for recommendations. Consider the following when attempting to reduce EMI:

- External field strength decreases rapidly with distance from source of the magnetic field.
- External leakage magnetic field of a three-phase transformer measures much less than that of a bank of three single-phase transformers of an equivalent power rating.
- Large electric motors constitute a source of substantial EMI.
- High-powered radio signals constitute a source of EMI.
- Maintain good screening of cables and cabinets.
- Consider and measure EMI fields of sites with main facility power running UNDER the floor or WITHIN the walls or ceilings of the scan room.
- Pay special attention to power substations and high-voltage power lines in proximity to the scan facility.
- If any concerns remain regarding excessive EMI fields, be sure to measure to confirm that your site meets all required specifications.

4.5 Equipment EMI “Envelopes”

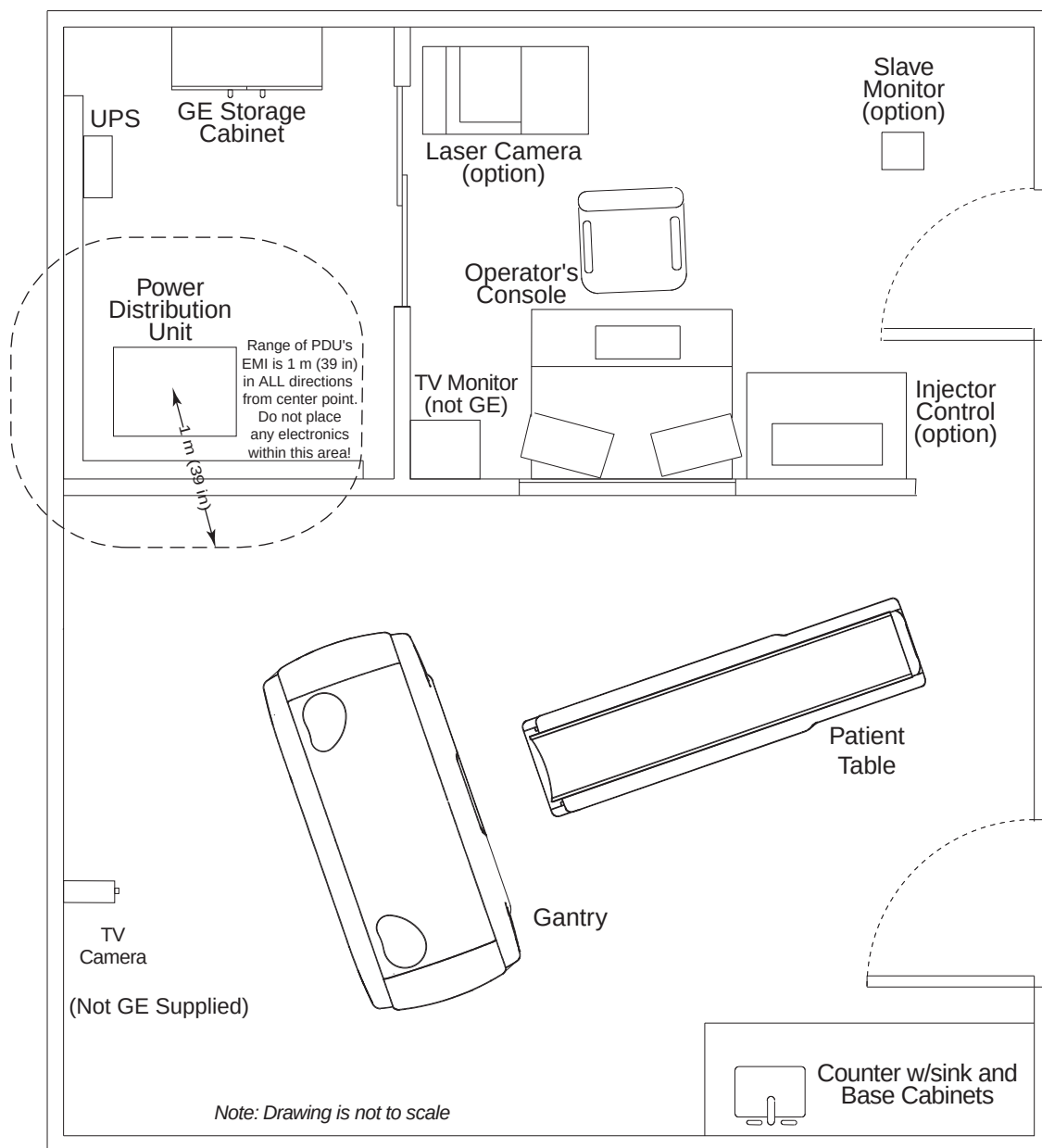


Figure 9-3 Sample Room Layout (Showing Approximate EMI Requirements)

Section 5.0: Electro-Magnetic Compatibility (EMC)

5.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2.1 (2004) EMC standard for medical devices. The LightSpeed VCT/Discovery CT750 HD is suitable to use in the electromagnetic environment, as per the limits and recommendations described in the following tables:

- Emission Compliance level and limits ([Table 9-4](#)).
- Immunity Compliance level and recommendations to maintain equipment clinical utility ([Table 9-5](#) and [Table 9-6](#)).

Note: This system complies with the EMC standard when used with supplied cables. If different cable lengths are required, contact a qualified GE service representative for advice.

5.2 Electromagnetic Emission

The LightSpeed VCT/Discovery CT750 HD is intended for use in the electromagnetic environment specified below. The customer or the user of the LightSpeed VCT/Discovery CT750 HD should assure that it is used in such an environment.

Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 1	The LightSpeed VCT/Discovery CT750 HD uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class A	When installed in such a shielded location, the LightSpeed VCT/Discovery CT750 HD is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Not applicable	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Not applicable	

Table 9-4 Electromagnetic Emissions

5.3 Electromagnetic Immunity

The LightSpeed VCT/Discovery CT750 HD is intended for use in the electromagnetic environment specified below. The customer or the user of the LightSpeed VCT/Discovery CT750 HD should assure that it is used in such an environment.

Immunity Test	EC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be a typical commercial or hospital environment
Surge IEC 61000-4-5	± 1 kV differential mode ± 2 kV common mode	± 1 kV differential mode ± 2 kV common mode	Mains power quality should be a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5 % U_T (> 95 % dip in U_T) for 5 sec	< 5 % U_T (> 95 % dip in U_T) for 5 sec	Mains power quality should be a typical commercial or hospital environment. If the user of the LightSpeed VCT/Discovery CT750 HD requires continued operation during power mains interruptions, it is recommended that the LightSpeed VCT/Discovery CT750 HD is powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Table 9-5 Electromagnetic Immunity

Immunity Test	EC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6	3 V _{RMS} 150 kHz to 80 MHz	3 V 150 kHz to 80 MHz	Do not use portable and mobile RF communications equipment closer to any part of the LightSpeed VCT/Discovery CT750 HD, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended Separation Distance: $d = \left[\frac{3,5}{3} \right] \sqrt{P}$ (see Table 9-6) $d = \left[\frac{3,5}{3} \right] \sqrt{P}$ 80 MHz to 800 MHz (see Table 9-6) $d = \left[\frac{7}{3} \right] \sqrt{P}$ 800 MHz to 2,5 GHz (see Table 9-6) where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,^a should be less than the compliance level in each frequency range.^b Interference may occur in the vicinity of equipment marked with the following symbol: 
Radiated RF IEC 61000-4-3 (Alternative method: Full range IEC 61000-4-21 test in lieu of Large, Permanently-Installed Equipment exemption)	3 V/m 150 kHz to 80 MHz	3 V/m 150 kHz to 80 MHz	

- ^a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the LightSpeed VCT/Discovery CT750 HD is used exceeds the applicable RF compliance level above, the LightSpeed VCT/Discovery CT750 HD should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the LightSpeed VCT/Discovery CT750 HD.

- ^b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

NOTE: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

NOTE: U_T is the AC mains voltage prior to application of the test level.

Table 9-5 Electromagnetic Immunity

The LightSpeed VCT/Discovery CT750 HD is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the LightSpeed VCT/Discovery CT750 HD can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the LightSpeed VCT/Discovery CT750 HD as recommended below, according to the maximum output power of the communications equipment.

Rated Maximum Output Power (P) of Transmitter Watts (W)	Separation Distance (Meters) by Frequency of Transmitter		
	150 kHz to 80 MHz	80 MHz to 800 MHz	800 MHz to 2.5 GHz
	$d = \left[\frac{3,5}{3}\right]\sqrt{P}$	$d = \left[\frac{3,5}{3}\right]\sqrt{P}$	$d = \left[\frac{7}{3}\right]\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.7	11.7	23.3

For transmitters rated at a maximum output power not listed above, the separation distance can be estimated using the equation in the corresponding column, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

Table 9-6 Recommended Separation Distances

As an example, keep a 1 W mobile phone (800 MHz to 2.5 GHz carrier frequency) at least 2.3 m from the LightSpeed VCT/Discovery CT750 HD (in order to avoid image interference risks.)

LIMITATIONS MANAGEMENT: Adhering to the distance separation recommended in [Table 9-6](#), between 150 KHz and 2.5 GHz, reduces disturbances recorded at the image level, but may not eliminate all disturbances. However, when installed and operated as specified, the system maintains its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

5.4 Installation Requirements and Environment Control

In order to minimize interference risks, the following requirements apply.

5.4.1 Cable Shielding and Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or panels, or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.

All interconnect cables to peripheral devices must be shielded and properly grounded, except when technologically prohibited. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

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Chapter 10

Radiation Protection Requirements

Section 1.0: Shielding Requirements



NOTICE Engage a **QUALIFIED RADIOLOGICAL HEALTH PHYSICIST** to review your scan room shielding requirements, taking into consideration:

- Scatter radiation levels within the scanning room (see [Figure 10-1](#)).
- Equipment placement.
- Weekly projected work-loads (number of patients/day technique (kvp*ma)).
- Materials used for construction of walls, floors, ceiling, doors, and windows.
- Activities in surrounding scan room areas.
- Equipment in surrounding scan room areas (e.g. film developer, film storage).
- Room size and equipment placement within the room relative to room size.

[Figure 10-1](#) depicts measured radiation levels within the scanning room, while scanning a 32 cm CTDI phantom and using a large filter, with the technique shown. Use the mAs, kV and aperture scaling factors shown in [Table 10-1](#) to adjust exposure levels to the scan technique used at the site.

Example (from [Figure 10-1](#)): The exposure level for a 120 kV, 800 mA, 1 sec. scan at 1270 mm (50 in.) away from the scan plane is: $10.4 \text{ mGy} \times 0.71 \times 800/100 = 59.2 \text{ mGy}$.

Note: Actual measurements can vary. Expected deviation equals $\pm 15\%$, except for the 5 mA and 1 mm techniques, where variation may be greater (up to a factor of 2), due to the inherent deviation in small values. The maximum deviation anticipated for tube output equals $\pm 40\%$.

Changed Parameter	Multiplication Factor
mAs	new mAs/100
80 kV	0.24
100 kV	0.45
120 kV	0.71
140 kV	1.00
1mm aperture	0.20
3 mm aperture	0.22
5 mm aperture	0.27
10 mm aperture	0.38
15 mm aperture	0.48
20 mm aperture	0.59
30 mm aperture	0.79
40 mm aperture	1.00

Table 10-1 Shielding Requirements Scaling



NOTICE This publication uses mGy (micrograys) to measure radiation levels. The conversion factor from mR to mGy (micrograys) is: **1 mR = 8.69 mGy**.

Note: Data in the figures included in this chapter may change prior to product release.

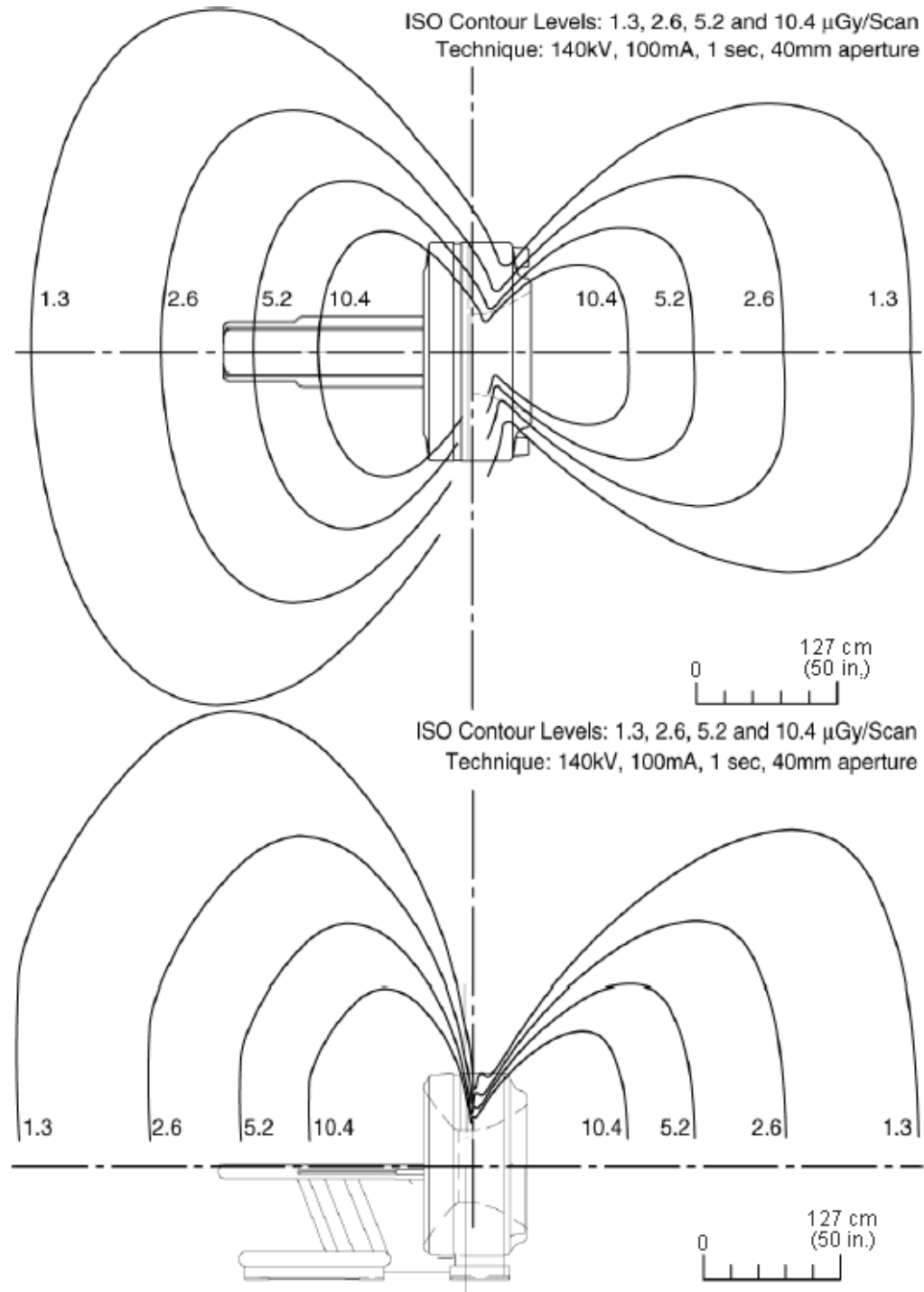
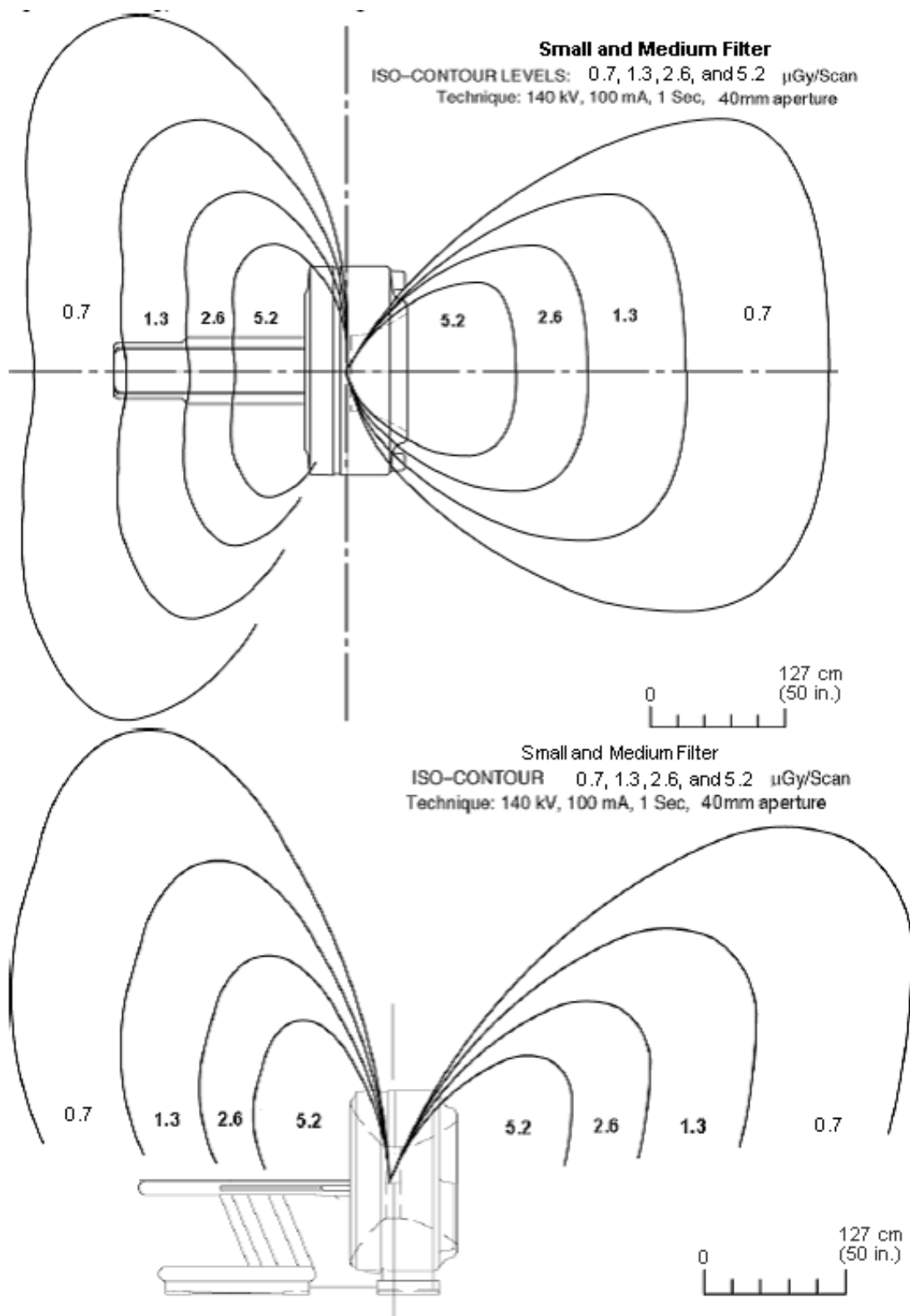


Figure 10-1 Typical Scatter Survey (Large Filter)

Figure 10-2 Typical Scatter Survey (Small and Medium Filter)



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Chapter 11

Network Requirements

Section 1.0: Network Connections

The network requirements listed in this chapter should allow you to connect the system to:

- Hospital/facility networks
- Filming cameras
- PACS
- Workstations
- Patient Information Systems

1.1 Network Type

LightSpeed VCT/Discovery CT750 HD systems require a broadband network connection.

1.2 Network Speed

The customer and the customer's IT contact should ensure that the site provides access to broadband using one of the following interface types:

- 100BASE-TX (100 Mbit/s)
- 1000BASE-T (1000 Mbit/s [1 Gbit/s]).

1.3 Network Cable Routing

The CT system connects to the facility's network through the console. To enable proper network cabling, the customer and the customer's IT contact should:

- Provide an RJ45 wall outlet within 2 m (79 in.) of the console location.
- Provide a patch cable, not to exceed 3.05 m (10 ft), to connect the operator console to a wall box. (See Notes on [Figure 13-2](#))
- Complete any cable duct-work or conduit installation that the customer site-unit might require to route connecting network cables to the workstation, camera, and operator console.
- Ensure that the run from the hospital/facility switch to the CT wall outlet does not exceed 88 m (290 ft). Bandwidth performance degrades significantly when the length exceeds 91 m (300 ft).

Section 2.0: Customer Broadband Responsibilities

2.1 Contact GE to Find Zone Broadband Specialist

Contact your GE PMI to obtain the name of the zone broadband specialist who will:

- Work with the Customer Champion to complete any identified infrastructure changes.
- Provide IP addresses for new CT equipment.
- Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption.
- Utilize an Internet Service Provider that supports static routing.

2.2 Provide GE with IT Contact Information for the Site

Provide your GE PMI with an accurate site address, telephone number, contact name, and e-mail address for the customer IT contact who will:

- Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments.
- Act as a focal point in assuring site broadband infrastructure meets GE Healthcare requirements for connection, as determined by a mutual assessment with the GE Healthcare connectivity team.
- Complete an equipment assessment with the GE Healthcare connectivity team to determine site readiness for broadband.

Chapter 12

Power Requirements

Be sure to communicate all necessary information in this chapter to the electrical contractor employed at the installation site.

Section 1.0: Introduction

The Power Distribution Unit (PDU) supplied with the LightSpeed VCT/Discovery CT750 HD system transforms and distributes power to all system components. The PDU constitutes the only power entry point required to operate the system. To minimize voltage regulation effects, keep power wiring between the facility main distribution panel and the PDU as short as possible.

When routing the power wiring, all three-phase wires and ground must run in the same conduit or raceway duct. Route power wires separate from the system control and signal cables, using a separate conduit or trough in a raceway duct. You may use a metallic conduit, floor duct, or surface raceway for running cables, depending upon local codes and practices. However, ensure that cable passageways are large enough to install additional cables with all other cables already installed. Do not use non-metallic conduit.

Section 2.0: System Input Power

2.1 Power Source Configuration

The LightSpeed VCT/Discovery CT750 HD CT scanner operates on a three-phase, solidly grounded four-wire wye or Delta power source. The neutral wire does not need to run to the system, (i.e., four-wire connection). If you are running a NEUTRAL wire, terminate it in the A1 box.

A dedicated feeder from the nearest Main Distribution Panel (MDP) should supply power to the scanner. In accordance with the National Electric Code (U.S.) and similar applicable national and local codes, the site MUST provide a protective disconnect device with LOCK-OUT and TAG-OUT provisions in the power line supplying the PDU, and MUST locate the protective disconnect device within 10 m (32 ft) of the PDU, visible to PDU service personnel. The disconnect device appears as A1 in the interconnection schematic diagrams.

2.2 Rating

The LightSpeed VCT/Discovery CT750 HD system operates on three-phase power that meets the following specifications:

- Voltage: 380 to 480 VAC
- Capacity: 150 kVA
- Frequency: 50 or 60 Hz $\pm$ 3 Hz
- Maximum power demand = 150 kVA @ 0.85 PF at a selected technique of 140 kV, 715 mA.
- Average (continuous) power demand at maximum duty cycle = 25 kVA.

The A1 disconnect device referenced above must provide overcurrent protection for the system and have at least one *Emergency Off* switch within the scan suite, near the console. The preferred disconnect utilizes undervoltage release control, rather than shunt trip devices. The rating of the A1 disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: Recommended Power Distribution System](#) for minimum rating requirements and suggested disconnect devices.

2.3 Regulation

Total load regulation, as measured at the PDU input terminals, must not exceed 6%. The capacity of the facility transformer and size and length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: Recommended Power Distribution System](#), for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges and Transients

Sags and surges of the power line must not exceed the absolute range limits shown in [Table 12-1](#). Limit maximum transient voltages to 1500 V peak.

2.6 Grounding

The customer's electrician needs to perform the following tasks:

- Bond metal conduit, raceway, or the armor of armored cable used to power the system to the PDU cabinet and to the A1 Disconnect
- Run a dedicated 1/0 (55 mm²) or larger insulated copper ground wire from the main distribution panel to the PDU with the phase wires.
- Run the ground wire with the three-phase wires from the power source to the A1 Disconnect and from A1 Disconnect to the PDU. Grounding does not require a neutral wire.

Note: The shield or armor of armored cable ALONE does NOT provide sufficient grounding.

Bond the ground wire to the intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0: Recommended Power Distribution System

In all cases, qualified personnel must verify that the transformer and feeder (at the point of take-off) and the run to the CT scanner meet all the requirements stated in this document.

3.1 Using a Dedicated Distribution Transformer (Recommended)

The recommended power distribution system for a CT Scanner is a dedicated feeder from the facility main isolation transformer. The minimum recommended transformer size for a dedicated distribution transformer provided for the scanner is 225 kVA, rated 2.4% regulation at unity power factor. [Table 12-2](#) shows the minimum recommended feeder size and overcurrent protection device based on line voltage for this configuration.

3.2 Using an Existing Distribution Transformer

If it proves necessary to power the scanner from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, avoid installation with other X-ray equipment that uses rapid film changers. These changers use a large number of high-powered, closely-spaced exposures, which may coincide with the CT scan and produce image artifacts.

3.3 System Power Requirements

Be sure that the site can meet all of the minimum power requirements listed below before installing the system:

- Maximum power demand = 150kVA @ 0.85 PF: at a Selected Technique of 140 kV, 715 mA.
- Continuous (average) power demand at maximum duty cycle = 30 kVA.
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 225 kVA, with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

Nominal line voltage MUST fall within ONE of these ranges.

Nominal Line Voltage	380	400	420	440	460	480
Hi-Line Limit, +10%	418	440	462	484	506	528
Lo-Line Limit, -10%	342	360	378	396	414	432
Continuous Line Current	38	36	34	33	31	30
Momentary Line Current	228	217	206	197	188	180
Maximum Line Current	253	241	229	219	209	200
Minimum Recommended Circuit Protection Rating	150	150	150	125	125	125

Table 12-1 Nominal Line Voltage Ranges

Feeder Length (Power Substation to A1 Disconnect)	Minimum Feeder Wire Size, AWG or MCM (sq. mm)/ VAC					
	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
15 m (50 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
30 m (100 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
46 m (150 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
61 m (200 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
76 m (250 ft)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)
91 m (300 ft)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)
107 m (350 ft)	4/0 (100)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)
122 m (400 ft)	250 (125)	4/0 (100)	3/0 (85)	3/0 (85)	3/0 (85)	2/0 (70)

Table 12-2 Minimum Feeder Wire Size

Note: In all cases the recommended ground wire is a 55 sq. mm (1/0) ground wire.

Sub-feeder Length (A1 to PDU)	Minimum Sub-feeder Wire, AWG or MCM (sq. mm)					
	380 VAC	400 VAC	420 VAC	440VAC	460VAC	480VAC
9.7536 m (32 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)

Table 12-3 Minimum Sub-Feeder Wire Size

The information in [Table 12-1](#), [Table 12-2](#), and [Table 12-3](#) (above) assumes the use of copper wire, rated 75 C and run in steel conduit. All ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002). The ampacity of the circuit protection device listed above determines the minimum feeder size, except where total source regulation limits require a larger size.



NOTICE

Power feeders running under the scan room floor, as well as power vault substations under the floor, above the scan suite, or in adjacent rooms, may cause excessive EMI fields. The responsibility for meeting all site EMI requirements rests with the customer.

Section 4.0: Ground System

The design of the CT750 HD scanner uses an equal potential grounding system. [Figure 12-1](#) shows the required ground system. Three primary grounding points exist, and include:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

The customer electrician should ground ALL patient-touchable metal surfaces to the same potential as the A1 Disconnect.

For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

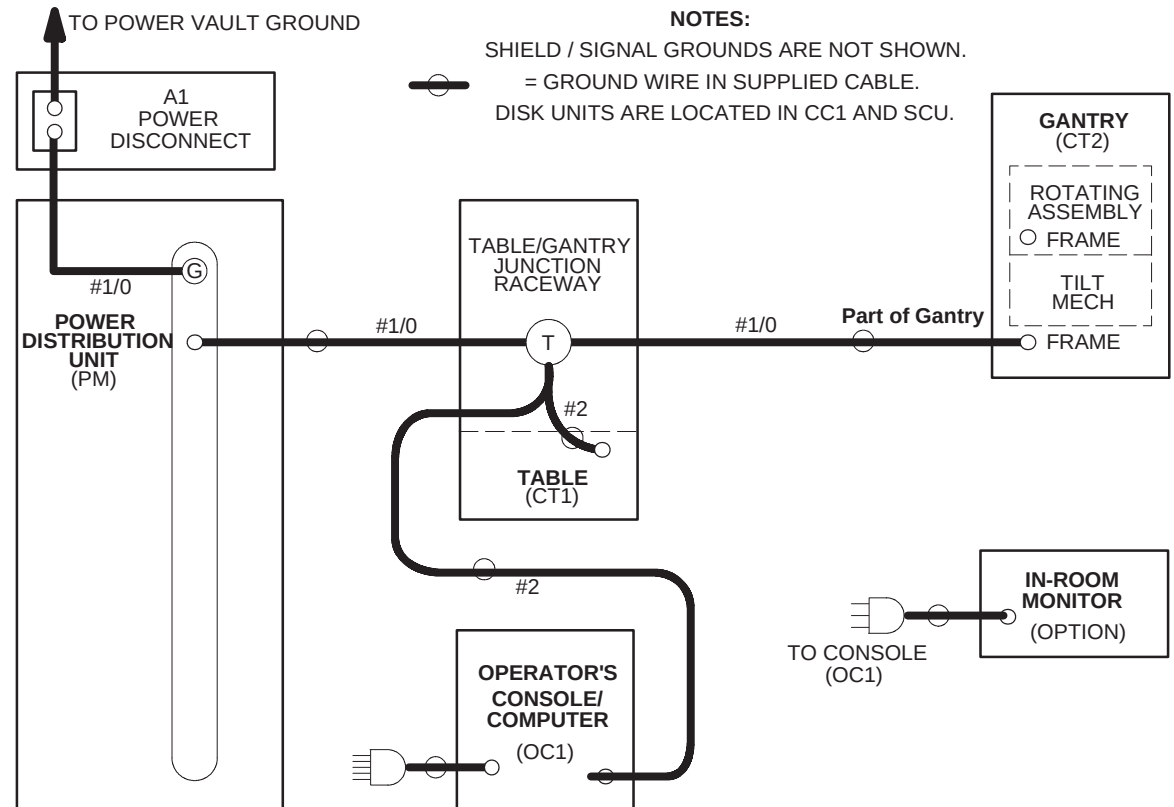


Figure 12-1 System Ground Map

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Chapter 13

Interconnection Data

Section 1.0: Introduction

The customer and the customer's electrical contractor should refer to the information in this section when establishing network and power interconnections for the system. Please note the following:

- [Figure 13-2](#) shows interconnection runs for a 50/60 Hz system.
- [Table 13-1](#) shows component designators for supplied equipment and options and wall power outlets.
- [Table 13-7](#) lists customer-installed wiring and supplied cables. The actual length of each run is less than the length of supplied cables to allow for routing inside the equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.
- [Table 13-2](#) and [Table 13-5](#) list details for connection to LightSpeed series equipment using standard (short) length and non-standard (long) length cables, respectively. Details appear for the following types of runs, when appropriate:
 - Flush-floor duct
 - Computer floor
 - Through-wall bushing
 - Junction box
 - Surface floor duct
 - Through-floor duct
 - Wall duct
 - Conduit
- To minimize the need for additional junction boxes, use either a cable raceway system or a raised computer floor. LightSpeed series systems use prefabricated cables with large plugs. Therefore, try to avoid conduit or pipe for cable runs.

Section 2.0: Component Designators

Designator	Applies To	Source
A1	Primary power disconnect	Contractor-supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor-supplied
OC1	Operator console/computer	System
PDU	Power Distribution Unit	System
SEO	System emergency off	Contractor-supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor-supplied
DS	Door Interlock Switch	Contractor-supplied
BBNC	Broadband network connection	Contractor-supplied

Table 13-1 Component Designators

Section 3.0: Interconnect Runs, Wiring, and Cables

3.1 GE Healthcare Supplied (HD Standard Length 2281840-10)

Run #	Length, Actual (Usable)		Part #	Description	UL Cable Information								Pull Size mm (inches)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	9 (6)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 0(.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	28 (20)	9 (6)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	11 (0.44) Dia
052	28 (20)	9 (6)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	120/208Y	90	14 0(.542)	5	8	56 (2.22) Dia
053	65 (60)	20 (18)	5121809-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (0.483)	4	10	56 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	43 (35)	13 (11)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 0(.608)	1	1/0	16 (0.62) Dia
056	71 (60)	22 (18.3)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (0.467)	1	2	12 (0.48) Dia
100	32.5 (20)	10 (6)	5120646-2	Signal, Gantry TGPU to PDU		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	71 (60)	22 (18)	5120645-2	Signal, Gantry TGPU to OC		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	68 (60)	21 (18)	2373436-3	Signal (LAN), Gantry to OC			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
103	68 (60)	21 (18)	5316594-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (0.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	
	68 (60)	21 (18)	5193969-6	Signal (LAN) (includes 2 cables)			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
		25 (24)	5169456-2	Signal, Gantry to Injector		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
			5199717-2	Signal, Gantry to RPM cable		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)

Table 13-2 GE Healthcare-Supplied Cables (Standard Run) - UL Information

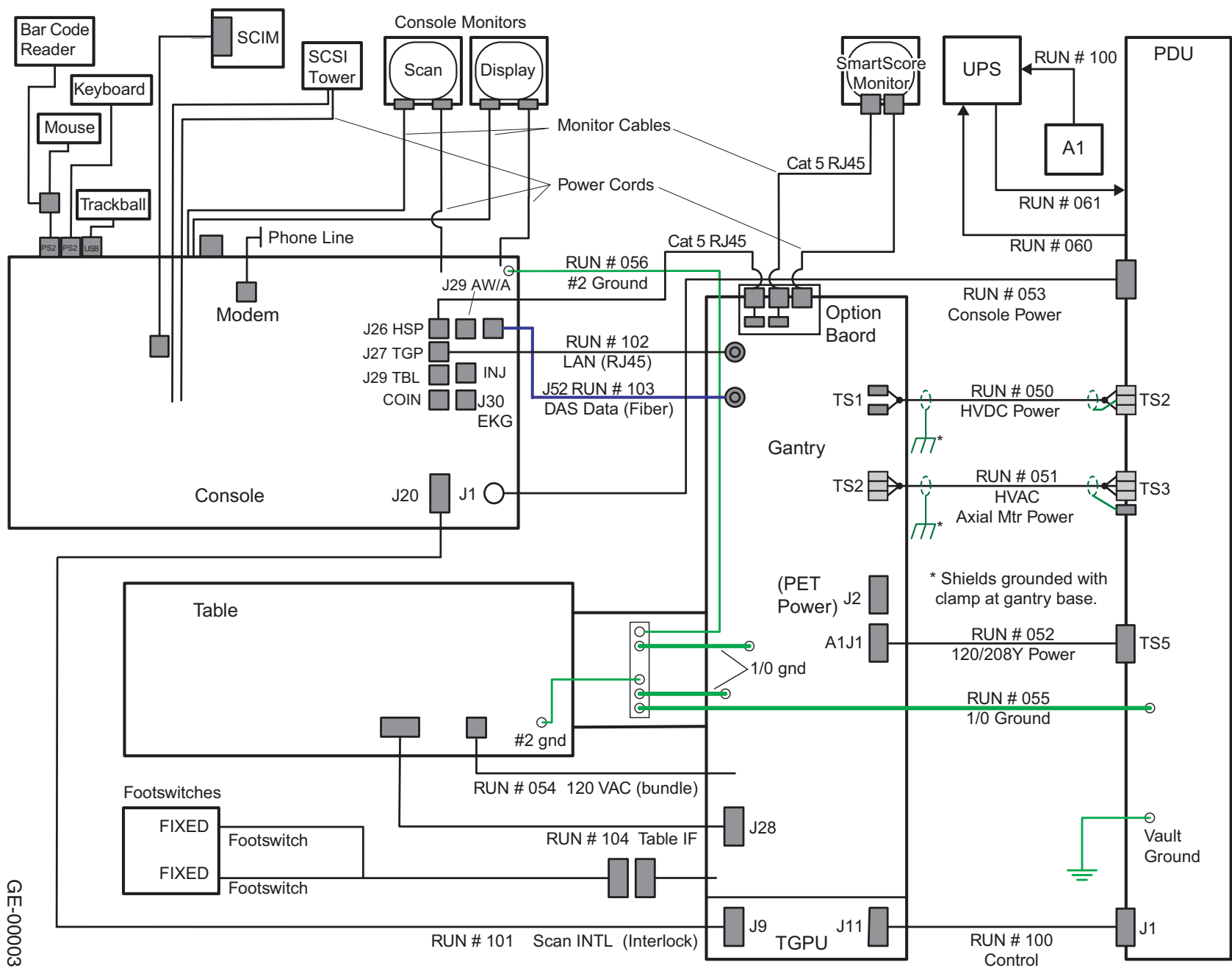


Figure 13-1 System Interconnect Diagram

3.2 GE Healthcare-Supplied (HD Optional, Long Run Length 2281840-9)

Run #	Length, Actual (Usable)		Part #	Description	UL Cable Information								Pull Size mm (inches)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	64 (55)	19.35 (17)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 0(.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	84 (75)	25.5 (23)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	11 (0.44) Dia
052	87	26.35	2343528	LVAC, PDU to Gantry	2587	FT4	600	120/208Y	90	14 0(.542)	5	8	56 (2.22) Dia
053	80 (75)	24 (23)	5121809	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (0.483)	4	10	56 (2.22) Dia
054	80 (75)	24.5 (23)	n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	87	26.35 (23)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 0(.608)	1	1/0	16 (0.62) Dia
056	64 (55)	19.35 (17)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (0.467)	1	2	12 (0.48) Dia
100	64	19.35 (17)	5120646	Signal, Gantry TGPU to PDU		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	64	19.35 (17)	5120645	Signal, Gantry TGPU to OC		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	70	21.4	2373436-2	Signal (LAN), Gantry to OC			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
103	92	28.15	5316594	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (0.39) Dia
104	100	30.48	n/a	Signal, Gantry to Table		FT-4	300		80		25	22	
	100	30.48	5193969-4	Signal (LAN) (includes 2 cables)			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
		25 (24)	5169456	Signal, Gantry to Injector		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
			5199717	Signal, Gantry to RPM cable		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)

Table 13-3 GE Healthcare-Supplied Cables (Optional, Long Run) - UL Information

3.3 GE Healthcare-Supplied (VCT 64 Standard Length 2281840-7)

Run #	Length, Actual (Usable)		Part #	Description	UL Cable Information								Pull Size mm (inches)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
055	28 (20)	9 (6)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (0.608)	1	1/0	16 (0.62) Dia
056	28 (20)	9 (6)	2371450-4	Ground, Console to Raceway	1283	VW-1 (FT-1)	600	0	105	11.9 (0.467)	1	2	12 (0.48) Dia
102	28 (20)	9 (6)	2373436-3	Gantry to Console LAN	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (0.234)	1	19	15 (0.59) Dia
	65 (60)	20 (18)	5125259-2	Fiber Optic, Gantry to Console 24 m, VCT HD			N/A	N/A			1	N/A	
			5121809-2	VCT Console Power	2587	FT-4	600	208Y/120	90	12.3 (0.483)	4	10	56 (2.22) Dia
	43 (35)	13 (11)	5120645-2	Cable Console to MSUB J9 - EMC Edition 2	UL	FT-4	300	<30VDC	80	11.2 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
	71 (60)	22 (18.3)	2343529-2	Cable, HVDC PDU to Gantry	2587	FT-4	600	350VDC	90	19 (0.751)	2	(2) 4 (1) 8	22 (0.87) Dia
	32.5 (20)	10 (6)	2343528-2	Cable, PDU to Gantry; 120 VAC Power	2587	FT-4	600	208Y/120	90	13.8 (0.542)	5	8	56 (2.22) Dia
	71 (60)	22 (18)	2343530-2	Cable, Axial Drive Power PDU to Gantry	Flexible Motor Supply Cable	FT-4	600	440Y/254	90	15.3 (0.604)	4	12	11 (0.44) Dia
	68 (60)	21 (18)	5120646-2	Cable, PDU to MSUB J11 - EMC Edition 2	UL	FT-4	300	<30VDC	80	13.3 (0.525)	37	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
	68 (60)	21 (18)	2373436-6	Cable LAN	UL	FT-4			60			24	

Table 13-4 GE Healthcare-Supplied Cables (Optional, Long Run) - UL Information

3.4 GE Healthcare-Supplied (VCT Optional, Long Run 2281840-6)

Run #	Length, Actual (Usable)		Part #	Description	UL Cable Information								Pull Size mm (inches)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
055	64 (55)	19.35 (17)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (0.608)	1	1/0	16 (0.62) Dia
056	84 (75)	25.5 (23)	2371450-3	Ground, Console to Raceway	1283	VW-1 (FT-1)	600	0	105	11.9 (0.467)	1	2	12 (0.48) Dia
102	87	26.35	2373436-2	Gantry to Console LAN	UL (RG-223/U)	FT-4	1900	<30VDC		5.9 (0.234)	1	19	15 (0.59) Dia
	80 (75)	24 (23)	5125259	Fiber Optic, Gantry to Console 24 m, VCT HD			N/A	N/A			1	N/A	
	80 (75)	24.5 (23)	5121809	VCT Console Power	2587	FT-4	600	208Y/120	90	12.3 (0.483)	4	10	56 (2.22) Dia
	87	26.35 (23)	5120645	Cable Console to MSUB J9 - EMC Edition 2	UL	FT-4	300	<30VDC	80	11.2 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
	64 (55)	19.35 (17)	2343529	Cable, HVDC PDU to Gantry	2587	FT-4	600	350VDC	90	19 (0.751)	2	(2) 4 (1) 8	22 (0.87) Dia
	64	19.35 (17)	2343528	Cable, PDU to Gantry; 120 VAC Power	2587	FT-4	600	208Y/120	90	13.8 (0.542)	5	8	56 (2.22) Dia
	64	19.35 (17)	2343530	Cable, Axial Drive Power PDU to Gantry	Flexible Motor Supply Cable	FT-4	600	440Y/254	90	15.3 (0.604)	4	12	11 (0.44) Dia
	70	21.4	5120646	Cable, PDU to MSUB J11 - EMC Edition 2	UL	FT-4	300	<30VDC	80	13.3 (0.525)	37	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
	92	28.15	2373436-4	Cable LAN	UL	FT-4			60			24	

Table 13-5 GE Healthcare-Supplied Cables (Optional, Long Run) - UL Information

3.5 UPS Wiring Cables

	Run #	Length Actual (Usable)		Part #	Description	UL Cable Information								Pull Size	
		ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	mm (inches)	
Standard	060	19 (15)	6 (5)	5125079	PDU to UPS	2587	FT4	600	+ & - 350VDC	90	19 (0.751)	3	4	22 (0.87) Dia	
	061	19 (15)	6 (5)	5125079-2	UPS to PDU	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	22 (0.87) Dia	
	110	45 (40)	13.7 (12.7)	5169224	A1 to UPS	2587	FT4	600	208Y/120	90	14 (0.542)	5	8	25 (1.0) Dia	
Long	060	62 (58)	19 (18)	5125079-5	PDU to UPS	2587	FT4	600	+ & - 350VDC	90	19 (0.751)	3	4	22 (0.87) Dia	
	061	62 (58)	19 (18)	5125079-5	UPS to PDU	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	22 (0.87) Dia	
	110	62 (58)	19 (18)	5169224-2	A1 to UPS	2587	FT4	600	208Y/120	90	14 (0.542)	5	8	25 (1.0) Dia	

3.6 A1 UPS

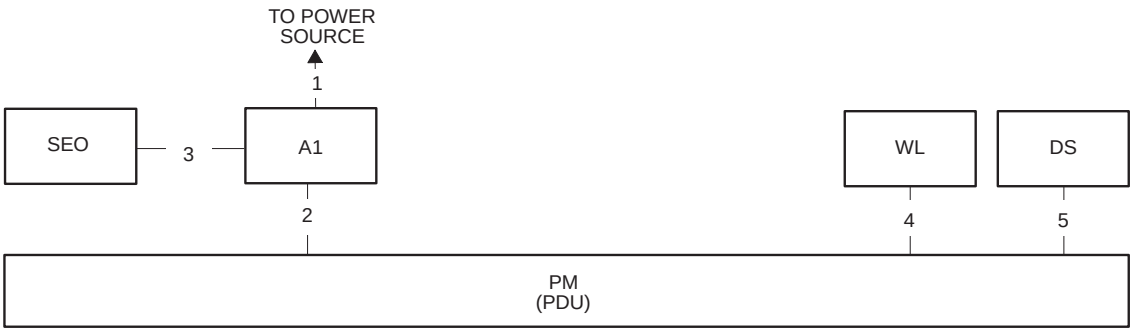
PDU Type & Model No.	Maximum Mom. kVA Rating	Required Main Disconnect (A1) Catalog No.		Optional Partial UPS Kit Catalog No.
		Europe & Asia (380-400V or 420V)	North America (440V or 460-480V)	
NGPDU-4 2326492-6	150kVA	E4502AF (150A) (incl. Auto Restart & Integrated UPS Control) (Additional A1 Disconnects available for Europe through the European Sales Team.)	E4502AE (125A) (incl. Auto Restart & Integrated UPS Control)	B7864PZ PowerWare 9355-15- 14GE (14.4 kVa - 40A) REQUIRES one of the A1 Panels shown at left or equal.

Table 13-6

3.7 Contractor/Customer-Supplied

Customer Installed Wiring		Description	Cables Supplied			Plug Pulling Dimensions		Wire and Cable Pigtails ft. (M.)	
Qty	Size AWG (mm ²)		Part No	LENGTH ft. (M.)	DIA. in (mm)	From	To	From	To
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER DISTRIBUTION UNIT (A1 - PM)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
-	-	NEUTRAL -- Not Required						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER DISTRIBUTION UNIT TO WARNING LIGHT CONTROL (PDU - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL TS6 1, 2, 3, 4, 5, 6, 7, 8							
RUN NO. 5 POWER DISTRIBUTION UNIT TO SCAN ROOM DOOR INTERLOCK (PDU - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK TS6 9, 10							
*	REFER TO Table 12-2 on page 102 FOR AWG (MM2) WIRE SIZES.								
RUN NO. n/a BBNC									
1	customer determined	Hospital Broadband Network Connection (Wall Jack: Placed on the wall behind the console.)							

Table 13-7 Runs 1, 2, 3, 4 and 5 Connections



- NOTES:**
- 1) Used for remote diagnostics - Option
 - 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
 - 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m
 - 4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the compliant cable/wire is recommended. For China market, China end-user shall purchase the power supply cable that has the CCC mark.

Only one phone connection is required for the system.

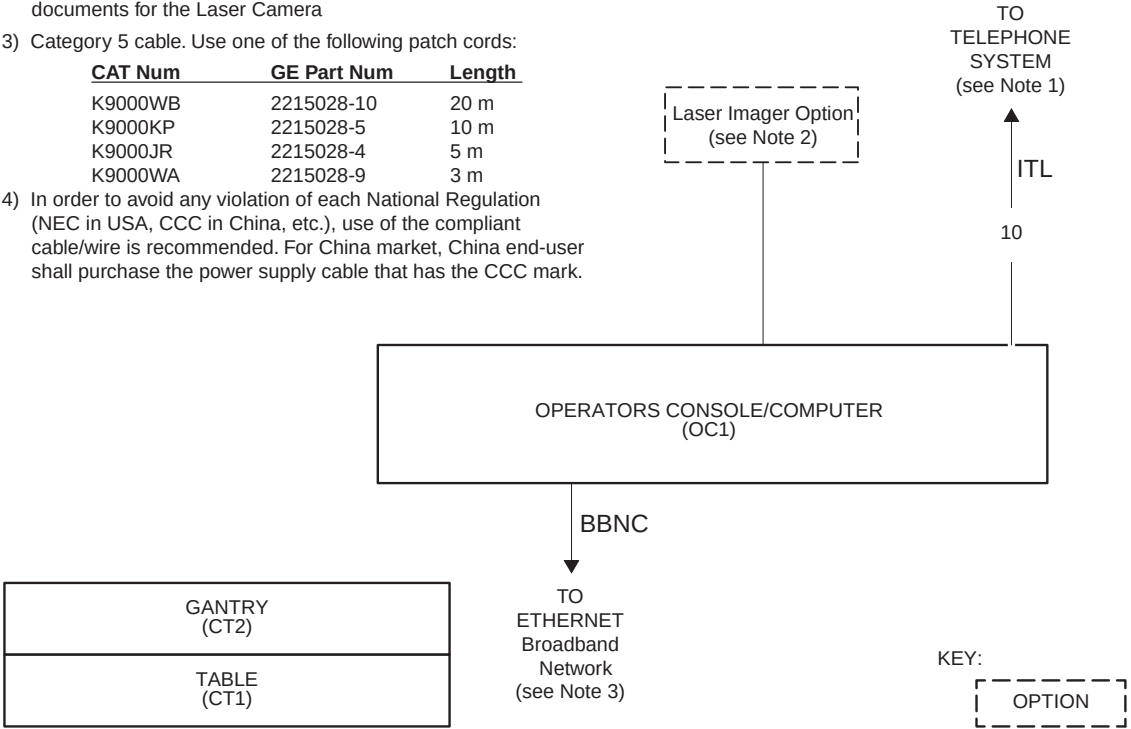


Figure 13-2 Interconnection Runs

Section 4.0: Contractor Supplied Components

Reference	Associated Equipment	Material/Labor Supplied by Customer Contractor	USA Vendor/CAT No. GE Catalog
A1 380 - 480V 50/60 Hz	Circuit Breaker with Magnetic Contactor	Three Pole, 380V - 480V, Combination breaker with magnetic contactor. Includes control transformer, optional UPS interface, On/Off controls and auto-restart feature, if GE-supplied.	Recommend*: - E4502AE (125A) - E4502AF (150A) Optional remote operator control available from GE Supply, Cat # GESCTR0CS1
BBNC (strongly recommended)	Broadband Network Connection	Broadband network connection wall jack, located within 1 m (39 in) of Operator Console location, for internal hospital networking and InSite Broadband connectivity. Cabling to conform to facility's IT standards.	
ITL (optional)	In-Suite Telephone Lines	Supply two (2) voice-grade telephone lines. One line must be a direct number from outside the facility—do not route this line through a telephone switchboard. Telephone line operating charges are paid by the customer.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	
		Room Warning Light Controller	E4500AM

*Refer to [Table 13-7 on page 112](#).

Table 13-8 Contractor-Supplied Components

Section 5.0 Primary Power Disconnect

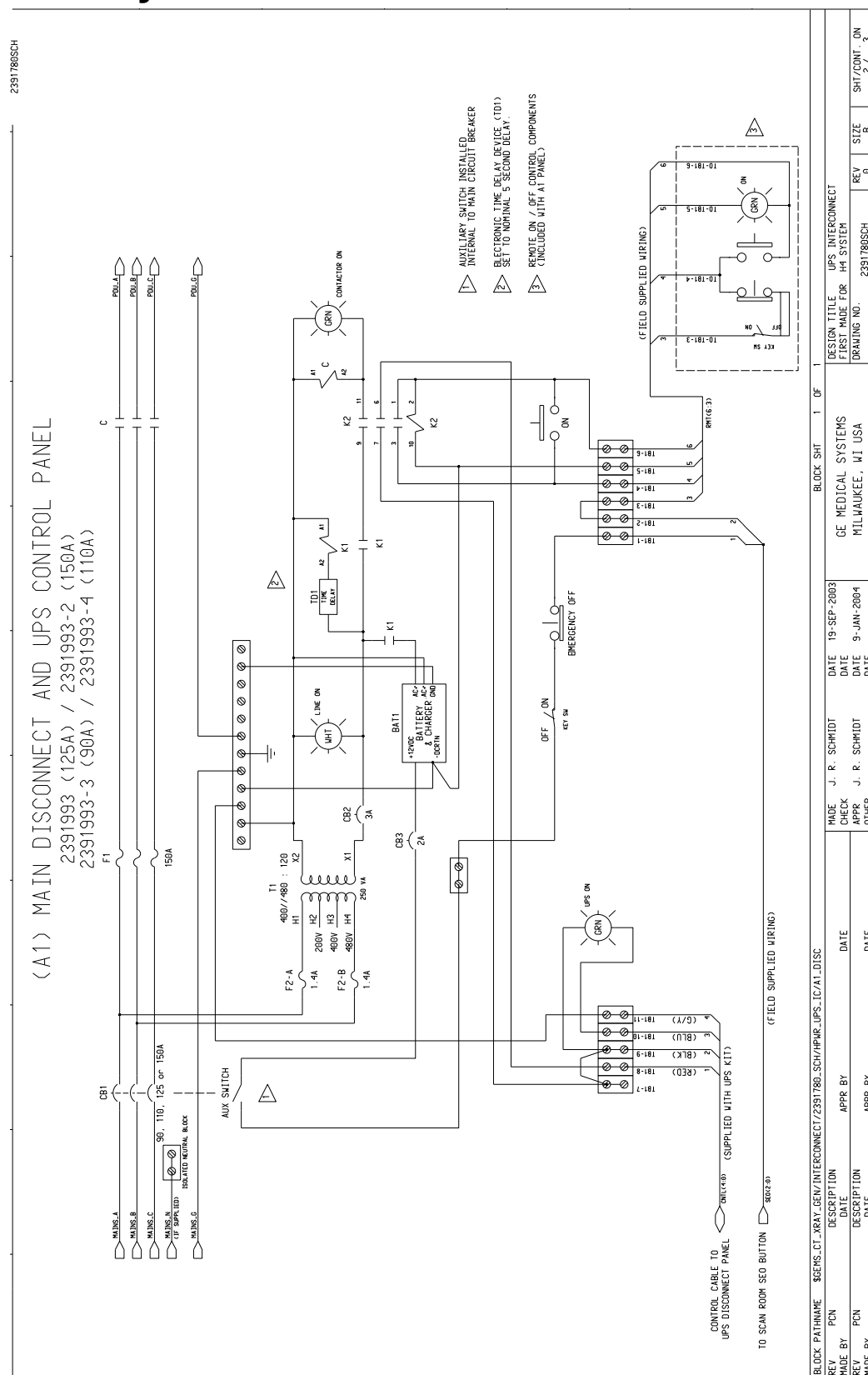


Figure 13-3 Typical Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor

Section 6.0

Scan Room Warning Light and Door Interlock

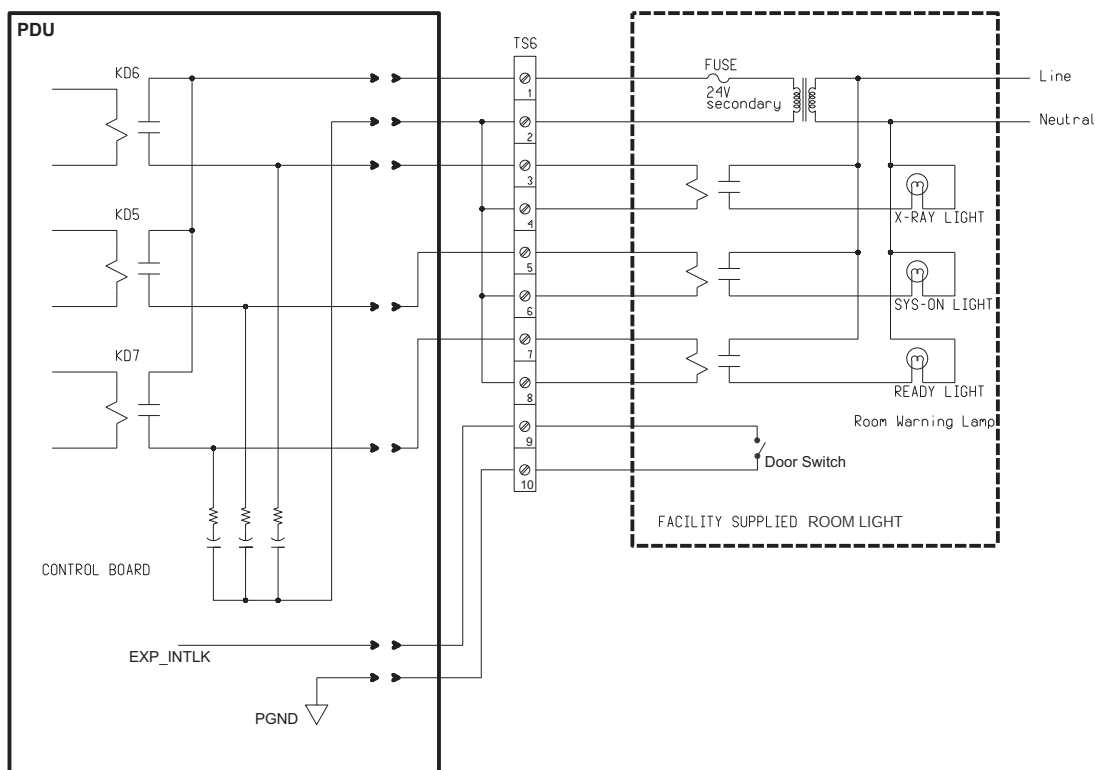


Figure 13-4 TS6 X-Ray Warning Light Connections

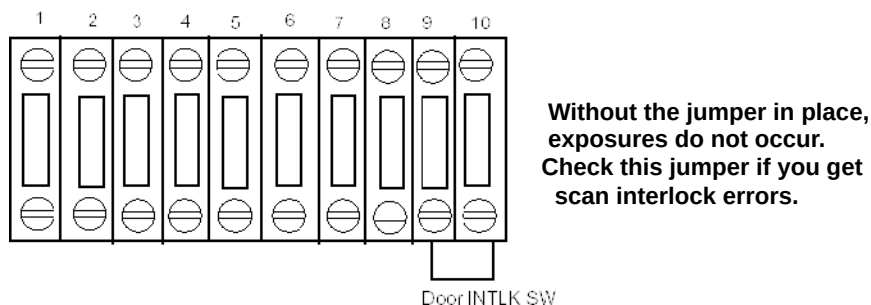


Figure 13-5 TS6 Room Door Interlock Connections - Without Door Interlock

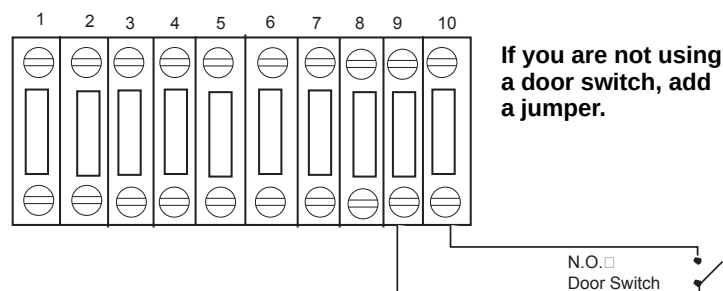


Figure 13-6 TS6 Room Door Interlock Connections - With Door Interlock

Chapter 14

Delivery and Storage Requirements

This chapter provides information necessary for planning a safe and successful delivery of the LightSpeed VCT/Discovery CT750 HD system from GE Healthcare to the receiving area of the installation site, and from the receiving area of that facility to the scan suite.

Section 1.0: Delivery to the Facility

Your Project Manager of Installation will determine the most appropriate means of transporting the system to your facility. However, the type of receiving area at the facility where the installation will occur determines, to a large extent, the method used to transport the system to that facility. When planning for delivery, facilities fall into two general categories: those with a loading dock, and those without a loading dock.

1.1 Loading Dock Deliveries

Facilities with a loading dock in their receiving area can generally accommodate delivery of the system by van. This is the preferred method of transporting the system to the installation site, as dock-to-dock shipment by van minimizes the possibility of dropping the gantry. Also, packing the CT system for van shipment involves minimum tear-down of components. This system is shipped Lean packed on pallets and dollies with approximately 10 units.

1.2 Ground (Non-Loading Dock) Deliveries

Facilities without a loading dock usually require ground delivery by either liftgate or tiltbed truck. Such deliveries require unloading the system components from the truck and then rolling them across smooth sidewalks or other paved surfaces into the facility.

1.2.1 Liftgate Truck

Delivery of the system by liftgate truck requires an appropriate capacity truck with a liftgate capable of lifting 3 tons. If using a rollback truck, the Project Manager of Installation should be on-site at the time of delivery to supervise this operation in person.

1.2.2 Tiltbed Truck

Delivery of the system by tiltbed truck also requires an appropriate capacity truck, capable of lifting 3 tons. Safe transport of the system by tiltbed truck requires securing the components to the truck to prevent damage during transportation. To avoid damage to the gantry or dolly when removing the gantry from a tiltbed truck, the Project Manager of Installation should direct the driver to attach straps to the lowest possible point on the dolly and lower the gantry at the slowest reasonable rate.

1.2.3 Forklift Truck

A forklift can be used to unload the gantry, provided that the *lifting option* is ordered and delivered. The system will arrive with a lifting skid attached to the gantry and table. This option cannot be added later as an on-site addition.

Section 2.0: Delivery to the Scan Suite

Once at the installation site, conveyance of the system into the scan suite may involve special considerations, such as vertical lifting, or transportation through stairwells, which involves additional planning by the Project Manager of Installation.

2.1 Lifting

Both vertical and horizontal lifting require professional riggers. The PMI should always notify CT engineering before attempting either lifting procedure and should make sure that the order includes the necessary lifting fixtures, as both vertical and horizontal fixtures must appear on the order for them to ship with the system.

If delivery requires vertical lifting, the PMI adds the appropriate identifier to the order. The gantry ships in a vertical lifting crate with lifting instructions for riggers.

If delivery requires horizontal lifting, the PMI adds the corresponding identifier to the order. The gantry ships in a horizontal lifting crate with lifting instructions for riggers.

2.1.1 Stairway Deliveries

Stairways with angles at or less than 45 degrees can accommodate delivery of system components. If the site requires delivery through stairwells, the PMI adds the appropriate identifier to the order to ensure proper packaging of the system, and notifies CT engineering before attempting the procedure. The components ship attached to special lifting skids with lifting instruction for riggers.

2.2 Floor Protection

GE recommends floor protection along the delivery path from the dock/receiving area to scan room.

2.3 Un-loading and un-packing the System

Retain the packaging surrounding the following components:

- Console-Shipped on a shock resistant skid. Do not remove the skid.
- UPS-Shipped on a shock resistant skid. Do not remove the skid.

Section 3.0: Dollies

3.1 Installations within the United States

Typically, domestic shipments (shipments within the United States) involve the use of dollies for moving the gantry, table, and operator console. After completing installation, return the dollies to GE using the shipping document found in Box #1.

3.2 Zero Clearance Dollies

Deliveries involving small elevators with a depth of at least 2692 mm (106 in.) require zero clearance dollies. Zero clearance dollies allow movement of the gantry in tight areas; avoid using them for normal dock or van deliveries. To order zero clearance dollies, go to: <http://www.umi-dollyshop.com>.

3.3 Tilting Table Dollies

Deliveries involving small elevators with a depth of at least 2438 mm (96 in.) require tilting table dollies. If storing the system prior to installation, do not order tilt dollies. If you are unable to obtain tilt dollies for delivery, substitute riggers in their place. A limited number of tilt dollies exist for U.S. deliveries. To order tilt dollies, go to: <http://www.umi-dollyshop.com>.

3.4 Installations Outside of the United States

Customers may purchase dollies (B7850LD) for shipments outside of the United States. After removing the system from the crates, DO NOT return dollies shipped outside of the US to GE Healthcare in Milwaukee, WI, USA. Instead, forward them to the local GE office or warehouse. Zero Clearance and Tilting Table dollies can be purchased through UMI, To buy tilt dollies, go to: <http://www.umi-dollyshop.com>.

Section 4.0: Gantry Delivery Considerations

4.1 Gantry Shipping State

The gantry ships with most covers installed, and the assembly mounted between two dollies (see [Figure 14-1](#)). Two side rails, bolted to the dollies, stabilize the dollies and protect the gantry. Use the dolly elevating casters to lift the gantry off its base and roll it into position.



Figure 14-1 Gantry with Shipping Dollies and Side Rails

4.2 Door Openings

Unobstructed door openings, for moving equipment into the building, must measure 1067 mm X 2083 mm (42 in. X 82 in.) minimum. Corridors with a width of 2439 mm (8 ft.) also prove helpful.

4.3 Elevator Requirements

When moving the gantry from the receiving location to the scanning room, pay special attention to elevator size and capacity. Removing side rails and one dolly after placing the gantry in the elevator reduces the gantry width/length and elevator depth requirements.

Due to gantry component weight differences all weights listed below are averages. This change can measure ± 18.14 kg (± 40 lb). Contact the elevator manufacturer if the gantry weight exceeds elevator capacity (see [Table 14-1](#)).

Configuration	Length	Width	Height	Weight
Dollies On, Side Rails On	2900 mm (114 in.)	1290 mm (51 in.)	2000 mm (79 in.)	1932 kg (4260 lb)
Dollies On, Side Rails Removed	2900 mm (114 in.)	1000 mm (39.4 in.)	2000 mm (79 in.)	1910 kg (4210 lb)
Dollies Off, Covers Off	2060 mm (81 in.)	860 mm (34 in.)	1850 mm (73 in.)	1864 kg (4110 lb)

Table 14-1 Size of Gantry & Dollies, with and without Side Rails

The minimum hallway and door size for a gantry with covers and dollies attached but side rails removed, is 1016 mm (40 in.). For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Section 5.0 Table Delivery Considerations

The table ships without side covers installed. Covers ship in four separate boxes; the table ships mounted between two dollies. Table shipping dimensions are 3200 mm (126 in.) long, 650 mm (25.6 in.) wide, and 1200 mm (47.2 in.) high.

Description	Length		Width		Height	
	mm	in	mm	in	mm	in
GT2000 w/Dollies	2997	118	762	30	1143	45
GT1700 w/Dollies	2489	98	762	30	1143	45
GT2000 w/o Dollies	2997	118	660	26	889	35
GT1700 w/o Dollies	2489	98	660	26	889	35

Table 14-2 GT Table Dimensions -- with and without dollies

Description	Length		Width		Height		Weight	
	mm	in	mm	in	mm	in	kg	lb
GT2000 Dollies	2210	87	762	30	127	5	132	290
GT1700 Dollies	210	87	762	30	127	5	132	290
GT2000 Tilting	762	30	762	30	889	35	136	300
GT1700 Tilting	762	30	762	30	889	35	132	291

Table 14-3 GT Table Dolly Dimensions

Description	Length		Width		Height		Weight	
	mm	in	mm	in	mm	in	kg	lb
GT2000	2997	118	762	30	1143	45	632	1390
GT1700	2489	98	762	30	1143	45	602	1325
GT2000 - Tilting (approx. dimensions)	2489 - 2921	98 - 115	660	26	1778 - 2032	70 - 80	636	1400

Table 14-4 GT Table Elevator Delivery Dimensions

Section 6.0 Console Delivery Considerations

The operator console ships without the keyboard table installed. The keyboard table ships with the operator console.

- The operator console ships on a skid equipped with ramps for unloading.
- Do not remove the operator console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) measure 1016 mm (40 in.) deep, 1346 mm (53 in.) wide, and 1041 mm (41 in.) high.

Note: The dimensions of the operator console alone (as shipped) measure 888 mm (34.95 in.) deep, 1238 mm (48.74 in.) wide, and 750 mm (29.5 in.) high.



NOTICE Do not lift the operator console by the monitor table top to remove it from the shipping skid.



Figure 14-2 Console Ready for Unloading from the Shipping Skid

Section 7.0: Storage Requirements



NOTICE Failure to adhere to storage requirements can result in equipment damage.

7.1 Short-term Storage (Less than Six Months)

If storing the CT system before installation for less than six months, store it in a temperature- and humidity-controlled warehouse. Protect it from weather, dirt, and dust.

Meeting the following requirements prevents rust and corrosion from forming on bearing surfaces due to condensation:

- Storage temperature should not exceed 4° to 27° C (40° to 80° F).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum rate of relative humidity change measures 5%/hr.
- Maximum rate of temperature change measures 3° C/hr. (5° F/hr.)
- Storage longer than 6 months is not recommended



NOTICE Between delivery qualifies as short-term storage. Van storage must meet the same specifications listed above.

7.2 Construction-Site Storage

When storing the CT system at a construction site be sure to adhere to the following storage requirements:

- Do not damage or puncture the shipping crate.
- Do not remove packaging until all construction is completed at the site and all dust created by the construction is removed.
- Maintain a storage temperature within the range of 10° to 32° C (50° to 90° F).
- Maintain a relative humidity (non-condensing) between 20% and 70%.

Section 8.0 Extreme Temperature Delivery and Storage



NOTICE Failure to adhere to extreme temperature requirements during delivery and storage can result in equipment damage.

Avoid extreme temperatures during system transportation and delivery.

Extreme temperatures consist of temperatures below -18° C (0° F), or above 49° C (120° F), without humidity control.

When transporting the CT system, prevent extended exposure of the system to temperatures or humidity outside of the following specifications:

- Temperature: -40° to +70° C (-40° to +158° F)
- Humidity: 5% to 95%



NOTICE Component freezing occurs when exposing the CT system to temperatures below -18° C (0° F) for a period longer than two (2) days. Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

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Chapter 15

Handling Requirements

Communicate the information in this chapter to any personnel who will transport, move, or otherwise handle the system components during transportation and delivery of the system.

Section 1.0: Transportation

To avoid dropping the gantry, it is recommended that the system is transported from GE Healthcare to the facility of the installation site, shipping dock-to-dock in a van. However, facilities without a loading dock may transport the system using liftgate or flatbed trucks, provided that no dropping or mishandling of the system occurs. These methods involve unloading system components from the truck and then rolling them across SMOOTH sidewalks or other paved surfaces.

Section 2.0: Handling Requirements

The design of the LightSpeed VCT/Discovery CT750 HD System does not tolerate dropping, shock, vibration, tipping, or hoisting. Be sure to communicate these handling requirements to all parties involved in transporting, moving, and handling system components.

2.1 Avoid Dropping

Never drop the gantry, console, table, or PDU. A drop from a height greater than 13 mm (0.5 in.) may cause structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not become apparent until after the system is installed.

2.2 Avoid Shocks and Vibrations

The design of the LightSpeed VCT/Discovery CT750 HD System, including the gantry, console, table, and PDU, do not tolerate excessive shock or vibration, which may occur during unloading. For example, rolling the console across a “washboard” style ramp may vibrate components, causing loose or broken connections. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive, or console failure) may not become evident until after the system is installed.

2.3 Avoid Tipping

All system components must remain upright at all times; avoid tipping them. Move the gantry by rolling it on its dollies ONLY, do NOT hoist it. Avoid tipping or lifting the gantry when moving it through hallways, doorways, elevators, etc.



NOTICE

Never lift the gantry with a forklift. Lifting the gantry requires engineering approval for each occurrence. Your GE PMI should contact CT Engineering for all special lifting requirements, as unauthorized gantry lifting can cause gantry bearing damage.

2.4 Inclines and Flat-bed Truck Removal

When moving the gantry down steep inclines, including removing the gantry from a flat-bed wrecker, attach the straps to the **LOWEST** possible point on the dolly, and lower the gantry at the **SLOWEST** reasonable rate, (see [Figure 15-1](#)).



Figure 15-1 Proper Gantry Strap Location

 **WARNING** SOME ASSEMBLIES MAY BE TOP-HEAVY. BE CAREFUL NOT TO TIP!

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